

Learning what we never observe

Maximum likelihood, expectation–maximization and hidden-variable models

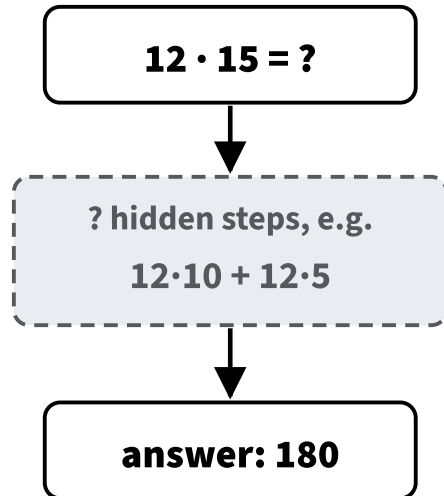
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In real data, the key variable is often never written down

 dashed box: never recorded (hidden)  solid box: in the data

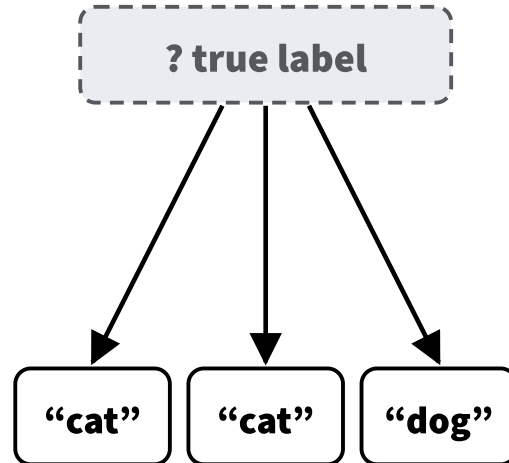
Learning to reason



data: question + answer

to learn from the data:
**good reasoning steps
from answers alone**

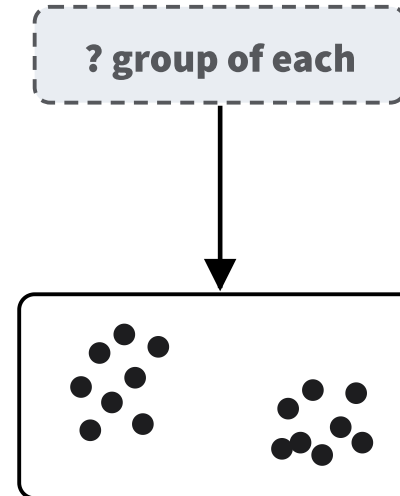
Cleaning labels



data: three annotators' labels
they disagree

to learn from the data:
**the true labels and
each annotator's skill**

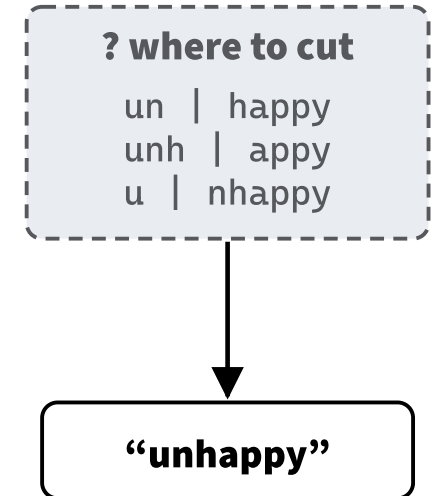
Fast vector search



data: the vectors

to learn from the data:
**the groups, so a query
checks its own group only**

Splitting words



data: the words of a text

to learn from the data:
**how often each piece
occurs, so where to cut**

All four use expectation-maximization (EM) or its hard-label form, k-means.

Two facts from earlier lectures that EM reuses

A coin shows 7 heads in 10 flips. What is our best guess of its heads chance?

1/2

7/10

3/10

Answer. 7/10: count heads, divide by flips. Today we prove this is the **maximum-likelihood** answer.



7 heads (H, blue) and 3 tails (T) in 10 flips

heads ÷ flips = 7/10

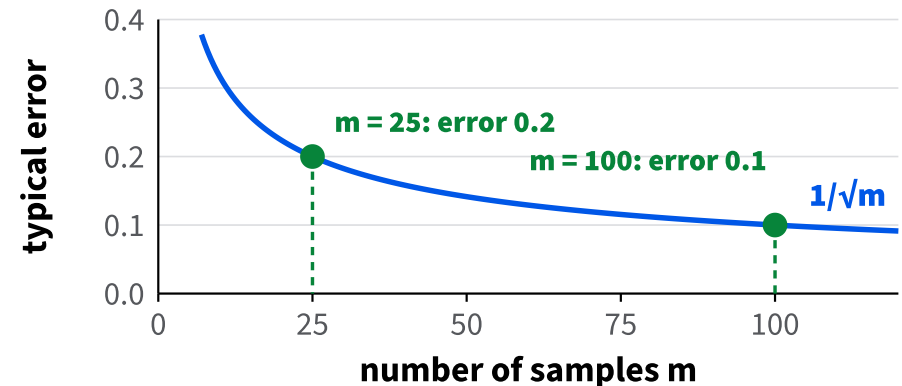
Monte Carlo (Lecture 4): with 4 times as many samples, the error shrinks by a factor of ...

4

1: no change

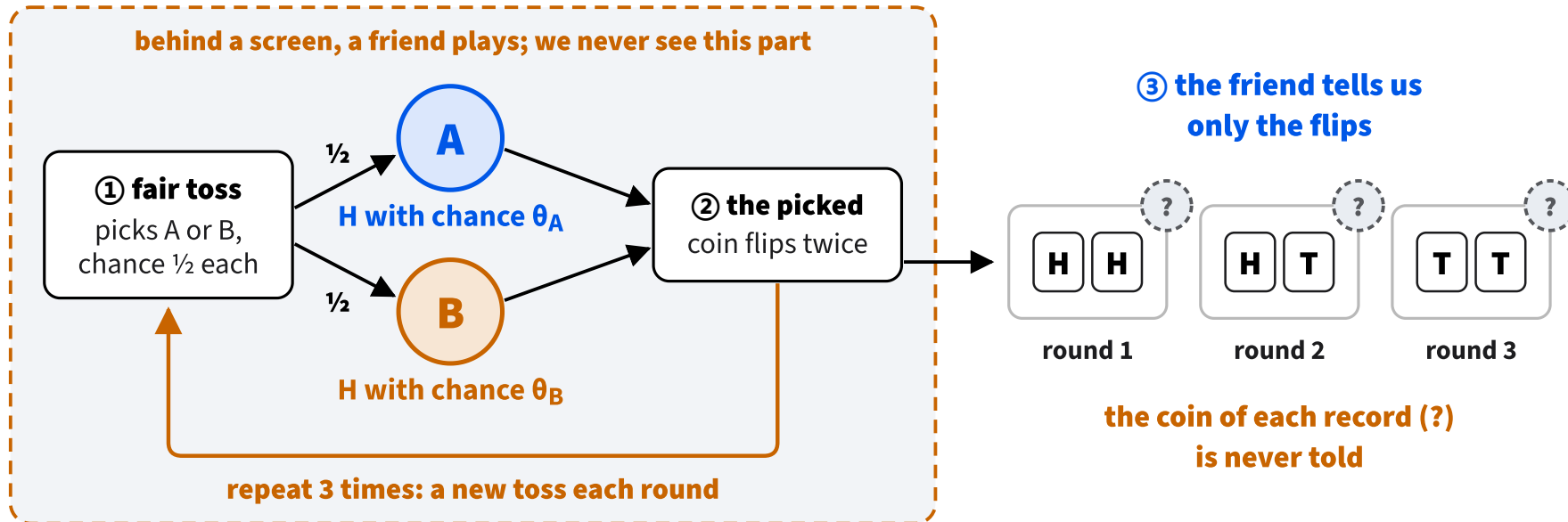
2

Answer. 2: with m samples the typical error is $\sigma/\sqrt{m}$. With spread $\sigma = 1$: $1/\sqrt{25} = 0.2$, $1/\sqrt{100} = 0.1$; so 4 times the samples halves the error, $0.2/0.1 = 2 = \sqrt{4}$.



Both return: **counting** in EM's maximization step, **sampling** for too many hidden cases.

The two-coin game: we see the flips, never the coin



We see

three records, **HH**, **HT**, **TT** (H = heads, T = tails); both flips of a record come from one coin.

Hidden

which coin made each record; a new fair toss each round, so the coins can differ.

Goal

estimate the heads chances θ_A, θ_B from the three records alone.

Each section: a brief example, the theory, then the **key numbers on these three records.**

Eight questions; each answer raises the next

Section	Question on the coins	Answer
1 Counting	If we are told each coin, what are θ_A, θ_B ?	count heads, divide
2 Hidden variables	Not told: how likely is a record, and which coin made it?	a sum; Bayes' rule
3 EM	Can we still count?	yes, with soft labels: split each record by its posterior
4 Why EM increases the likelihood	Does every round improve the fit?	yes: each round raises a lower bound that touches the log-likelihood
5 Mixtures	What if coins become bell curves?	the same EM, now clustering points
6 Hidden chains	What if the coin switches between flips?	sum-product on a chain
7 Monte Carlo EM	Too many hidden cases to add up?	sample them
8 Trusting the fit	Is the answer right, and the only one?	restart from several guesses; check the data can pin down every unknown



Each section: a brief example, the whole theory, one key coin calculation, a connection question; full calculations in the Appendix.

The machine-learning applications (label smoothing, tokenizers, reasoning models, ...) are in the **Week 5 lab**.

By the end of today you can do six things

1 • Count

fit a model when nothing is hidden, and say why counting is the **maximum-likelihood** answer.

You can answer: why is 7 heads in 10 flips fitted by 7/10?

2 • Infer

compute a record's probability with a hidden variable, and the **posterior** of that variable.

You can answer: which coin probably made HH?

3 • Run EM

by hand: E-step (expectation: **soft labels**) and M-step (maximization: count and divide).

You can answer: how do soft labels become new heads chances?

4 • Prove

that no EM round lowers the log-likelihood (how well θ fits the data), with **one inequality**.

You can answer: can an EM round ever make the fit worse?

5 • Extend

EM to **bell curves**, coins that **switch** between flips, and **drawn** hidden cases.

You can answer: what does EM add up when the coin can switch?

6 • Judge

a fit: a start that gets stuck, the names A and B swapped, or **more unknowns than the data can fix**.

You can answer: why may EM swap the names A and B?

Six skills, one example: three records of two coin flips.

We see x , never z , and want to learn θ_A, θ_B

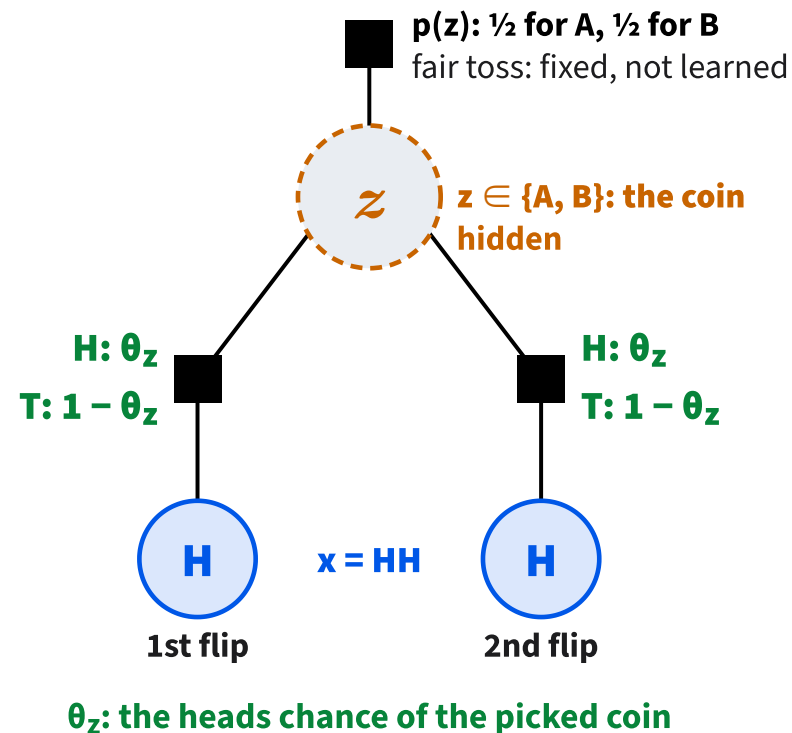
Symbol	Meaning	On the coins
x	observed record	the two flips, e.g. $x = \text{HH}$
z	hidden variable	the coin, $z \in \{A, B\}$
$\theta = (\theta_A, \theta_B)$	parameters	heads chances of A, B (unknown)
$p_\theta(x, z)$	joint probability of record and coin	$p_\theta(\text{HH}, A) = \frac{1}{2} \cdot \theta_A \cdot \theta_A$

Take a starting guess (EM will improve it later): A leans heads, $\theta_A = \frac{3}{4}$; B leans tails, $\theta_B = \frac{1}{4}$.

$$p_\theta(\text{HH}, A) = \underbrace{\frac{1}{2}}_{\text{pick A}} \cdot \underbrace{\frac{3}{4}}_{\text{1st H by A}} \cdot \underbrace{\frac{3}{4}}_{\text{2nd H by A}} = \frac{9}{32}.$$

Lecture 2's factor graph of one record

■ square: a factor (a chance) · ○ circle: a variable



The subscript θ in p_θ means "computed with these parameters"; the toss chance $\frac{1}{2}$ is fixed.

HH has chance 5/16 in all; Bayes' rule gives coin A 9/10 of it

$$p_{\theta}(x) = \sum_z p_{\theta}(x, z)$$

$$p_{\theta}(\text{HH}) = \frac{9}{32} + \frac{1}{32} = \frac{10}{32} = \frac{5}{16}$$

$$p_{\theta}(z \mid x) = \frac{p_{\theta}(x, z)}{p_{\theta}(x)}$$

$$p_{\theta}(A \mid \text{HH}) = \frac{9/32}{10/32} = \frac{9}{10}$$

Likelihood of a record: add the joint over the hidden coin.

9 HH squares of A plus 1 of B, each worth 1/32.

Posterior (Bayes' rule): which coin, given the flips.

9 of the 10 HH squares are A's.

coin A picked
chance $\frac{1}{2} \cdot \text{heads } \frac{3}{4}$

	H	H	H	T
H				
H				
H				
T				

9 squares are HH

coin B picked
chance $\frac{1}{2} \cdot \text{heads } \frac{1}{4}$

	H	T	T	T
H				
T				
T				
T				

1 square is HH

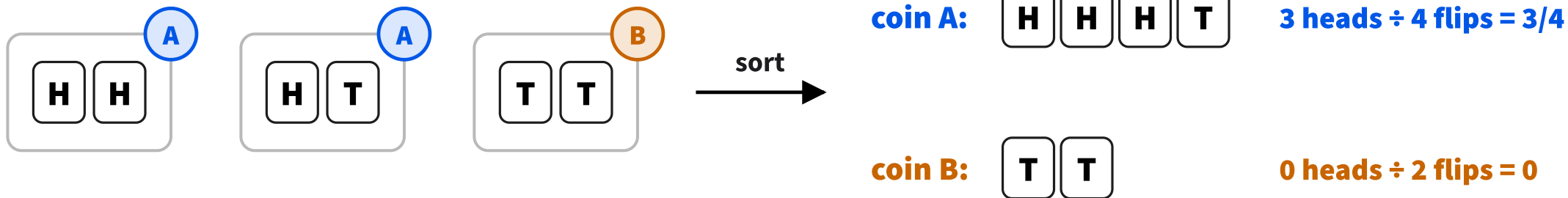
each row (1st flip) or column (2nd flip) is $\frac{1}{4}$,
so a square = $\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$

Log-likelihood: each round has its own fair toss, so the records are independent given θ , and $p_{\theta}(\text{all three}) = p_{\theta}(\text{HH}) \cdot p_{\theta}(\text{HT}) \cdot p_{\theta}(\text{TT})$; the log turns the product into a sum: $\ell(\theta) = \ln p_{\theta}(\text{HH}) + \ln p_{\theta}(\text{HT}) + \ln p_{\theta}(\text{TT})$.

Maximum likelihood: pick the θ with the largest $\ell(\theta)$. First, the easy case: the coin is told.

If the friend tells us the coin, we just count

Goal: learn θ_A, θ_B , the heads chances of coins A and B, from the records HH, HT, TT. First the easy case: the friend also tells us which coin made each record.



the friend now says which coin made each record

estimate = heads ÷ flips, coin by coin

Recall the game (p. 4)

A fair toss picks coin A or B, flipped twice. Normally we see **only the flips**; for now the friend also names the coin.

Estimate

heads ÷ flips: $\theta_A = \frac{3}{4}$ and $\theta_B = \frac{0}{2} = 0$.

Two questions

Why is counting the **best** guess? Can coin B really **never** land heads?

With the coin told, learning is counting: heads ÷ flips, coin by coin.

The likelihood scores each heads chance by the flips it explains

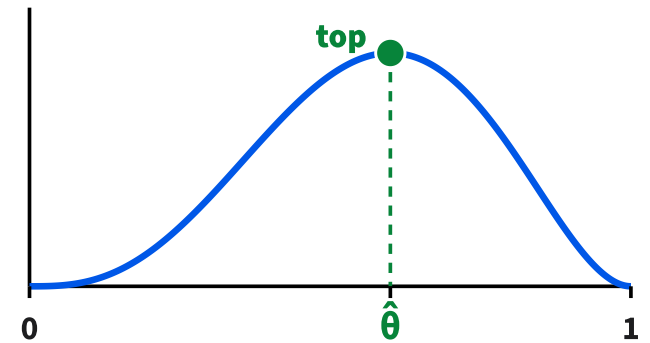
Definitions. A coin with heads chance θ shows h heads and t tails in independent flips. The **likelihood** $L(\theta)$ is the probability of these flips, read as a function of the unknown θ . The **maximum-likelihood estimate** $\hat{\theta}$ is the θ with the largest likelihood.

$$\begin{aligned}
 L(\theta) &= \underbrace{\theta \cdots \theta}_{h \text{ heads}} \cdot \underbrace{(1 - \theta) \cdots (1 - \theta)}_{t \text{ tails}} && \text{independent flips multiply} \\
 &= \theta^h \cdot (1 - \theta)^t && \text{count the factors} \\
 \ell(\theta) &= \ln L(\theta) = h \cdot \ln \theta + t \cdot \ln(1 - \theta) && \text{log of a product = sum of logs} \\
 \hat{\theta} &= \arg \max_{\theta} L(\theta) = \arg \max_{\theta} \ell(\theta) && \text{ln only rises: same top}
 \end{aligned}$$

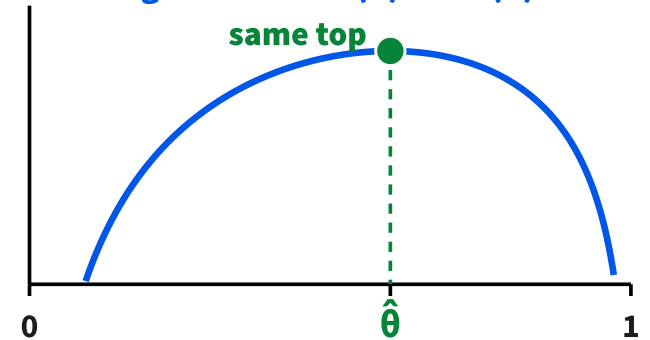
In words: the likelihood multiplies one factor per flip; its log ℓ adds them, and $\arg \max$ names the θ at the top.

Why it matters: a sum is easier to differentiate than a product, so we find $\hat{\theta}$ by setting the slope of ℓ to zero.

$$\text{likelihood } L(\theta) = \theta^h \cdot (1 - \theta)^t$$



$$\text{log-likelihood } \ell(\theta) = \ln L(\theta)$$



ln only rises: same peak $\hat{\theta}$

Maximum likelihood picks the θ that makes the flips we saw most probable.

For any coin, the counted fraction maximizes the likelihood

Theorem 1 (maximum likelihood by counting). Independent flips of a coin with heads chance θ show h heads and t tails. Then $\ell(\theta) = h \cdot \ln \theta + t \cdot \ln(1 - \theta)$ is largest at $\hat{\theta} = \frac{h}{h+t}$.

$$\ell'(\theta) = h \cdot \frac{1}{\theta} + t \cdot \frac{-1}{1-\theta}$$

$$= \frac{h}{\theta} - \frac{t}{1-\theta}$$

$$\ell'(\theta) = 0 \iff h \cdot (1-\theta) = t \cdot \theta$$

$$\iff h = (h+t) \cdot \theta$$

$$\iff \theta = \frac{h}{h+t}$$

$$\ell''(\theta) = -\frac{h}{\theta^2} - \frac{t}{(1-\theta)^2} < 0$$

$$h = 0: \ell = t \cdot \ln(1-\theta) \text{ falls}$$

$$t = 0: \ell = h \cdot \ln \theta \text{ rises}$$

$$(\ln \theta)' = 1/\theta; (\ln u)' = u'/u, \\ u = 1 - \theta, u' = -1$$

simplify

interior top ($h, t > 0$); times $\theta \cdot (1 - \theta) > 0$

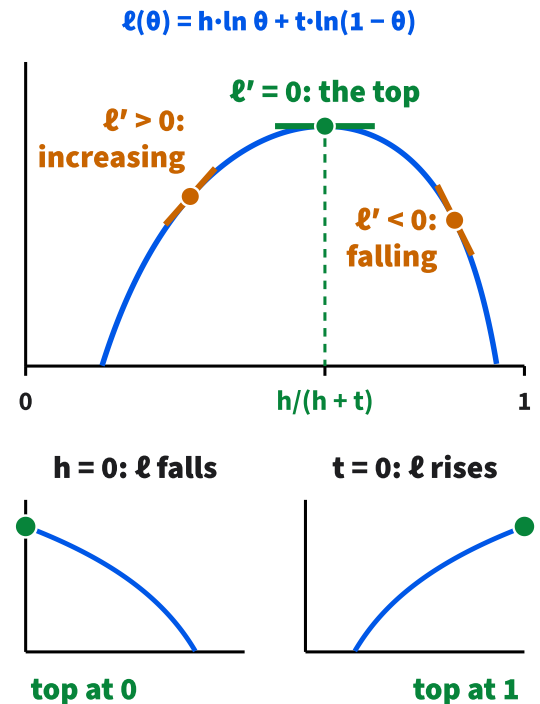
expand $h - h \cdot \theta = t \cdot \theta$, add $h \cdot \theta$

divide by $h + t$

Check: curves down: one top

Check: top at $\theta = 0 = \frac{0}{0+t}$

Check: top at $\theta = 1 = \frac{h}{h+0}$. ■



In words: the best heads chance is the fraction of heads seen. **Why it matters:** with each flip's coin known, learning needs no search: count, then divide.

Whatever the counts, the likelihood peaks at heads $\div$ flips.

A zero count makes maximum likelihood claim certainty

↳ Theorem 1 (p. 11) with no heads ($h = 0$) in t flips gives $\hat{\theta} = \frac{0}{0+t} = 0$.

What does this fitted coin predict for a later record x that contains a head?

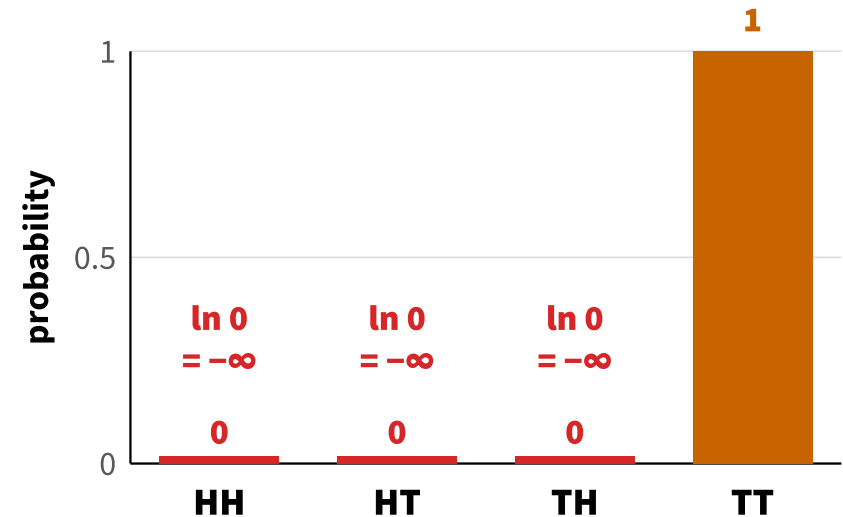
$p(H \mid \hat{\theta}) = \hat{\theta} = 0$	one head, fitted coin
$p(x \mid \hat{\theta}) = \hat{\theta} \cdot (\text{other factors}) = 0$	flips multiply
$\ln p(x \mid \hat{\theta}) = \ln 0 = -\infty$	log-likelihood of x

Weak evidence: even a fair coin shows t tails in a row with chance $(\frac{1}{2})^t$, which is not rare when t is small.

In words: "no heads seen" is read as "heads impossible".

Why it matters: one later head makes the log-likelihood $-\infty$. The fix: keep θ uncertain with a prior, and average.

fitted $\hat{\theta} = 0$: chance of the next two flips



every record with a head is called impossible;
all the probability sits on tails only

Maximum likelihood turns an unseen outcome into an impossible one.

Averaging over the posterior gives the chance of the next head

Definitions. Treat θ as unknown. The **uniform prior** makes every value in $[0, 1]$ equally likely before any flip: $p(\theta) = 1$. The **posterior** $p(\theta \mid \text{flips})$ is what we believe after the flips; the **posterior mean** is its average, $\mathbb{E}[\theta \mid \text{flips}] = \int_0^1 \theta \cdot p(\theta \mid \text{flips}) d\theta$. Below, u is an integration variable on $[0, 1]$.

$$p(\theta \mid \text{flips}) = \frac{p(\theta) \cdot L(\theta)}{\int_0^1 p(u) \cdot L(u) du}$$

Bayes' rule

$$= \frac{\theta^h \cdot (1 - \theta)^t}{\int_0^1 u^h \cdot (1 - u)^t du}$$

$$p(\theta) = 1, \\ L(\theta) = \theta^h \cdot (1 - \theta)^t$$

$$p(\text{next H} \mid \text{flips}) = \int_0^1 p(\text{next H} \mid \theta) \cdot p(\theta \mid \text{flips}) d\theta$$

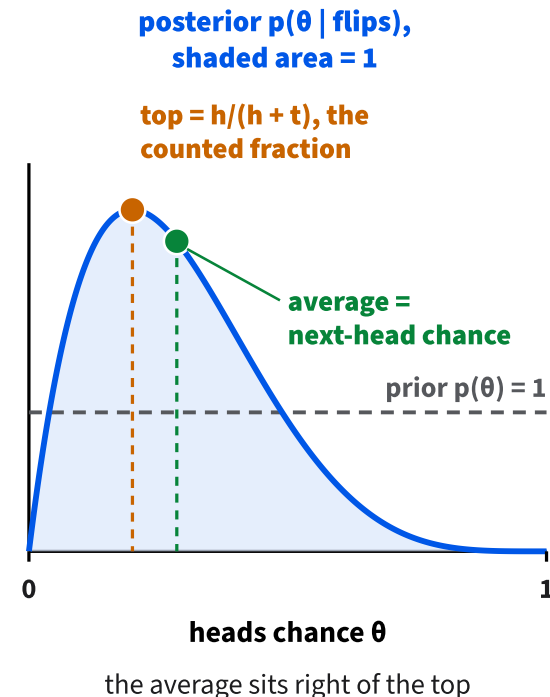
total probability

$$= \int_0^1 \theta \cdot p(\theta \mid \text{flips}) d\theta = \mathbb{E}[\theta \mid \text{flips}]$$

$p(\text{next H} \mid \theta) = \theta$

In words: the next-head chance is the heads chance averaged over what we **still believe**.

Why it matters: every $\theta > 0$ that keeps posterior weight lifts the average above 0.



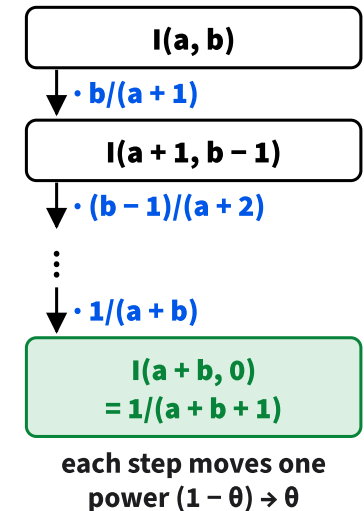
The posterior's top is the counted fraction; its average is the next-head chance.

The Beta integral turns powers of θ and $1 - \theta$ into factorials

Both integrals in the posterior mean have the form $\int_0^1 \theta^a \cdot (1 - \theta)^b d\theta$, the area under the likelihood of a heads and b tails. **Beta integral** for whole numbers $a, b \geq 0$: $I(a, b) = \int_0^1 \theta^a \cdot (1 - \theta)^b d\theta = \frac{a! b!}{(a + b + 1)!}$. If $b = 0$, $I(a, 0) = \frac{1}{a+1}$ directly; for $b \geq 1$, integrate by parts:

$$\begin{aligned}
 I(a, b) &= \left[\frac{\theta^{a+1}}{a+1} \cdot (1 - \theta)^b \right]_0^1 - \int_0^1 \frac{\theta^{a+1}}{a+1} \cdot (-b \cdot (1 - \theta)^{b-1}) d\theta \\
 &= \left[\dots \right]_0^1 + \frac{b}{a+1} \cdot I(a+1, b-1) \\
 &= \frac{b}{a+1} \cdot I(a+1, b-1) \\
 &= \frac{b}{a+1} \cdot \frac{b-1}{a+2} \cdots \frac{1}{a+b} \cdot I(a+b, 0) \\
 &= \frac{a! b!}{(a+b)!} \cdot \frac{1}{a+b+1} = \frac{a! b!}{(a+b+1)!} \blacksquare
 \end{aligned}$$

$\int u' \cdot v = [u \cdot v] - \int u \cdot v'$,
 $u' = \theta^a$, $v = (1 - \theta)^b$
 minus times minus;
 the integral is $I(a+1, b-1)$
 bracket = 0: $\theta^{a+1} = 0$ at 0,
 $(1 - \theta)^b = 0$ at 1 ($b \geq 1$)
 repeat $(a, b) \rightarrow (a+1, b-1)$,
 b times; $I(a+b, 0) = 1/(a+b+1)$
 tops: $b!$; bottoms: $(a+b)!/a!$;
 $(a+b)! (a+b+1) = (a+b+1)!$



the b factors multiply to $a! b!/(a+b)!$

In words: each step trades one factor $(1 - \theta)$ for one θ and a known fraction. **Why it matters:** both integrals in the posterior mean become factorials, which cancel.

The Beta integral turns every posterior integral into factorials.

Laplace's rule: the posterior mean adds one head and one tail

Theorem 2 (Laplace's rule). Put a uniform prior on θ , then see h heads and t tails. The posterior mean is $\mathbb{E}[\theta \mid \text{flips}] = \frac{h+1}{h+t+2}$.

$$\begin{aligned}
 \mathbb{E}[\theta \mid \text{flips}] &= \int_0^1 \theta \cdot \frac{\theta^h \cdot (1-\theta)^t}{\int_0^1 u^h \cdot (1-u)^t du} d\theta \\
 &= \frac{\int_0^1 \theta \cdot \theta^h \cdot (1-\theta)^t d\theta}{\int_0^1 \theta^h \cdot (1-\theta)^t d\theta} \\
 &= \frac{I(h+1, t)}{I(h, t)} \\
 &= \frac{(h+1)! t! / (h+t+2)!}{h! t! / (h+t+1)!} \\
 &= \frac{(h+1)!}{h!} \cdot \frac{(h+t+1)!}{(h+t+2)!} \\
 &= (h+1) \cdot \frac{1}{h+t+2} = \frac{h+1}{h+t+2} \blacksquare
 \end{aligned}$$

posterior mean;
Bayes with prior 1

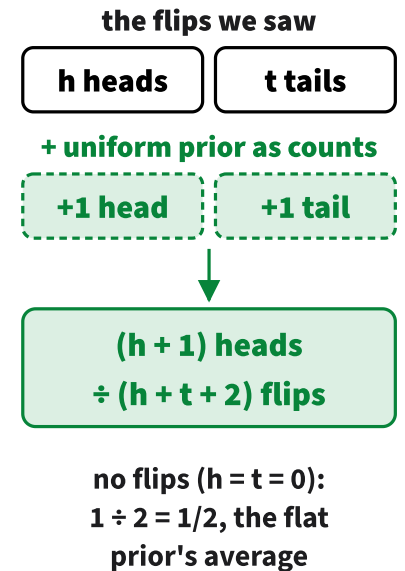
the denominator is a constant: take it out

$\theta \cdot \theta^h = \theta^{h+1}$:
top $(a, b) = (h+1, t)$

Beta integral, top and below

divide; $t!$ cancels

$$\frac{(h+1)!}{h!} = h+1, \quad \frac{(h+t+1)!}{(h+t+2)!} = \frac{1}{h+t+2}$$



In words: add one imagined head and one imagined tail, then count. **Why +1 and +1?** With no flips ($h = t = 0$): $\frac{0+1}{0+2} = \frac{1}{2}$, the flat prior's average, so **the imagined head and tail are the uniform prior written as counts**.

Laplace's rule is counting with one imagined head and one imagined tail.

Laplace's rule pulls the count toward 1/2 by a weight ε

Corollary of Theorem 2. Over $n = h + t \geq 1$ flips, Laplace's value is a mix of the **counted fraction** $\frac{h}{n}$ and the even guess $\frac{1}{2}$:

$$\frac{h+1}{n+2} = (1 - \varepsilon) \cdot \frac{h}{n} + \varepsilon \cdot \frac{1}{2}, \quad \varepsilon = \frac{2}{n+2}.$$

$$\begin{aligned} (1 - \varepsilon) \cdot \frac{h}{n} + \varepsilon \cdot \frac{1}{2} &= \frac{n}{n+2} \cdot \frac{h}{n} + \frac{2}{n+2} \cdot \frac{1}{2} & 1 - \varepsilon &= \frac{n}{n+2} \\ &= \frac{h}{n+2} + \frac{1}{n+2} & (1 - \varepsilon) \cdot \frac{h}{n} &= \frac{h}{n+2}, \\ &= \frac{h+1}{n+2} & \varepsilon \cdot \frac{1}{2} &= \frac{1}{n+2} \end{aligned}$$

same denominator

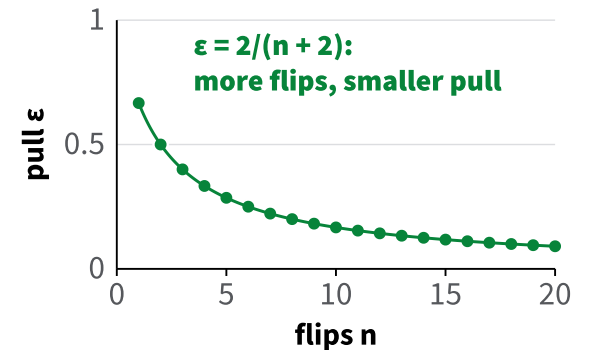
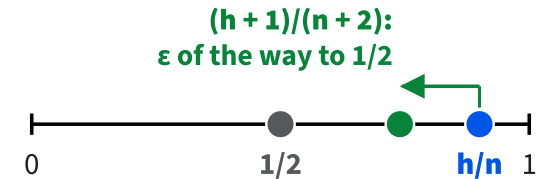
$$\begin{aligned} \frac{h+1}{n+2} - \frac{h}{n} &= (1 - \varepsilon) \cdot \frac{h}{n} + \varepsilon \cdot \frac{1}{2} - \frac{h}{n} & \text{the mix equals } (h+1)/(n+2); \\ & & \text{subtract } h/n \\ &= \varepsilon \cdot \left(\frac{1}{2} - \frac{h}{n} \right) & (1 - \varepsilon) \cdot h/n - h/n &= -\varepsilon \cdot h/n; \\ & & \text{factor out } \varepsilon \end{aligned}$$

In words: Laplace's value $\frac{h+1}{n+2}$ sits ε of the way from the count to $\frac{1}{2}$.

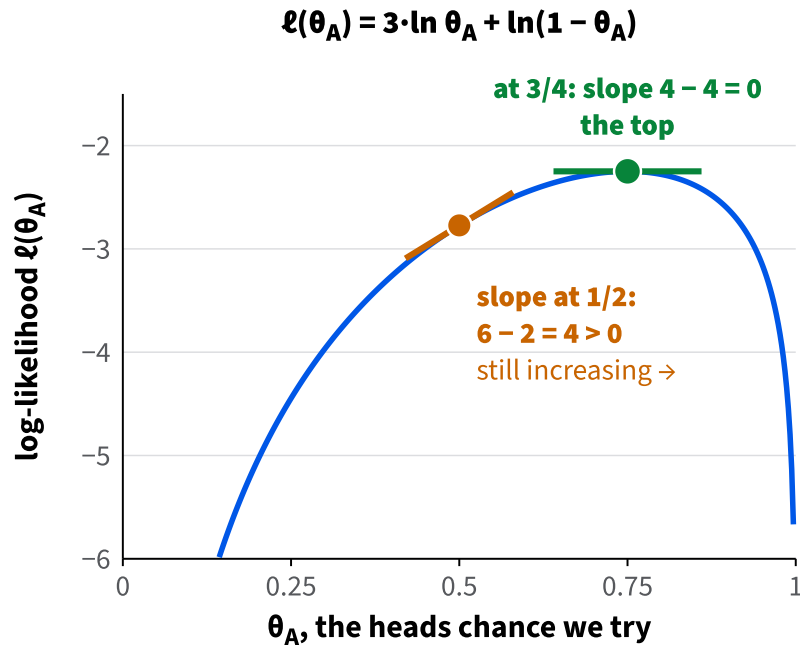
Why it matters: more flips shrink ε , so the data soon **outweigh** the two imagined flips.

Laplace moves the count $\varepsilon = \frac{2}{n+2}$ of the way to $\frac{1}{2}$; more flips, smaller pull.

one coin: n flips, h heads



Coin A's log-likelihood has slope zero at $\theta_A = 3/4$



↳ Theorem 1 (p. 11) applied to coin A, which gave H, H, H, T ($h = 3, t = 1$); independent flips multiply, so $L(\theta_A) = \theta_A^3 \cdot (1 - \theta_A)$. Theorem 1's slope: $\ell'(\theta) = \frac{h}{\theta} - \frac{t}{1-\theta}$.

$$\hat{\theta}_A = \frac{h}{h+t} = \frac{3}{3+1} = \frac{3}{4} \quad \text{Theorem 1, } h = 3, t = 1$$

$$\ell'\left(\frac{3}{4}\right) = \frac{3}{3/4} - \frac{1}{1/4} = 4 - 4 = 0 \quad \text{check: slope zero at the top}$$

Left of the top the log-likelihood still increases:

$$\ell'\left(\frac{1}{2}\right) = \frac{3}{1/2} - \frac{1}{1/2} = 6 - 2 = 4 > 0.$$

Maximum likelihood gives $\theta_A = \frac{3}{4}$, the counted fraction 3 of 4.

Counting says coin B can never land heads

counting only

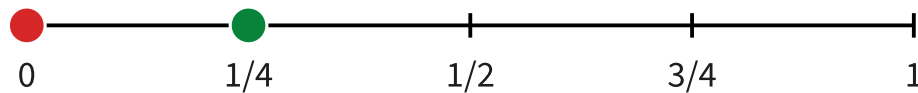
$$\boxed{T} \boxed{T} \longrightarrow 0 \text{ heads} \div 2 \text{ flips} = 0$$

with one imagined head and one imagined tail

$$\boxed{T} \boxed{T} + \boxed{H} \boxed{T} \longrightarrow 1 \text{ head} \div 4 \text{ flips} = 1/4$$

0: heads impossible

1/4: heads rare, still possible



estimate of θ_B , coin B's heads chance

↳ Theorem 1's zero-count case (p. 12) on coin B, which showed T, T: $\theta_B = \frac{0}{0+2} = 0$.

Too sure. Suppose B is flipped again and shows HT:

$$p(\text{HT} \mid B) = \theta_B \cdot (1 - \theta_B) = 0 \cdot 1 = 0.$$

Its log-likelihood is $\ln 0 = -\infty$: the model calls a record we could easily see **impossible**.

Weak evidence: a fair coin shows TT with chance $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Fix (Theorem 2, Laplace's rule) (p. 15): add one imagined head and one imagined tail (dashed):

$$\theta_B = \frac{0 + 1}{2 + 2} = \frac{1}{4}, \quad p(\text{HT} \mid B) = \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{16}.$$

Two tails cannot prove heads impossible; imagined flips keep every outcome possible.

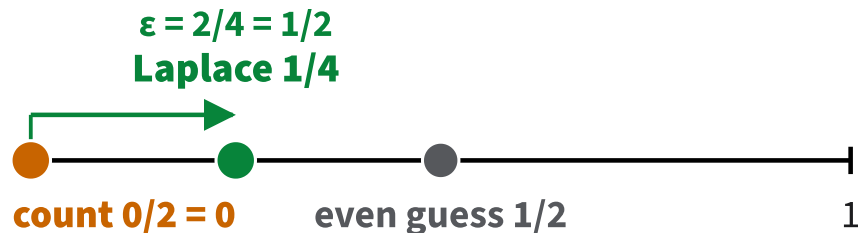
Laplace's rule on the coins: B rises from 0 to $1/4$, A drops to $2/3$

↳ Theorem 2 (p. 15) and its corollary (p. 16) applied to both coins:
Laplace's value $\frac{h+1}{n+2}$ is the count $\frac{h}{n}$ moved the weight $\varepsilon = \frac{2}{n+2}$ of the way to $\frac{1}{2}$.

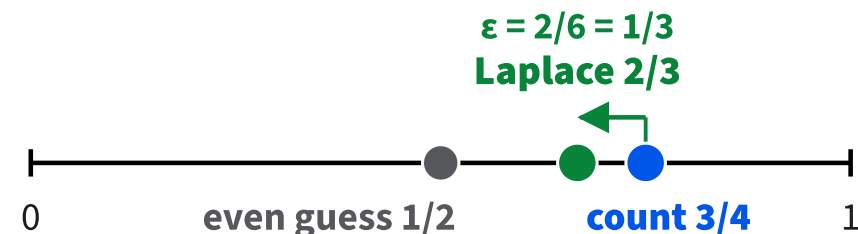
	coin B: T, T	coin A: H, H, H, T
count $\frac{h}{n}$	$\frac{0}{2} = 0$	$\frac{3}{4}$
weight $\varepsilon = \frac{2}{n+2}$	$\frac{2}{2+2} = \frac{1}{2}$	$\frac{2}{4+2} = \frac{1}{3}$
Laplace $\frac{h+1}{n+2}$	$\frac{0+1}{2+2} = \frac{1}{4}$	$\frac{3+1}{4+2} = \frac{2}{3}$

Coin A has more flips (4, not 2), so its pull is smaller: $\frac{1}{3} < \frac{1}{2}$.

coin B: T, T (n = 2 flips)



coin A: H, H, H, T (n = 4 flips)



Laplace moves the count ε of the way
toward the even guess $1/2$

Two flips: Laplace moves coin B halfway to $\frac{1}{2}$. Four flips: coin A only a third.

Next: the friend stops telling us the coin

we are here

Counting

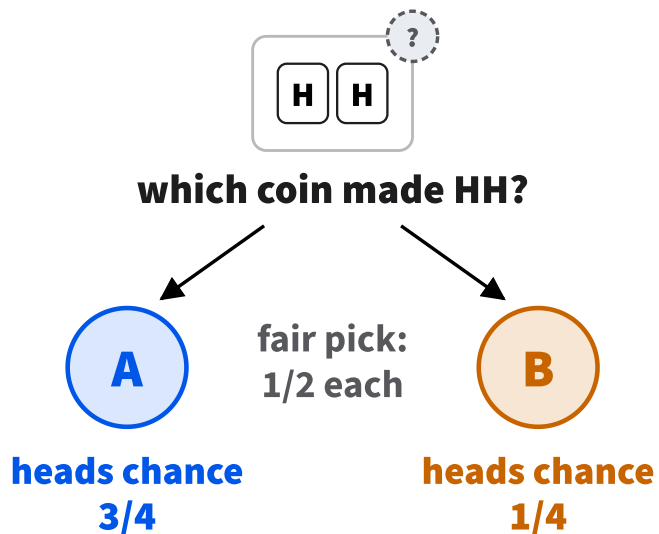
Hidden
variables

next

EM

Why EM
increases ℓ

Mixtures

Hidden
chainsMonte Carlo
EMTrusting the
fit

Counting needs each flip's coin. Now the friend reports only "HH".

Working guess from now on: $\theta_A = \frac{3}{4}$ (A's counted fraction (p. 17), kept because 4 flips make the pull small), $\theta_B = \frac{1}{4}$ (B's Laplace value (p. 19): heads stays possible), fair pick. How likely is HH?

9/16

5/16

1/16

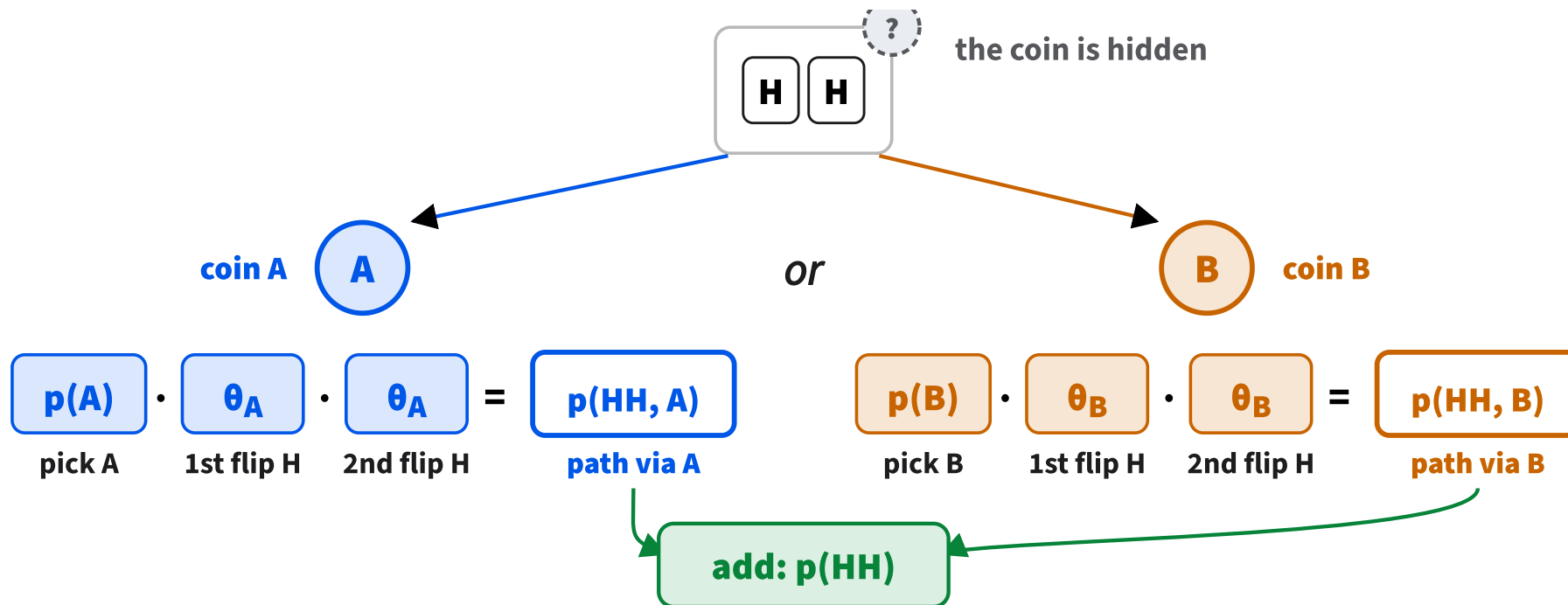
Answer. $\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{9}{32} + \frac{1}{32} = \frac{10}{32} = \frac{5}{16}$: a sum over the two coins, each weighted by its pick chance $\frac{1}{2}$.

A hidden coin turns one probability into a sum over the coins.

With the coin hidden, the chance of HH adds two paths

Goal: learn θ_A, θ_B from HH, HT, TT, now that the friend no longer says which coin was used.

Recall the game (p. 4): a fair toss picks coin A or coin B; θ_A, θ_B are their heads chances. Working guess: $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$.



Why add? HH came from A **or** from B, never both, so the chances of the two paths add.

$p(HH)$ = path via A + path via B: a sum over the hidden coin.

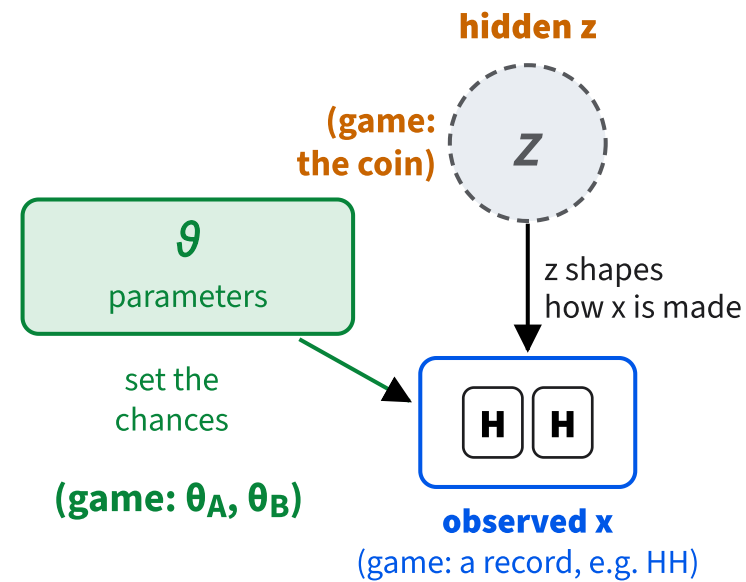
In general: an observed x , a hidden z , parameters θ

Definition (hidden-variable model). A model $p_\theta(x, z)$ gives the chance of an observed record x **together with** a hidden variable z ; θ are its parameters. Each value of z is one **path** to x . The data are N independent records $x^{(1)}, \dots, x^{(N)}$.

name	formula	what it does
marginal likelihood	$p_\theta(x) = \sum_z p_\theta(x, z)$	adds the paths
log-likelihood	$\ell(\theta) = \sum_{n=1}^N \ln p_\theta(x^{(n)})$	adds the records' logs
posterior	$p_\theta(z x) = \frac{p_\theta(x, z)}{p_\theta(x)}$	path ÷ total

Why "marginal"? Adding over z is called marginalizing z out: what is left is the chance of x alone.

Goal: the θ with the largest $\ell(\theta)$, the **maximum-likelihood estimate**.



$$p_\theta(x, z) = p(z) \cdot p_\theta(x | z)$$

$p(z)$: chance of the hidden value

$p_\theta(x | z)$: chance of x given z

one value of z = one path to x

Three formulas: add the paths, take logs, divide for the posterior.

With z told, the log splits into one term per parameter

↳ Recall Theorem 1 (counting) (p. 11): a coin with h heads and t tails has best heads chance $\frac{h}{h+t}$. It needs each coin's flips **separately**. In the game, $z \in \{A, B\}$ is the coin and $\theta = (\theta_A, \theta_B)$.

z told: the log of the complete data

$$\begin{aligned} \sum_n \ln p_\theta(x^{(n)}, z^{(n)}) &= \sum_n [\ln p(z^{(n)}) + \ln p_\theta(x^{(n)} | z^{(n)})] && \text{log of the product } p(z) \cdot p_\theta(x | z) \\ &= \sum_n \ln p(z^{(n)}) + \sum_{z^{(n)}=A} \ln p_\theta(x^{(n)} | A) + \sum_{z^{(n)}=B} \ln p_\theta(x^{(n)} | B) && \text{group the records by coin} \\ &= \underbrace{\sum_n \ln p(z^{(n)})}_{\text{no } \theta} + \underbrace{\sum_{z^{(n)}=A} \ln p_{\theta_A}(x^{(n)} | A)}_{\text{only } \theta_A} + \underbrace{\sum_{z^{(n)}=B} \ln p_{\theta_B}(x^{(n)} | B)}_{\text{only } \theta_B} && \begin{aligned} &\text{each coin uses only its own chance:} \\ &p_\theta(x | A) = p_{\theta_A}(x | A), \\ &p_\theta(x | B) = p_{\theta_B}(x | B) \end{aligned} \end{aligned}$$



In words: with z told, each parameter has its own sum of logs, so each is maximized alone.

Why it matters: this is why Theorem 1 works coin by coin: θ_A is fitted to A's flips only, θ_B to B's flips only.

z told $\Rightarrow$ separate terms $\Rightarrow$ count each coin's flips (Theorem 1).

A log of a sum does not split, so counting no longer works

↳ Now z is hidden: record n enters through $p_\theta(x) = \sum_z p_\theta(x, z)$, with paths $\underbrace{p(A) \cdot p_{\theta_A}(x^{(n)} | A)}_{a_n: \text{only } \theta_A}$ via A and $\underbrace{p(B) \cdot p_{\theta_B}(x^{(n)} | B)}_{b_n: \text{only } \theta_B}$ via B.

z hidden: one log holds every parameter

$$\begin{aligned}
 \ell(\theta) &= \sum_n \ln p_\theta(x^{(n)}) && \text{definition of } \ell \\
 &= \sum_n \ln(a_n + b_n) && \text{marginal likelihood: add the paths} \\
 &\neq \sum_n \ln(a_n \cdot b_n) && \text{a sum is not a product} \\
 &= \sum_n [\ln a_n + \ln b_n] && \text{a log splits products: } \ln a + \ln b = \ln(a \cdot b)
 \end{aligned}$$

z told: one path per record, a product

$$\ln(a \cdot b) = \ln a + \ln b$$

splits: one term per parameter

vs

z hidden: two paths per record, a sum

$$\ln(a + b) \neq \ln a + \ln b$$

one log holds θ_A and θ_B

In words: with z hidden, all parameters share one log of a sum, and it does not split.

Why it matters: ℓ has no term with θ_A alone, so Theorem 1's per-coin counting (p. 11) has nothing to act on.

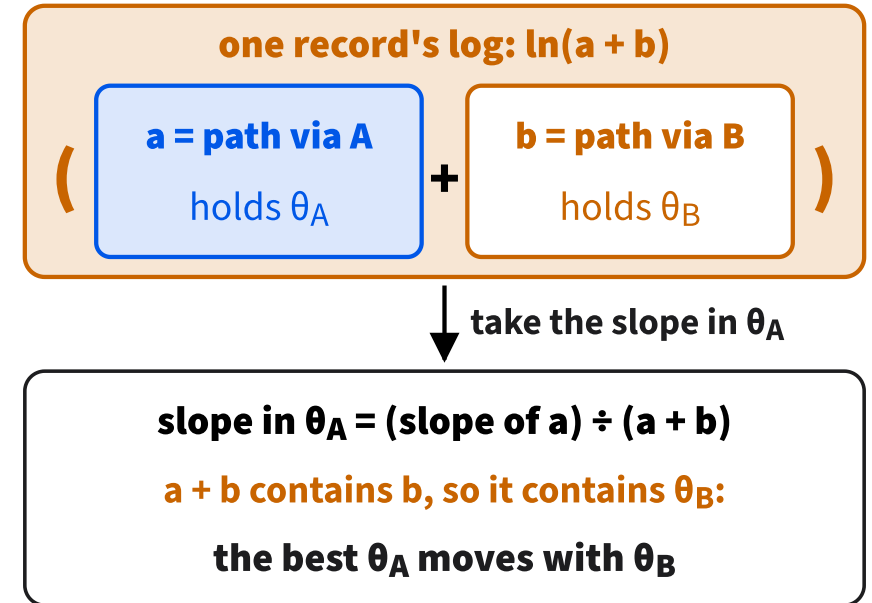
A log of a sum is not a sum of logs: ℓ has no separate term per parameter.

The slope of ℓ in θ_A still holds θ_B , so the parameters are tied

↳ Recall $\ell(\theta) = \sum_n \ln(a_n + b_n)$ (p. 24): path a_n holds only θ_A , path b_n only θ_B . To increase ℓ , take its slope in θ_A .

z hidden: the slope of a log of a sum

$$\begin{aligned} \frac{\partial \ell}{\partial \theta_A} &= \sum_n \frac{1}{a_n + b_n} \cdot \frac{\partial(a_n + b_n)}{\partial \theta_A} && \text{chain rule } (\ln f)' = \frac{f'}{f} \\ &= \sum_n \frac{\partial a_n / \partial \theta_A + \partial b_n / \partial \theta_A}{a_n + b_n} && \text{slope of a sum = sum of slopes} \\ &= \sum_n \frac{\partial a_n / \partial \theta_A}{a_n + b_n} && b_n \text{ holds only } \theta_B, \text{ so } \partial b_n / \partial \theta_A = 0; \\ &&& \text{the denominator still holds } \theta_B \\ &= \sum_n \underbrace{\frac{a_n}{a_n + b_n}}_{p_\theta(A|x^{(n)})} \cdot \frac{\partial \ln a_n}{\partial \theta_A} && a' = a \cdot (\ln a)'; \\ &&& \text{path } \div \text{ total} = \text{posterior} \end{aligned}$$



In words: the slope keeps the whole sum $a_n + b_n$ in its denominator.

Why it matters: the slope in θ_A contains θ_B , so the best θ_A depends on θ_B and Theorem 1 cannot be used coin by coin.

Hidden $z \Rightarrow$ a log of a sum $\Rightarrow$ the parameters are tied together: no counting formula.

The slope of a log of a sum is a posterior-weighted slope

↳ Recall (p. 25): $\frac{\partial \ell}{\partial \theta_A} = \sum_n p_\theta(A | x^{(n)}) \cdot \frac{\partial \ln a_n}{\partial \theta_A}$, the posterior of A · the slope of $\ln a_n$. Theorem 3 says this holds for every model.

Theorem 3 (Fisher's identity). Let θ_j be any one entry of θ (in the game, θ_A or θ_B). Then

$$\frac{\partial}{\partial \theta_j} \ln p_\theta(x) = \sum_z p_\theta(z | x) \cdot \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z).$$

In words: the slope of the log of a sum is each path's slope, weighted by its posterior.

Why a slope? To increase ℓ toward its maximum, we need its slope in each parameter.

Why it matters: one path's log splits, $\ln p_\theta(x, z) = \ln p(z) + \ln p_\theta(x | z)$, so its slope is as easy as counting; the weights are the posteriors, which we can compute.

slope of $\ln p_\theta(x)$ in one parameter
a log of a sum: hard to handle directly

=

$p_\theta(z_1 | x)$
· slope of
 $\ln p_\theta(x, z_1)$
one path, no sum
weight · easy slope

+

$p_\theta(z_2 | x)$
· slope of
 $\ln p_\theta(x, z_2)$
one path, no sum
weight · easy slope

+ ... (one term per value of z)

one path splits into separate logs:

$$\ln p_\theta(x, z) = \ln p(z) + \ln p_\theta(x | z)$$

its slope is as easy as counting

game: $z_1 = \text{coin A}$, $z_2 = \text{coin B}$

Slope of a log of a sum = posterior-weighted single-path slopes; those split like counting.

Proof of Theorem 3: chain rule, then Bayes' rule

Proof. One record x . Recall the marginal likelihood (p. 22) $p_\theta(x) = \sum_z p_\theta(x, z)$, posterior $p_\theta(z | x) = p_\theta(x, z) / p_\theta(x)$.

$$\frac{\partial}{\partial \theta_j} \ln p_\theta(x) = \frac{1}{p_\theta(x)} \cdot \frac{\partial}{\partial \theta_j} p_\theta(x)$$

$$\text{chain rule } (\ln f)' = \frac{f'}{f}$$

$$= \frac{1}{p_\theta(x)} \cdot \frac{\partial}{\partial \theta_j} \sum_z p_\theta(x, z)$$

$$\text{marginal likelihood } f = p_\theta(x) = \sum_z p_\theta(x, z)$$

$$= \frac{1}{p_\theta(x)} \cdot \sum_z \frac{\partial}{\partial \theta_j} p_\theta(x, z)$$

the slope of a sum is the sum of the slopes

$$= \frac{1}{p_\theta(x)} \cdot \sum_z p_\theta(x, z) \cdot \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z)$$

$$\text{for each path } g = p_\theta(x, z): (\ln g)' = \frac{g'}{g}, \text{ so } g' = g \cdot (\ln g)'$$

$$= \sum_z \frac{p_\theta(x, z)}{p_\theta(x)} \cdot \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z)$$

$$\text{move } \frac{1}{p_\theta(x)} \text{ inside the sum}$$

$$= \sum_z p_\theta(z | x) \cdot \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z)$$

Bayes: path $\div$ total = posterior. ■

In words: differentiate the log, write each path's slope via its own log; path $\div$ total is the posterior.

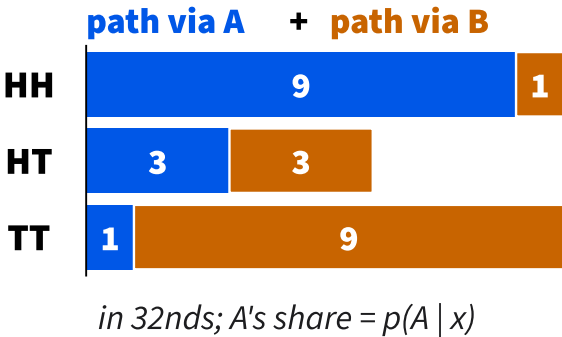
Why it matters: the right side differentiates only **one-path logs**, never a log of a sum; expectation–maximization (EM) builds on it.

Six steps, each a rule from calculus or probability: the identity holds for every model $p_\theta(x, z)$.

Two paths give $p(\text{HH}) = 5/16$; Bayes' rule gives coin A 9/10

↳ Definitions (p. 22) and Theorem 3 (p. 26) applied to the coins, at the working guess $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$: a path is $\frac{1}{2}$ (pick the coin) · the chance of each flip; $p(x)$ adds the two paths; the posterior is path ÷ total; ℓ adds the logs.

record	path via A	path via B	$p(x)$	$p(A x)$	$\ln p(x)$
HH	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{10}{32}$	$\frac{9}{10}$	-1.163
HT	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{32}$	$\frac{6}{32}$	$\frac{3}{6} = \frac{1}{2}$	-1.674
TT	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{10}{32}$	$\frac{1}{10}$	-1.163
total	add the logs: $\ell = -4.000$				



$$p(\text{HH}) = \frac{9}{32} + \frac{1}{32} = \frac{10}{32} = \frac{5}{16}, \quad p(A | \text{HH}) = \frac{9}{32} \div \frac{10}{32} = \frac{9}{10}$$

add the paths; path ÷ total

$$\frac{\partial}{\partial \theta_A} \ln p(\text{HH}) = \frac{9}{10} \cdot \underbrace{\frac{2}{\theta_A}}_{\text{slope of } \ln \frac{1}{2} + 2 \cdot \ln \theta_A = 8/3} + \frac{1}{10} \cdot \underbrace{0}_{\text{no } \theta_A} = 2.4$$

Theorem 3

$$= \frac{\theta_A}{p(\text{HH})} = \frac{3/4}{10/32} = 2.4 \checkmark$$

directly, chain rule

Two paths: 5/16; Bayes: 0.9 A / 0.1 B; these weights turn the path slopes into 2.4 (Theorem 3).

Our guess scores $\ell = -4.000$, just below a top of the map

↳ Definition (p. 22) applied to the coins, with $N = 3$ records:

$$\ell(\theta) = \sum_{n=1}^N \ln p_{\theta}(x^{(n)}).$$

Every pair (θ_A, θ_B) gets its own **log-likelihood** ℓ ; the map colours them all. A higher ℓ means the data are more likely.

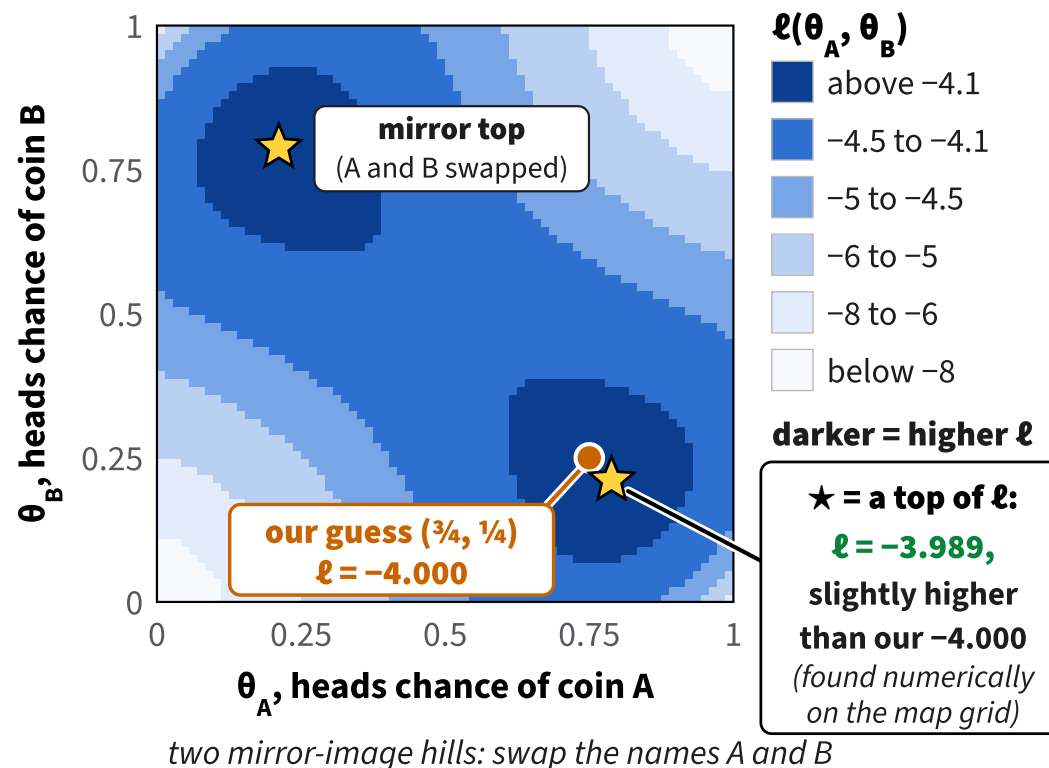
our guess $(\frac{3}{4}, \frac{1}{4})$

$\ell = -4.000$

a top ★ $(0.789, 0.211)$

$\ell = -3.989$

Learning = finding the (θ_A, θ_B) at a top of the map (★), the **maximum-likelihood estimate**. Swapping the names A and B gives the mirror top.



Ours scores $\ell = -4.000$, close to a top but not on it. How do we increase ℓ ?

Told, counting fits θ_A alone; hidden, the best θ_A moves with θ_B

↳ The theory's two chains applied to the coins. Recall Theorem 1 (counting) (p. 11): best heads chance $\frac{h}{h+t}$ from h heads and t tails of one coin.

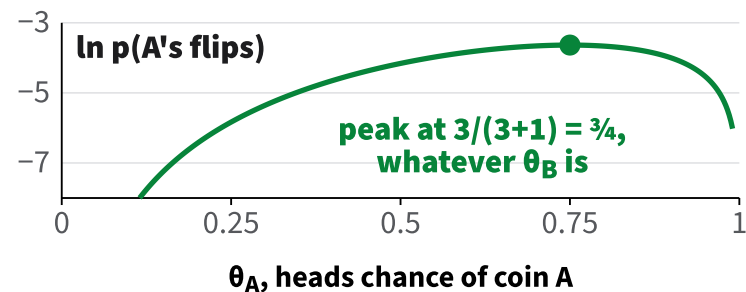
Coin told: the log splits

Say HH and HT came from A, so A's flips are H, H, H, T. Their log is $\ln p(\text{HH}, A) + \ln p(\text{HT}, A)$
 $= 2 \cdot \ln \frac{1}{2} + 3 \cdot \ln \theta_A + \ln(1 - \theta_A)$: no θ_B , so **counting** gives $\theta_A = \frac{3}{3+1} = \frac{3}{4}$, whatever θ_B is.

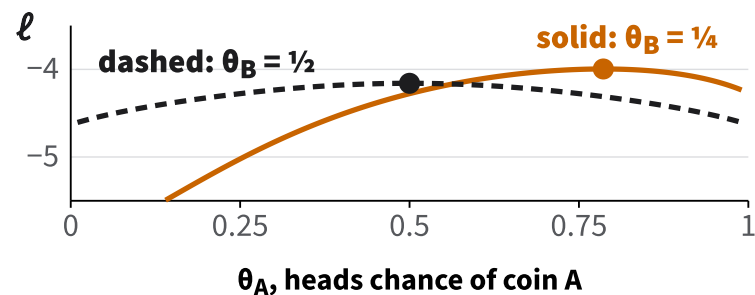
Coin hidden: one log of a sum

ℓ contains $\ln p(\text{HH}) = \ln \left(\frac{1}{2} \cdot \theta_A^2 + \frac{1}{2} \cdot \theta_B^2 \right)$, one log holding both parameters, so ℓ does not split. In the figure (ℓ against θ_A), changing θ_B from $\frac{1}{4}$ to $\frac{1}{2}$ moves the best θ_A (dot) from about 0.79 to 0.5.

Coin told (say HH and HT were from A):
A's flips are H, H, H, T

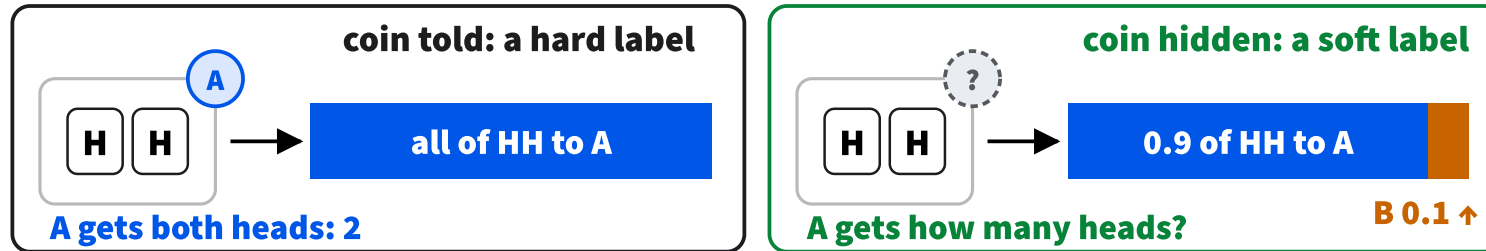


Coin hidden: the best θ_A (dot) moves with θ_B



Hidden coin $\Rightarrow$ a log of a sum $\Rightarrow$ the best θ_A depends on θ_B : no counting formula.

Next: count with soft labels



The posterior says HH is 0.9 coin A (p. 28). How many heads should HH give to coin A?

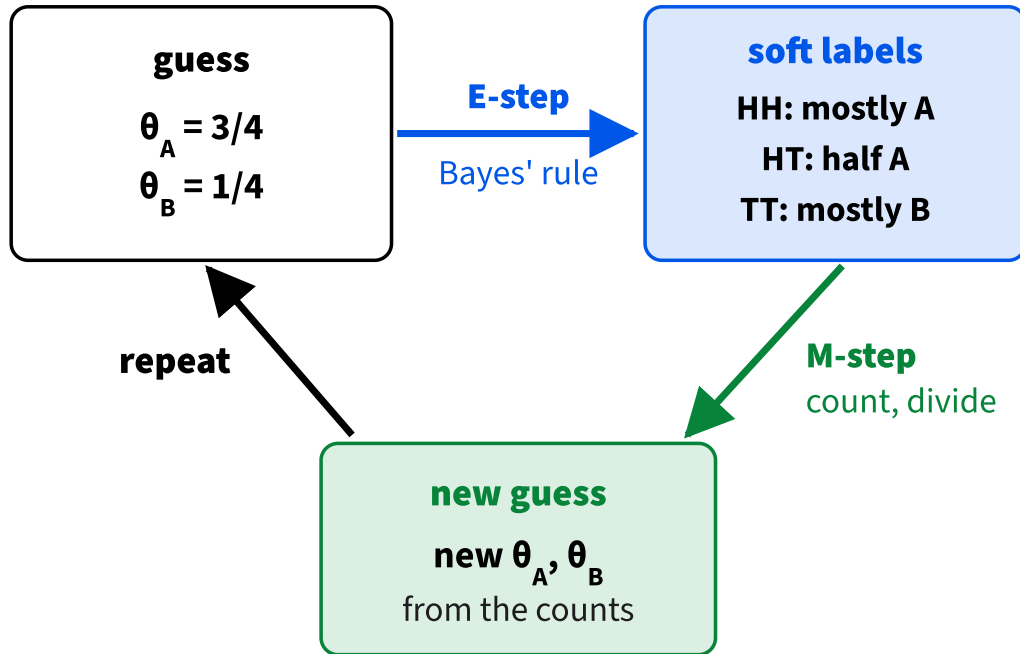
1.8 2 0.9

Answer. $1.8 = 0.9 \cdot 2$ heads. Computing the 0.9 is the E-step (expectation); counting fractional heads and dividing is the M-step (maximization) of expectation–maximization (EM).

Soft-label every record, count fractions, divide: expectation–maximization.

EM: guess the coins, soft-label the records, count, repeat

one EM round on the records HH, HT, TT



the flips never change; only the guess does

Goal: the θ_A, θ_B with the largest **log-likelihood** $\ell = \ln p(\text{HH}) + \ln p(\text{HT}) + \ln p(\text{TT})$, where $p(\text{HH})$ is the model's chance of record HH.

A circle: with θ we could label the records; with labels we could count θ . **Break the circle with a guess.**

E-step (expectation): Bayes' rule turns the guess into **soft labels**, each record's chance of coin A.

M-step (maximization): **count heads ÷ flips** for each coin, a record counting only by its share.

Why a guess is fine: each round corrects the guess using only the flips; whether a round can ever make the fit worse is the question this section ends on.

Each round needs only the flips and the last guess, never the true coin.

EM for any hidden-variable model: posterior, then weighted fit

Definition (expectation-maximization; Dempster, Laird & Rubin, 1977). Records $x^{(1)}, \dots, x^{(N)}$, hidden z , model $p_\theta(x, z)$. Start from a guess θ_0 ; for $t = 0, 1, 2, \dots$:

$$q_t^{(n)}(z) = p_{\theta_t}(z \mid x^{(n)}) = \frac{p_{\theta_t}(x^{(n)}, z)}{\sum_{z'} p_{\theta_t}(x^{(n)}, z')}$$

$$\theta_{t+1} = \arg \max_{\theta} \sum_n \sum_z q_t^{(n)}(z) \cdot \ln p_{\theta}(x^{(n)}, z)$$

E-step: posterior soft labels, record n 's chance of each value of z . **Bayes' rule:** one path $\div$ all paths of the record.

M-step: maximize each path's log, weighted by its soft label.

SYMBOL		IN THE COIN GAME
$x^{(n)}$	$\Leftrightarrow$	record n : HH, HT, TT
z	$\Leftrightarrow$	the coin, A or B
θ_t	$\Leftrightarrow$	guess after t rounds; θ_0 the start
$q_t^{(n)}(z)$	$\Leftrightarrow$	soft label of record n
$p_\theta(x, z)$	$\Leftrightarrow$	one path: pick coin z , then its flips

Why "expectation": the M-step maximizes

$\sum_n \mathbb{E}_{z \sim q_t^{(n)}} [\ln p_\theta(x^{(n)}, z)]$, the **expected complete-data log-likelihood** (record and coin together). It is not ℓ itself but an easy stand-in: every term is the log of one path, with no log of a sum, so it splits into counts. Whether raising it always raises ℓ is Section 4's question.

E: infer the hidden part. M: fit as if it were seen, with weights.

The M-step objective collects expected heads and tails

↳ Soft labels $q^{(n)}(z)$ held fixed (round index dropped). Record $x^{(n)}$ has $h^{(n)}$ heads, $2 - h^{(n)}$ tails; θ_z means θ_A if $z = A$, θ_B if $z = B$. A path is $p_\theta(x^{(n)}, z) = \frac{1}{2} \cdot \theta_z^{h^{(n)}} \cdot (1 - \theta_z)^{2-h^{(n)}}$: pick z , then its flips (pick chance held at $\frac{1}{2}$ for now).

$$\begin{aligned}
 \sum_n \sum_z q^{(n)}(z) \cdot \ln p_\theta(x^{(n)}, z) &= \sum_n \sum_z q^{(n)}(z) \cdot \ln \left(\frac{1}{2} \cdot \theta_z^{h^{(n)}} \cdot (1 - \theta_z)^{2-h^{(n)}} \right) && \text{(insert the path)} \\
 &= \sum_n \sum_z q^{(n)}(z) \cdot \left[\ln \frac{1}{2} + h^{(n)} \cdot \ln \theta_z + (2 - h^{(n)}) \cdot \ln(1 - \theta_z) \right] && \text{(product} \rightarrow \text{sum of logs)} \\
 &= \sum_n \sum_z q^{(n)}(z) \cdot \ln \frac{1}{2} + \sum_n \sum_z q^{(n)}(z) \cdot h^{(n)} \cdot \ln \theta_z && \text{(multiply out the bracket)} \\
 &\quad + \sum_n \sum_z q^{(n)}(z) \cdot (2 - h^{(n)}) \cdot \ln(1 - \theta_z) \\
 &= \ln \frac{1}{2} \cdot \sum_n \sum_z q^{(n)}(z) + \sum_z \ln \theta_z \cdot \sum_n q^{(n)}(z) \cdot h^{(n)} && \text{(swap the order of the sums;} \\
 &\quad + \sum_z \ln(1 - \theta_z) \cdot \sum_n q^{(n)}(z) \cdot (2 - h^{(n)}) && \text{ln } \theta_z \text{ does not depend on } n) \\
 &= N \cdot \ln \frac{1}{2} + \sum_z [H_z \cdot \ln \theta_z + T_z \cdot \ln(1 - \theta_z)] && (\sum_z q^{(n)}(z) = 1; \text{ name sums)} \\
 &= \text{const} + \underbrace{H_A \cdot \ln \theta_A + T_A \cdot \ln(1 - \theta_A)}_{\text{coin A's part: only } \theta_A} + \underbrace{H_B \cdot \ln \theta_B + T_B \cdot \ln(1 - \theta_B)}_{\text{coin B's part: only } \theta_B} && \text{(write out } z = A, B)
 \end{aligned}$$

$H_A = \sum_n q^{(n)}(A) \cdot h^{(n)}$, $T_A = \sum_n q^{(n)}(A) \cdot (2 - h^{(n)})$
 expected heads and tails **credited to A**; H_B, T_B the same
 with $q^{(n)}(B)$.

In words: Theorem 1's form (p. 11), **heads**·ln θ + **tails**·ln(1 - θ), once per coin, with fractional counts.

Why it matters: θ_A, θ_B share no term: each coin is fitted **on its own**.

No log of a sum remains: the M-step objective is expected counts times logs.

Slope zero: each coin's part peaks at expected heads ÷ flips

↳ Coin A's part: $f(\theta_A) = H_A \cdot \ln \theta_A + T_A \cdot \ln(1 - \theta_A)$; H_A, T_A = A's expected heads, tails.

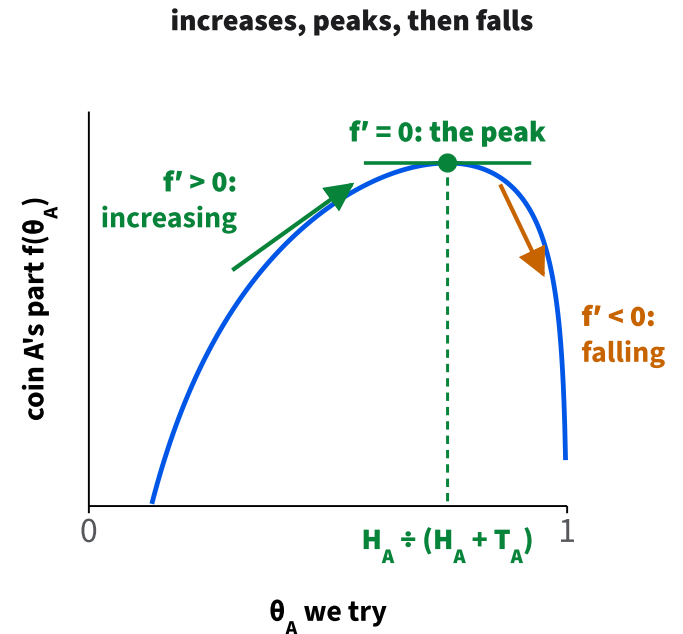
$$\begin{aligned}
 f'(\theta_A) &= \frac{H_A}{\theta_A} - \frac{T_A}{1 - \theta_A} && ((\ln \theta)' = \frac{1}{\theta}, (\ln(1 - \theta))' = \frac{-1}{1 - \theta}) \\
 &= \frac{H_A \cdot (1 - \theta_A) - T_A \cdot \theta_A}{\theta_A \cdot (1 - \theta_A)} && (\text{common denominator}) \\
 &= \frac{H_A - (H_A + T_A) \cdot \theta_A}{\theta_A \cdot (1 - \theta_A)} && (\text{multiply out the top}) \\
 &= \frac{H_A + T_A}{\theta_A \cdot (1 - \theta_A)} \cdot \left(\frac{H_A}{H_A + T_A} - \theta_A \right) && (\text{factor out } H_A + T_A)
 \end{aligned}$$

The first factor is positive, so $f' > 0$ left of $\frac{H_A}{H_A + T_A}$ and $f' < 0$ right of it: f **increases, then falls**.

$$\theta_A^{\text{new}} = \frac{H_A}{H_A + T_A} = \frac{\text{expected heads by A}}{\text{expected flips by A}}, \quad \theta_B^{\text{new}} = \frac{H_B}{H_B + T_B}$$

In words: Theorem 1 (p. 11) with $h = H_A, t = T_A$: **no search needed**.

The M-step is count-and-divide, with soft labels as fractional counts.



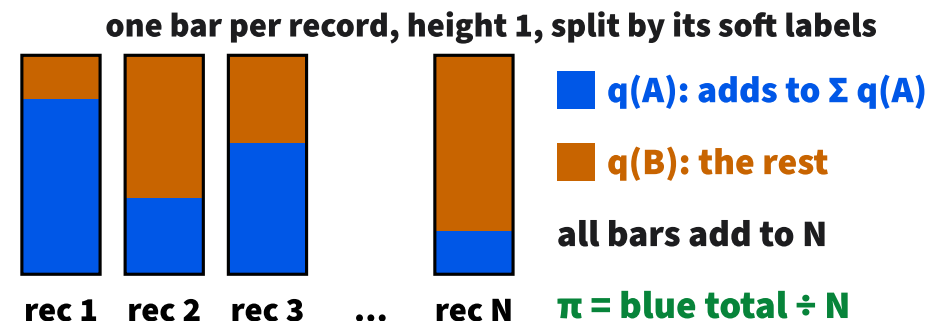
The pick chance π is counted too: records by A ÷ records

↳ So far the pick chance was held at $\frac{1}{2}$, which gave the constant $N \cdot \ln \frac{1}{2}$; learning $\pi = p(A)$ replaces that constant. A path's log splits, $\ln p_\theta(x^{(n)}, z) = \ln p(z) + \ln p_\theta(x^{(n)} | z)$, and only the pick factor $p(z)$, π if $z = A$ and $1 - \pi$ if $z = B$, holds π . Soft labels $q^{(n)}(z)$ held fixed.

$$\begin{aligned}
 \sum_n \sum_z q^{(n)}(z) \cdot \ln p(z) &= \sum_n [q^{(n)}(A) \cdot \ln \pi + q^{(n)}(B) \cdot \ln(1 - \pi)] && \text{(write out } z = A, B: p(A) = \pi, p(B) = 1 - \pi) \\
 &= \left(\sum_n q^{(n)}(A) \right) \cdot \ln \pi + \left(\sum_n q^{(n)}(B) \right) \cdot \ln(1 - \pi) && \text{(swap the sums)} \\
 \Rightarrow \pi &= \frac{\sum_n q^{(n)}(A)}{\sum_n q^{(n)}(A) + \sum_n q^{(n)}(B)} && \text{(slope zero, as for } \theta_A \text{ in Theorem 1's form,} \\
 &= \frac{\sum_n q^{(n)}(A)}{N} && \text{with } h = \sum_n q^{(n)}(A), t = \sum_n q^{(n)}(B)) \\
 &&& (\sum_z q^{(n)}(z) = 1 \text{ for each of the } N \text{ records})
 \end{aligned}$$

In words: π = **expected records by A ÷ records**, the same count-and-divide as θ_A .

Why it matters: the θ terms do not hold π , so θ_A, θ_B keep their formulas; **every table**, root included, is count-and-divide.



Learning the pick chance adds one more count: soft labels summed, divided by N .

On a tree model, the M-step is count-and-divide with fractions

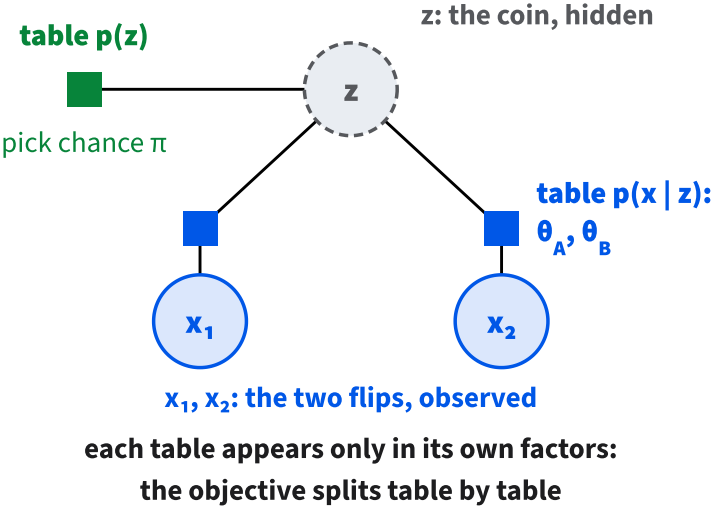
↳ Every table we have met, θ_A , θ_B and π , is count-and-divide. In general:

Corollary (of Theorem 1) (p. 11). If $p_\theta(x, z)$ is a product of tables on a tree factor graph, the M-step sets each table entry to

$$\theta_{\text{entry}} = \frac{\text{expected count of its case}}{\text{expected count of its condition}}.$$

In words: each record adds soft label · (times the case occurs in it).

entry of the coin model	case ÷ condition	formula
$\theta_A = p(H \mid A)$	flips by A showing H ÷ flips by A	$\frac{H_A}{H_A + T_A}$
$\theta_B = p(H \mid B)$	flips by B showing H ÷ flips by B	$\frac{H_B}{H_B + T_B}$
$\pi = p(A)$, pick chance	records by A ÷ records	$\frac{\sum_n q^{(n)}(A)}{N}$



Why: each table, a **conditional distribution** $p(\text{child} \mid \text{parent})$, appears only in its own factors and sums to 1 on its own, so the objective splits table by table.

Why it matters: a chain is a tree, so the hidden chain of Section 6 is refilled the same way.

Every table of a tree model is refilled by counting, with soft labels as fractions.

Where EM stops: a fixed point, $\theta^* = M(\theta^*)$

The round map. One EM round turns a guess θ' into

$$M(\theta') = \arg \max_{\theta} \sum_n \sum_z p_{\theta'}(z \mid x^{(n)}) \cdot \ln p_{\theta}(x^{(n)}, z),$$

so $\theta_{t+1} = M(\theta_t)$. A **fixed point** is a θ^* with $M(\theta^*) = \theta^*$.

Claim. If the rounds settle, $\theta_t \rightarrow \theta^*$, and M is continuous, then θ^* is a fixed point:

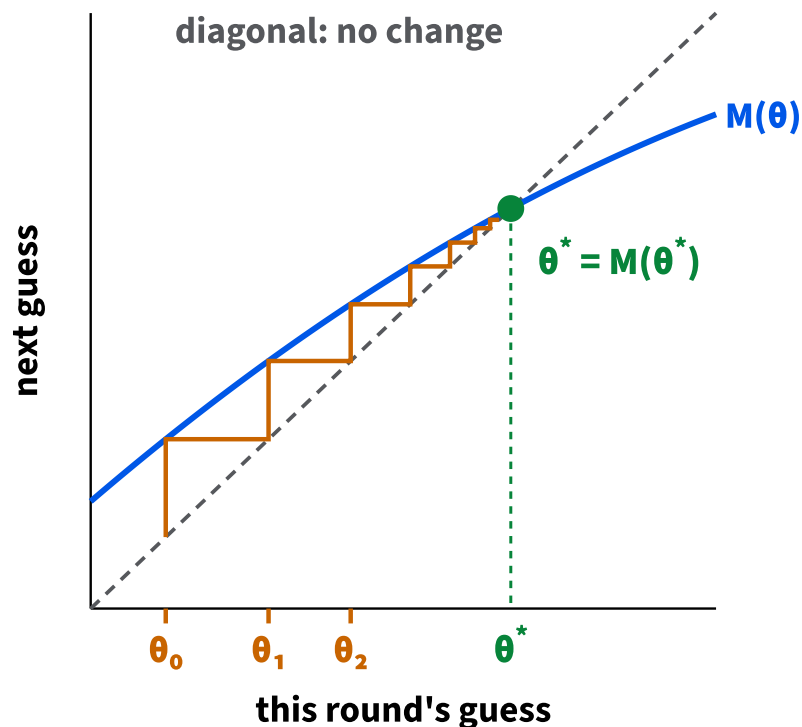
$$\begin{aligned} \theta^* &= \lim_{t \rightarrow \infty} \theta_{t+1} && \text{(the rounds settle at } \theta^*) \\ &= \lim_{t \rightarrow \infty} M(\theta_t) && \text{(one round: } \theta_{t+1} = M(\theta_t)) \\ &= M\left(\lim_{t \rightarrow \infty} \theta_t\right) = M(\theta^*) && (M \text{ continuous}) \end{aligned}$$

In words: EM stops where one more round gives back the same guess.

Why it matters: to find the end, solve the equation $\theta = M(\theta)$ instead of running rounds.

EM's end point solves $\theta = M(\theta)$: one more round changes nothing.

round map M: the staircase stops at the diagonal



At a fixed point inside, every slope of ℓ is zero

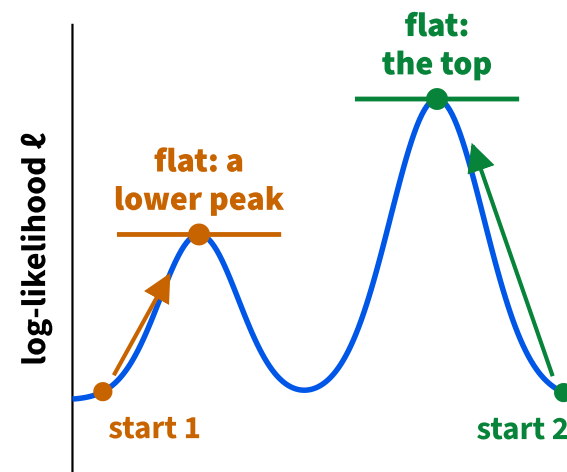
↳ $\ell(\theta) = \sum_n \ln p_\theta(x^{(n)})$ is the log-likelihood; θ_j is any entry of θ . Let $\theta^* = M(\theta^*)$ lie inside the range of θ , with soft labels $q_*^{(n)}(z) = p_{\theta^*}(z | x^{(n)})$. Recall Theorem 3 (p. 26) (Fisher's identity): $\frac{\partial}{\partial \theta_j} \ln p_\theta(x) = \sum_z p_\theta(z | x) \cdot \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z)$.

$$\begin{aligned}
 \frac{\partial \ell}{\partial \theta_j}(\theta^*) &= \sum_n \frac{\partial}{\partial \theta_j} \ln p_\theta(x^{(n)}) \Big|_{\theta^*} && \text{(slope of a sum)} \\
 &= \sum_n \sum_z p_{\theta^*}(z | x^{(n)}) \cdot \frac{\partial}{\partial \theta_j} \ln p_\theta(x^{(n)}, z) \Big|_{\theta^*} && \text{(Fisher's identity, each record)} \\
 &= \frac{\partial}{\partial \theta_j} \left[\sum_n \sum_z q_*^{(n)}(z) \cdot \ln p_\theta(x^{(n)}, z) \right]_{\theta^*} && \text{(labels } q_*^{(n)} \text{ held fixed)} \\
 &= 0 && \text{(its peak is } M(\theta^*) = \theta^*, \text{ inside)}
 \end{aligned}$$

In words: at EM's end, ℓ is **flat in every direction**.

Why it matters: flat is **not always the top**: a lower peak is flat too, so the start guess can decide where EM ends.

EM increases ℓ up to the flat point near its start



parameter θ
zero slope: a peak,
not always the highest

EM stops at a point where the log-likelihood has zero slope.

When is a flat point the top? Gibbs' inequality can certify it

↳ A flat point need not be the top. When ℓ is an **average of logs of shares**, one inequality settles it.

Gibbs' inequality. For distributions p and s on the same values k (every $p_k > 0$),

$$\sum_k p_k \cdot \ln s_k \leq \sum_k p_k \cdot \ln p_k, \quad \text{with equality only when } s = p.$$

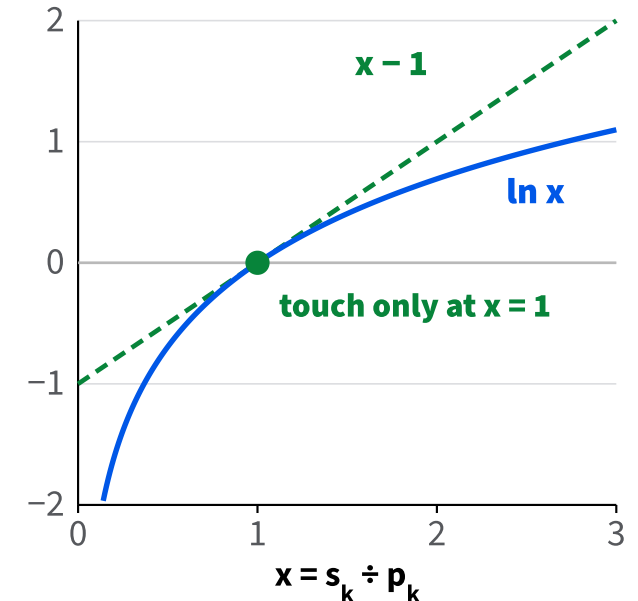
$$\begin{aligned} \sum_k p_k \cdot \ln s_k - \sum_k p_k \cdot \ln p_k &= \sum_k p_k \cdot \ln \frac{s_k}{p_k} && \text{(log of a quotient)} \\ &\leq \sum_k p_k \cdot \left(\frac{s_k}{p_k} - 1 \right) && (\ln x \leq x - 1; \text{ equal only at } x = 1) \\ &= \sum_k s_k - \sum_k p_k && \text{(multiply out)} \\ &= 1 - 1 = 0 && \text{(both sum to 1)} \end{aligned}$$

In words: the average log of a model share is **largest when the model's shares are the data's shares**.

Why it matters: it can certify that a flat point is the **global top**, not a lower peak.

Weighted by the data's shares, an average log share peaks only at those shares.

$\ln x$ never rises above $x - 1$



so equality needs $s_k = p_k$ for every k

E-step by hand: Bayes' rule splits each record between the coins

↳ **Definition (E-step) (p. 33)** on the coins. Start guess $\theta_0 = (\frac{3}{4}, \frac{1}{4})$: $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$; the friend picks each coin with chance $\frac{1}{2}$.

Bayes' rule, from the definition:

$$\underbrace{p(A \mid \text{record})}_{\text{share to A, the posterior}} = \frac{\text{path via A}}{\text{path via A} + \text{path via B}}$$

A path multiplies its steps, e.g. HH via A:

$$\underbrace{\frac{1}{2}}_{\text{pick A}} \cdot \underbrace{\frac{3}{4}}_{\text{H by A}} \cdot \underbrace{\frac{3}{4}}_{\text{H by A}} = \frac{9}{32}$$

record	path via A	path via B	share to A
HH	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{9}{9+1} = 0.9$
HT	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{32}$	$\frac{3}{3+3} = 0.5$
TT	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{1+9} = 0.1$

both paths, in 32nds

■ via A

■ via B



share to A: $9/10 = 0.9$



share to A: $3/6 = 0.5$



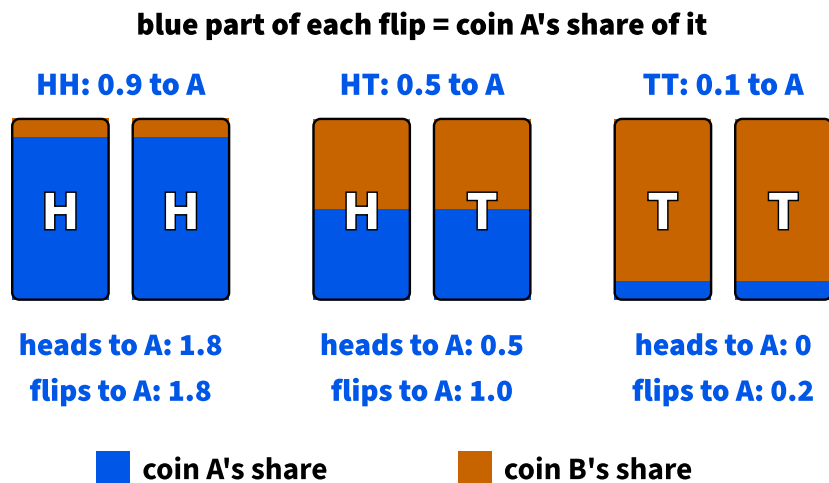
share to A: $1/10 = 0.1$

share = via A ÷ both

E-step output: soft labels 0.9, 0.5, 0.1 for coin A; coin B gets the rest.

M-step by hand: the guess moves to (0.767, 0.233)

↳ Count-and-divide (p. 35), $\theta_A = \frac{H_A}{H_A + T_A}$, on the coins: A is credited its share (0.9, 0.5, 0.1) (p. 41) of each record's heads and flips.



blue totals: heads 2.3, flips 3

record	heads to A share · heads	flips to A share · 2
HH	$0.9 \cdot 2 = 1.8$	$0.9 \cdot 2 = 1.8$
HT	$0.5 \cdot 1 = 0.5$	$0.5 \cdot 2 = 1.0$
TT	$0.1 \cdot 0 = 0$	$0.1 \cdot 2 = 0.2$
total	2.3 = H_A	3.0 = $H_A + T_A$

Coin B is credited the other shares, $1 - 0.9$, $1 - 0.5$, $1 - 0.1 = 0.1$, 0.5 , 0.9 .

$$\theta_A = \frac{\text{heads credited to A}}{\text{flips credited to A}} = \frac{2.3}{3} = 0.767$$

$$\theta_B = \frac{0.1 \cdot 2 + 0.5 \cdot 1 + 0.9 \cdot 0}{2 \cdot (0.1 + 0.5 + 0.9)} = \frac{0.7}{3} = 0.233$$

One round moved the guess from $(\frac{3}{4}, \frac{1}{4})$ to (0.767, 0.233).

Each round, θ_A increases: 0.750, 0.767, 0.777, ..., 0.789

↳ The round map (p. 38) $\theta_{t+1} = M(\theta_t)$ on the coins, again and again.

By symmetry $\theta_B = 1 - \theta_A$, HT's share stays 0.5 and A is credited 3 flips every round. One round:

$$\theta_A^{\text{new}} = \frac{\overbrace{2 \cdot q_{\text{HH}}}^{\text{HH's heads}} + \overbrace{1 \cdot 0.5}^{\text{HT's}}}{3}, \quad q_{\text{HH}} = \frac{\theta_A^2}{\theta_A^2 + \theta_B^2}$$

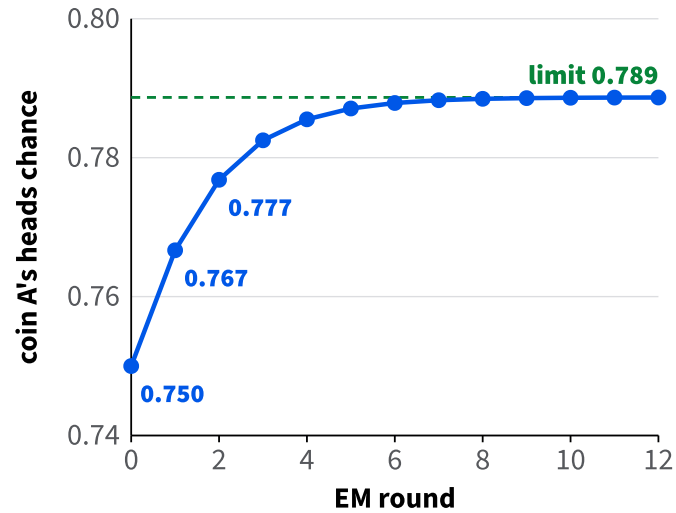
round	θ_A	q_{HH} , share to A	new θ_A
1	$\frac{3}{4}$	$\frac{9}{9+1} = 0.900$	$\frac{2 \cdot (0.900) + 0.5}{3} = 0.767$
limit	$\frac{3+\sqrt{3}}{6} = 0.789$	$\frac{0.789^2}{0.789^2 + 0.211^2} = 0.933$	$\frac{2 \cdot (0.933) + 0.5}{3} = 0.789$

At the limit a round returns the same θ_A : EM has reached a **fixed point**. Is it the best possible fit, or only a flat point?

All steps: Appendix, Section 3 (p. 120).

round 12 · $\theta_A = 0.789$ · $\theta_B = 0.211$

At 0.789 a round gives back 0.789: EM has reached a fixed point.



Why 0.789 is the global top: no pair of coins scores above -3.989

↳ **Gibbs' inequality (p. 40) applied to the coins.** Write $s_2 = p(\text{HH})$, $s_1 = p(\text{HT}) + p(\text{TH}) = 2 \cdot p(\text{HT})$, $s_0 = p(\text{TT})$: the model's chances of 2, 1, 0 heads, which add to 1. The data have one record of each, so the data's shares are $p_k = \frac{1}{3}$.

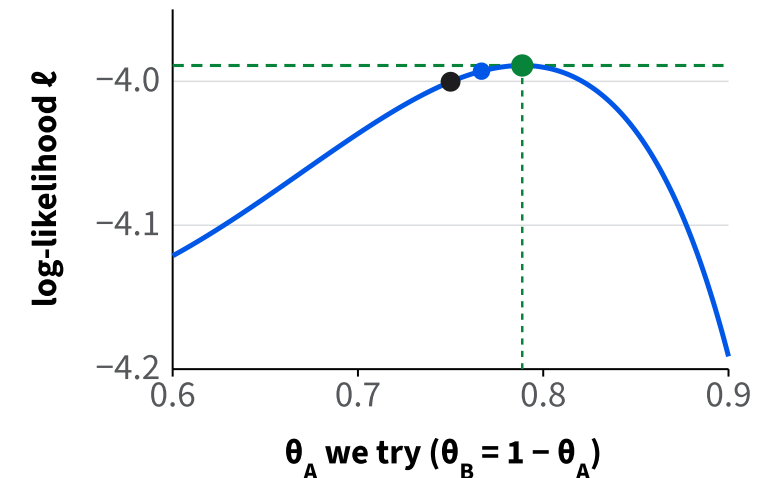
$$\begin{aligned}
 \ell(\theta) &= \ln p(\text{HH}) + \ln p(\text{HT}) + \ln p(\text{TT}) && \text{(the three records)} \\
 &= \ln s_2 + \ln \frac{s_1}{2} + \ln s_0 && (p(\text{HT}) = \frac{s_1}{2}) \\
 &= 3 \cdot \left(\frac{1}{3} \cdot \ln s_2 + \frac{1}{3} \cdot \ln s_1 + \frac{1}{3} \cdot \ln s_0 \right) - \ln 2 && \text{(pull out 3 and } \ln 2) \\
 &\leq 3 \cdot \left(\frac{1}{3} \cdot \ln \frac{1}{3} + \frac{1}{3} \cdot \ln \frac{1}{3} + \frac{1}{3} \cdot \ln \frac{1}{3} \right) - \ln 2 && \text{(Gibbs, } p_k = \frac{1}{3}) \\
 &= 3 \cdot \ln \frac{1}{3} - \ln 2 = -3.989 && \text{(for every } \theta)
 \end{aligned}$$

At the limit $\theta_A = \frac{3+\sqrt{3}}{6} = 0.789$, $\theta_B = \frac{3-\sqrt{3}}{6} = 0.211$: $\theta_A \cdot \theta_B = \frac{9-3}{36} = \frac{1}{6}$, so

$$\begin{aligned}
 s_1 &= 2 \cdot \theta_A \cdot \theta_B = \frac{1}{3}, \\
 s_2 &= s_0 = \frac{1}{2} \cdot (\theta_A^2 + \theta_B^2) = \frac{1}{2} \cdot ((\theta_A + \theta_B)^2 - 2 \cdot \theta_A \cdot \theta_B) = \frac{1}{2} \cdot \left(1 - \frac{1}{3}\right) = \frac{1}{3}.
 \end{aligned}$$

Every share equals the data's share, so the " $\leq$ " is an equality: $\ell = -3.989$ **reaches the ceiling** that no pair of coins can pass.

ℓ increases to the bound and touches it



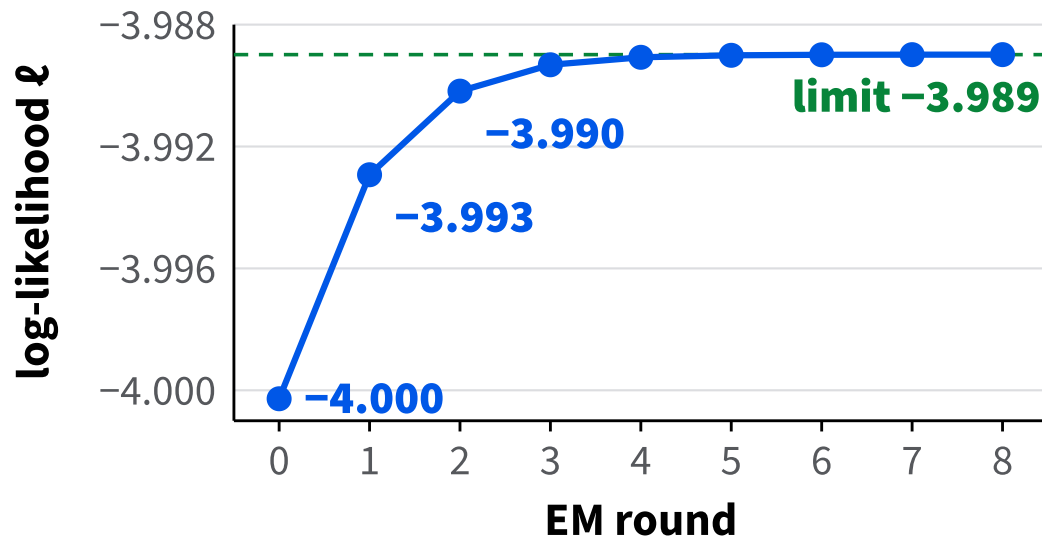
- start 0.750: $\ell = -4.000$
- round 1, 0.767: $\ell = -3.993$
- fixed point 0.789: $\ell = -3.989$
- Gibbs bound $3 \cdot \ln(1/3) - \ln 2$

Gibbs' inequality gives a ceiling for every θ ; at 0.789 EM touches it, so the fixed point is the global top.

Next: does every EM round improve the fit?



ℓ after each round on HH, HT, TT



On the coins, ℓ rose in every round (figure), from -4.000 to -3.989 where $\theta_A = 0.789$ (p. 44).

All steps: Appendix, Section 3 (p. 120).

For any model and any start, can an EM round lower ℓ ?

yes, sometimes

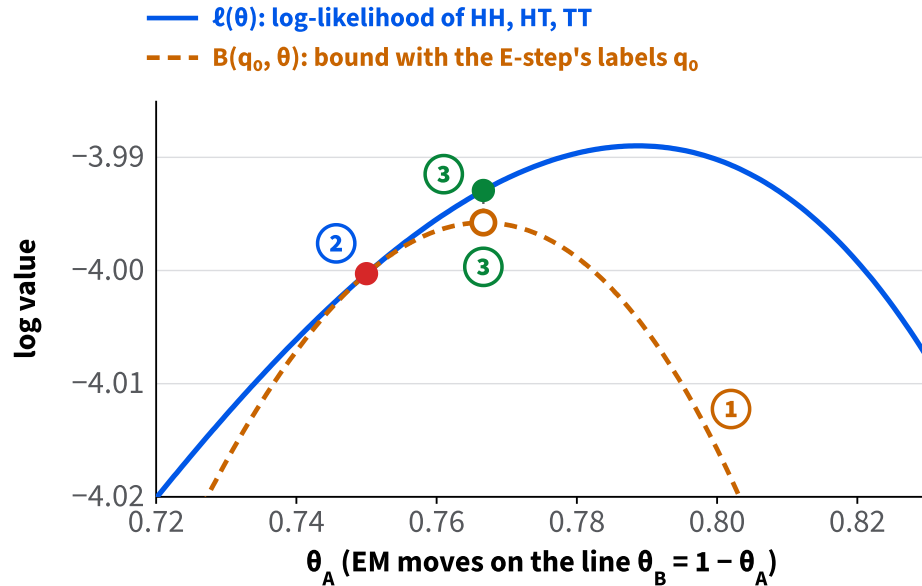
never

only at the start

Answer. Never. Section 4 proves it with one picture and one inequality.

A guarantee, not luck: EM increases the likelihood.

EM builds a floor under ℓ , then jumps to the floor's top



round **1**: θ_A moves from the red dot to the circle ($\theta_A = 0.767$)

Goal: show that EM's log-likelihood ℓ , the sum of $\ln p(\text{record})$, never goes down.

1 · The bound B (dashed curve)

With the E-step's labels $q_0(z)$ (chance of coin A: HH 0.9, HT 0.5, TT 0.1) (p. 41), define $B(q_0, \theta) = \sum_{\text{records } x} \sum_{z=A,B} q_0(z) \cdot \ln \frac{p_\theta(x, z)}{q_0(z)}$, with $p_\theta(x, z)$ = chance of picking coin z and seeing x . It is a **floor**: $B \leq \ell$ (Lemma 1).

2 · Touch (red dot)

At the current guess θ_0 the labels are the posteriors, and the bound **touches** ℓ .

3 · Below, then top

Elsewhere B lies **below** ℓ . The M-step jumps to B 's top (circle); ℓ is **higher still** (green dot).

Touch at the current estimate, below everywhere else, jump to the top: ℓ can only go up.

Lemma 1: for every guess, $\ln p(x) = \text{bound} + \text{gap}$

↳ Symbols: record x ; hidden z (the coin); guess $q(z)$, any chances over z adding to 1; posterior $p_\theta(z | x) = p_\theta(x, z)/p_\theta(x)$; log-likelihood $\ell(\theta) = \sum_n \ln p_\theta(x^{(n)})$; bound $B(q, \theta) = \sum_z q(z) \cdot \ln \frac{p_\theta(x, z)}{q(z)}$.

$$\ln p_\theta(x) = \ln p_\theta(x, z) - \ln p_\theta(z | x)$$

$$= \sum_z q(z) \cdot [\ln p_\theta(x, z) - \ln p_\theta(z | x)]$$

$$= \sum_z q(z) \cdot \left[\ln \frac{p_\theta(x, z)}{q(z)} + \ln \frac{q(z)}{p_\theta(z | x)} \right]$$

$$= \sum_z q(z) \cdot \ln \frac{p_\theta(x, z)}{q(z)} + \sum_z q(z) \cdot \ln \frac{q(z)}{p_\theta(z | x)}$$

$$= \underbrace{B(q, \theta)}_{\text{bound}} + \underbrace{D_{\text{KL}}(q \| p_\theta(\cdot | x))}_{\text{gap}}$$

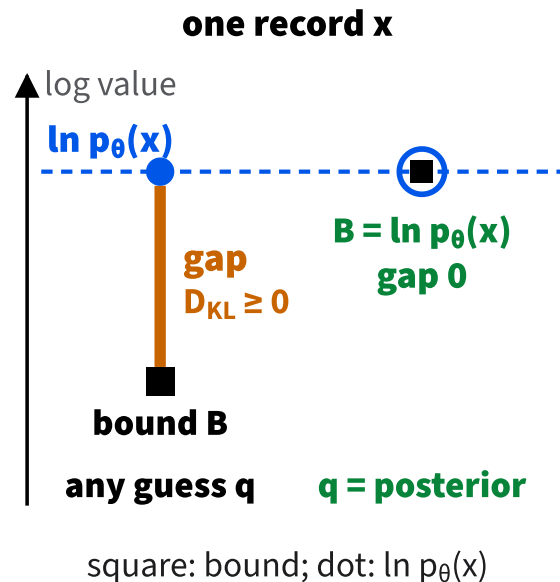
posterior's definition, in logs;
true for every z

no z on the left: its q -average
(weights adding to 1) is itself

subtract and add $\ln q(z)$

split the sum in two

name the two sums



The second sum is the **Kullback–Leibler divergence** of the guess from the posterior. **In words:** for **any** guess, the log-likelihood splits exactly into the bound and a gap.

Why it matters: the bound uses only $p_\theta(x, z)$, one path at a time, so it is easy to maximize; the gap is what we give up by using it.

For any guess, $\ln p(x)$ splits exactly into bound + gap.

The gap is never negative because $\ln y \leq y - 1$

↳ To show: $D_{\text{KL}} \geq 0$. Write $y = p_\theta(z | x) / q(z)$, posterior ÷ guess.

$$\begin{aligned}
 -D_{\text{KL}} &= -\sum_z q(z) \cdot \ln \frac{q(z)}{p_\theta(z | x)} && \text{the gap's definition} \\
 &= \sum_z q(z) \cdot \ln \frac{p_\theta(z | x)}{q(z)} && \text{flip fraction and sign} \\
 &\leq \sum_z q(z) \cdot \left(\frac{p_\theta(z | x)}{q(z)} - 1 \right) && \ln y \leq y - 1; q(z) \geq 0 \\
 &= \sum_z (p_\theta(z | x) - q(z)) && q(z) \text{ cancels} \\
 &= \sum_z p_\theta(z | x) - \sum_z q(z) \leq 1 - 1 = 0 && \text{posterior adds to } \leq 1, \text{ guess to } 1
 \end{aligned}$$

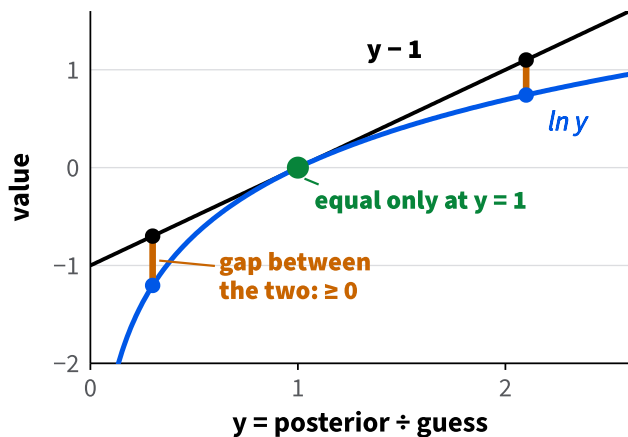
Equality needs every $y = 1$, that is $q = \text{posterior}$. ■ (Sums skip z with $q(z) = 0$, which contribute 0; over the rest the posterior adds to at most 1.)

Same fact, picture form (Jensen's inequality): $\ln$ bends down, so mean of logs $\leq$ log of mean): $B(q, \theta) = \sum_z q(z) \cdot \ln \frac{p_\theta(x, z)}{q(z)} \leq \ln \sum_z q(z) \cdot \frac{p_\theta(x, z)}{q(z)} = \ln p_\theta(x)$.

In words: gap ≥ 0 , 0 only at the posterior. **Why it matters:** for every guess, the bound is a **floor** under $\ln p_\theta(x)$.

Lemma 1 (bound and gap). For every guess q : (i) $\ln p_\theta(x) = B(q, \theta) + D_{\text{KL}}(q \| p_\theta(\cdot | x))$; (ii) $D_{\text{KL}} \geq 0$, and $D_{\text{KL}} = 0$ exactly when $q = p_\theta(z | x)$. Hence $B(q, \theta) \leq \ln p_\theta(x)$ for every guess; with one guess $q^{(n)}$ per record, $\sum_n B(q^{(n)}, \theta) \leq \ell(\theta)$.

$\ln y$ stays below its tangent line $y - 1$



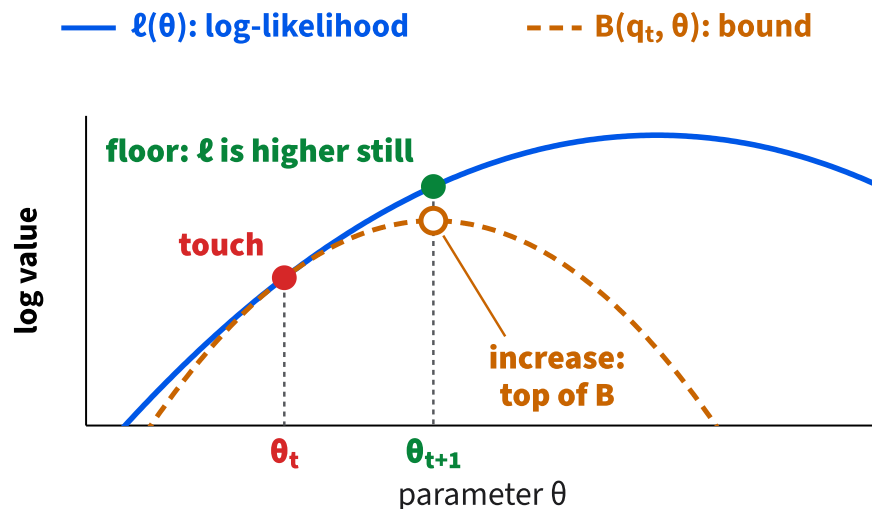
$\ln$ lies below its tangent, so the gap is ≥ 0 and vanishes only at the posterior: Lemma 1 holds.

Theorem 4: no EM round lowers the log-likelihood

Recall EM (p. 33), round t , records $x^{(n)}$:

E-step $q_t^{(n)}(z) = p_{\theta_t}(z \mid x^{(n)})$, the posterior.

M-step $\theta_{t+1} = \arg \max_{\theta} \sum_n \sum_z q_t^{(n)}(z) \cdot \ln p_{\theta}(x^{(n)}, z)$.



Why it matters: EM needs no step size and never overshoots; we can stop when ℓ stops rising.

Theorem 4 (EM increases the likelihood; Dempster, Laird & Rubin, 1977). With Lemma 1's bound (p. 47) at the E-step's q_t ,

$$B(q_t, \theta) = \sum_n \sum_z q_t^{(n)}(z) \cdot \ln \frac{p_{\theta}(x^{(n)}, z)}{q_t^{(n)}(z)},$$

every round t satisfies

$$\ell(\theta_{t+1}) \underset{\text{floor}}{\geq} B(q_t, \theta_{t+1}) \underset{\text{increase}}{\geq} B(q_t, \theta_t) \underset{\text{touch}}{=} \ell(\theta_t).$$

In words: Read right to left: equal at θ_t , the M-step raises B , ℓ stays above B .

Predict: if the M-step only raises its objective, without maximizing it, can ℓ fall?

yes no

Answer. No: the increase link only needs B to rise. This variant is **generalized EM**.

Touch, increase, floor: every EM round keeps or raises ℓ .

Proof of Theorem 4: touch, increase, floor

↳ Touch and floor use Lemma 1 (p. 47); increase uses the M-step's definition. Write $c_t = \sum_n \sum_z q_t^{(n)}(z) \cdot \ln q_t^{(n)}(z)$: it does not depend on θ .

$$\begin{aligned}
 \ell(\theta_{t+1}) &= B(q_t, \theta_{t+1}) + \sum_n D_{\text{KL}}(q_t^{(n)} \parallel p_{\theta_{t+1}}(\cdot \mid x^{(n)})) \\
 &\geq B(q_t, \theta_{t+1}) \\
 &= \sum_n \sum_z q_t^{(n)}(z) \cdot \ln p_{\theta_{t+1}}(x^{(n)}, z) - c_t \\
 &\geq \sum_n \sum_z q_t^{(n)}(z) \cdot \ln p_{\theta_t}(x^{(n)}, z) - c_t \\
 &= B(q_t, \theta_t) \\
 &= \ell(\theta_t) - \sum_n D_{\text{KL}}(q_t^{(n)} \parallel p_{\theta_t}(\cdot \mid x^{(n)})) \\
 &= \ell(\theta_t) \quad \blacksquare
 \end{aligned}$$

Lemma 1 at θ_{t+1} , guess q_t ,
summed over n

floor: each $D_{\text{KL}} \geq 0$

$$\ln \frac{p}{q} = \ln p - \ln q$$

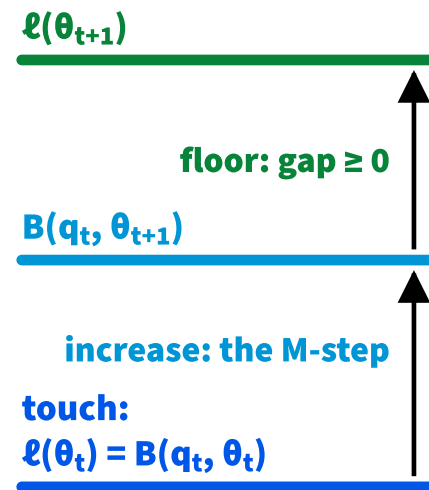
increase: the sum's max, at θ_{t+1} ,
 $\geq$ its value at θ_t

the same split, backwards

Lemma 1 at θ_t

touch: $q_t = \text{posterior}$, so $D_{\text{KL}} = 0$

read bottom to top



each link is $\geq$ or $=$

In words: start equal at θ_t ; the M-step raises the bound; ℓ stays above the bound. **Why it matters:** Lemma 1 holds for **every** guess, also the old q_t at the new θ_{t+1} .

Chain the three lines: $\ell(\theta_{t+1}) \geq B(q_t, \theta_{t+1}) \geq B(q_t, \theta_t) = \ell(\theta_t)$.

Where EM stops, ℓ is flat, but not always at a peak

↳ EM **stops** at θ_t when the M-step returns it: $\theta_{t+1} = \theta_t$, inside the allowed range. Take any one entry θ_j of θ . Theorem 3 (Fisher's identity) (p. 26): $\frac{\partial}{\partial \theta_j} \ln p_\theta(x) = \sum_z p_\theta(z | x) \cdot \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z)$.

$$\left. \frac{\partial \ell}{\partial \theta_j} \right|_{\theta_t} = \sum_n \sum_z p_{\theta_t}(z | x^{(n)}) \cdot \left. \frac{\partial}{\partial \theta_j} \ln p_{\theta_t}(x^{(n)}, z) \right|_{\theta_t}$$

Theorem 3, each record

$$= \sum_n \sum_z q_t^{(n)}(z) \cdot \left. \frac{\partial}{\partial \theta_j} \ln p_{\theta_t}(x^{(n)}, z) \right|_{\theta_t}$$

E-step: $q_t^{(n)}$ = posterior at θ_t

$$= \frac{\partial}{\partial \theta_j} \left[\sum_n \sum_z q_t^{(n)}(z) \cdot \ln p_{\theta_t}(x^{(n)}, z) \right]_{\theta_t}$$

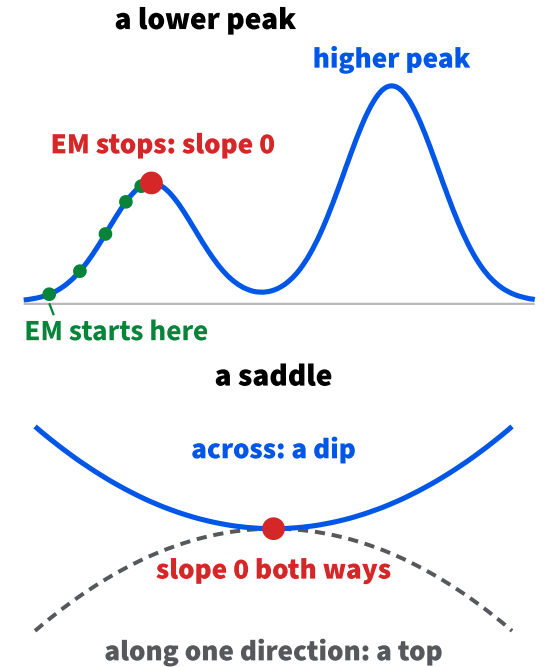
q_t is fixed numbers (no θ),
so ∂ moves outside the sums

$$= 0$$

θ_t maximizes the M-step sum
inside the allowed range: slope 0

Where EM stops. A point that EM returns to itself is a **stationary point** of ℓ : every slope is 0. It may be the highest peak, a lower peak, or a **saddle**: a top in one direction, a dip in another.

In words: EM stops where ℓ is flat. **Why it matters:** Theorem 4 (p. 49) forbids going down, not stopping early; start EM from several different guesses.



Increasing ℓ is guaranteed; reaching the highest peak is not.

On HH every guess is a floor; only the posterior touches $\ln p(\text{HH})$

↳ Lemma 1 (p. 47) applied to HH at $\theta = (\frac{3}{4}, \frac{1}{4})$. Recall its paths (p. 28) $\frac{9}{32}$ via A and $\frac{1}{32}$ via B, so $\ln p(\text{HH}) = \ln \frac{5}{16} = -1.163$.

$$B(q) = q(A) \cdot \ln \frac{9/32}{q(A)} + q(B) \cdot \ln \frac{1/32}{q(B)}$$

Predict: which guess $q(A)$ gives HH the highest floor?

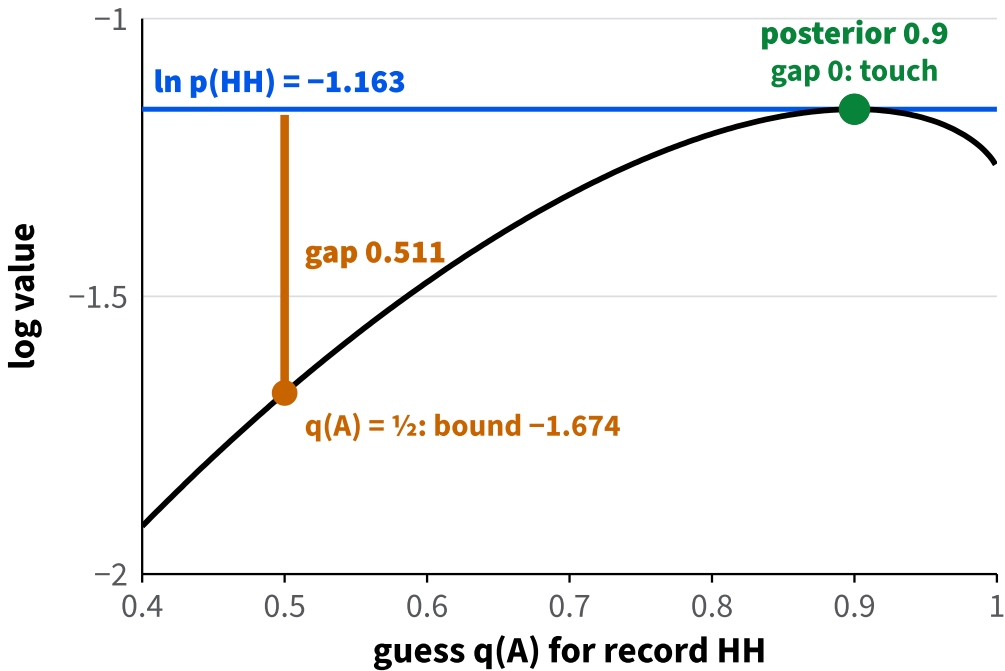
Answer. **0.9, HH's posterior**: its floor reaches $\ln p(\text{HH})$ itself.

guess $q(A)$	bound B	gap = $-1.163 - B$
$\frac{1}{2}$	-1.674	0.511
0.9 (the posterior)	-1.163	0

Why **0.9 touches**: path ÷ posterior = $p(\text{HH})$ for **each** coin, so $B = \ln p(\text{HH})$.

All steps: Appendix, Section 4 (p. 121).

black: bound $B(q)$ for every guess $q = (q(A), 1 - q(A))$



guess $q(A) = 0.90$: bound -1.163 + gap 0.000 = -1.163

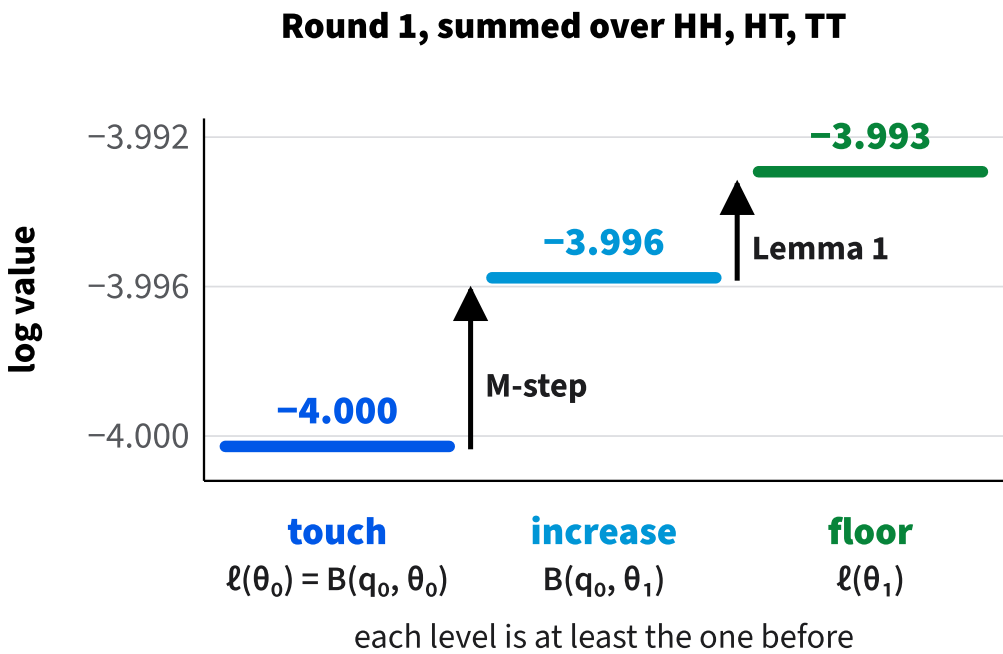
Every guess is a floor; only the posterior touches, and that is the E-step's guess.

Round 1 on the coins: ℓ goes up from -4.000 to -3.993

↳ Theorem 4 (p. 49) on one EM round from $\theta_0 = (\frac{3}{4}, \frac{1}{4})$ to $\theta_1 = (0.767, 0.233)$ (both from Section 3) (p. 42).

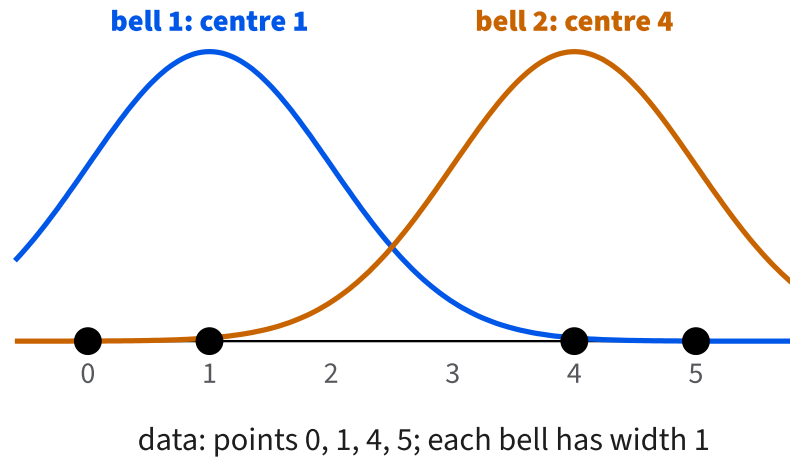
step	value
start: $\ell(\theta_0)$; the floor touches here	-4.000
M-step: the top of the floor	-3.996
new $\ell(\theta_1)$, above the floor	-3.993

All steps: Appendix, Section 4 (p. 122).



Each step is at least the one before, so ℓ went up.

Next: replace the coins by bell curves



Nothing in the proofs of Lemma 1 (p. 48) and Theorem 4 (p. 50) used coins: only an observed x , a hidden z , and a model $p_{\theta}(x, z)$.

The points 0, 1, 4, 5 come from two bell curves. In EM, what plays the hidden coin?

each point

each point's bell

the bell's width

Answer. The bell: hidden, picked first, then the point is drawn from it. This is a Gaussian mixture.

Same E-step, same M-step, same increase in ℓ : now EM clusters points.

Same game, new records: numbers on a line instead of coin flips

Goal: find hidden groups in data with the same EM.

A friend picks **group 1 or group 2** by a fair toss, then draws a number from that group's **bell**: a normal (Gaussian) distribution with standard deviation (width)

1. We see only the numbers **0, 1, 4, 5**.

COIN GAME

coin A or B, chance $\frac{1}{2}$

record HH

θ_A , guess $\frac{3}{4}$

hidden: which coin

$\Leftrightarrow$

$\Leftrightarrow$

$\Leftrightarrow$

$\Leftrightarrow$

MIXTURE

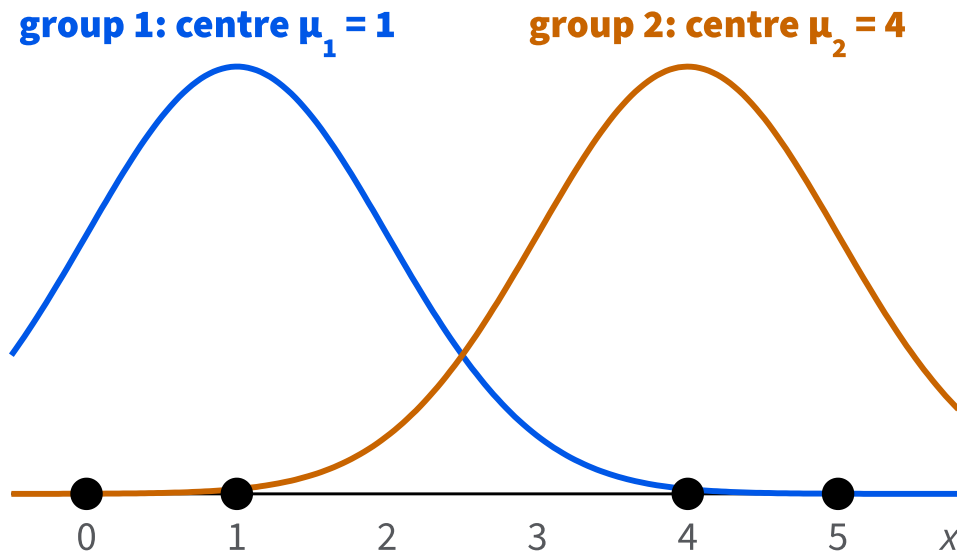
group 1 or 2, chance $\frac{1}{2}$

one number, e.g. 1

centre μ_1 , guess 1

hidden: which group

Unknowns: the centres μ_1, μ_2 . Working guess $\mu_1 = 1, \mu_2 = 4$, as $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$ for the coins.



data: points 0, 1, 4, 5

which group made each point is hidden
like the coins: pick a group ($\frac{1}{2}, \frac{1}{2}$),
then draw a point from its bell

Coin $\Leftrightarrow$ group, flips $\Leftrightarrow$ a number: each EM step carries over one by one.

EM for mixtures: Bayes' rule, then weighted averages

Gaussian mixture. Pick group k with chance π_k ; draw x from the bell $\mathcal{N}(x \mid \mu_k, \sigma^2) = \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot e^{-(x-\mu_k)^2/2\sigma^2}$ (centre μ_k , standard deviation, or width, σ), for $k = 1, \dots, K$. Data $x_1, \dots, x_N$; the hidden group of point n is z_n , as the hidden coin was.

$$p(x) = \sum_k p(z = k) \cdot p(x \mid z = k) \quad \text{sum over the hidden group}$$

$$= \sum_k \pi_k \cdot \mathcal{N}(x \mid \mu_k, \sigma^2) \quad \text{pick chance} \cdot \text{bell}$$

$$r_{nk} = p(z_n = k \mid x_n) = \frac{\pi_k \cdot \mathcal{N}(x_n \mid \mu_k, \sigma^2)}{\sum_j \pi_j \cdot \mathcal{N}(x_n \mid \mu_j, \sigma^2)} \quad \text{E-step: Bayes' rule}$$

$$= \frac{e^{-(x_n - \mu_k)^2/2\sigma^2}}{\sum_j e^{-(x_n - \mu_j)^2/2\sigma^2}} \quad \text{if all } \pi_j \text{ are equal: } \pi_j, 1/(\sqrt{2\pi} \cdot \sigma) \text{ cancel}$$

$$\mu_k = \frac{\sum_n r_{nk} \cdot x_n}{\sum_n r_{nk}}, \quad \pi_k = \frac{1}{N} \cdot \sum_n r_{nk} \quad \text{M-step: weighted averages}$$

In words: r_{nk} = posterior chance that group k made point n ; new centre = share-weighted average.

Why it matters: both steps are closed-form, one division per point and one average per group, so a round is cheap.

The coins' two steps, with bell heights in place of $\theta^h \cdot (1 - \theta)^t$ (h heads, t tails).

The weighted average maximizes the weighted log-likelihood

↳ Group 1's weighted log-likelihood $\ell_1(\mu) = \sum_n r_n \cdot \ln \mathcal{N}(x_n | \mu, 1)$: point n counts with its share $r_n = r_{n1}$ (group 1's share of point n). **EM's M-step maximizes this.** For simplicity every bell has width $\sigma = 1$, so $\ln \mathcal{N}(x | \mu, 1) = -\frac{1}{2} \cdot \ln 2\pi - \frac{1}{2} \cdot (x - \mu)^2$.

$$\ell_1(\mu) = \sum_n r_n \cdot \left(-\frac{1}{2} \cdot \ln 2\pi - \frac{1}{2} \cdot (x_n - \mu)^2 \right)$$

log of each bell

$$= c - \frac{1}{2} \cdot \sum_n r_n \cdot (x_n - \mu)^2$$

$$c = -\frac{1}{2} \cdot \ln 2\pi \cdot \sum_n r_n: \text{no } \mu$$

$$\frac{d\ell_1}{d\mu} = \sum_n r_n \cdot (x_n - \mu)$$

$$\frac{d}{d\mu} \left[-\frac{1}{2} \cdot (x_n - \mu)^2 \right] = x_n - \mu$$

$$= \sum_n r_n \cdot x_n - \mu \cdot \sum_n r_n$$

split the sum

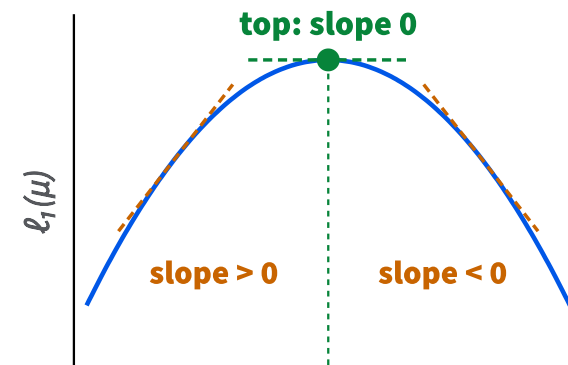
$$0 = \sum_n r_n \cdot x_n - \mu \cdot \sum_n r_n$$

at the top the slope is 0

$$\mu = \frac{\sum_n r_n \cdot x_n}{\sum_n r_n}$$

divide by $\sum_n r_n > 0$;

$$\frac{d^2\ell_1}{d\mu^2} = -\sum_n r_n < 0: \text{a top}$$



$$\mu = \sum r_n x_n \div \sum r_n$$

a downward parabola in μ :
its one top is where the slope is 0

In words: the best centre is the average of the points, each point counted r_n times.

Why it matters: the same algebra on $h \cdot \ln \theta + t \cdot \ln(1 - \theta)$ gives the coins' $\theta = h/(h + t)$ (p. 11): weighted heads $\div$ weighted flips.

The M-step is maximum likelihood with fractional counts, for coins and bells alike.

k-means: join the nearest centre, then move to the average

k-means (Lloyd, 1982), the most widely used clustering method, repeats two steps until nothing changes. Its score is $J = \sum_n (x_n - \mu_{k(n)})^2$, the sum of (point – its centre)², where $k(n)$ is the label of point n ; smaller is tighter.

Assign

each point joins the **nearest** centre,
 $k(n) = \arg \min_k |x_n - \mu_k|$: a hard label, weight 1 for that centre and 0 for the others, never a fraction.

Assign: hard labels



dashed: each point joins its nearest centre (triangle)

Move

each centre goes to the **plain average** of the points that joined it: $\mu_k = \frac{1}{|G_k|} \cdot \sum_{n \in G_k} x_n$, with G_k the set of points labelled k .

Move: plain averages



each centre (triangle) moves to the average of its points (diamond)

Two hard steps: a nearest-centre test for each point, then a plain average for each group.

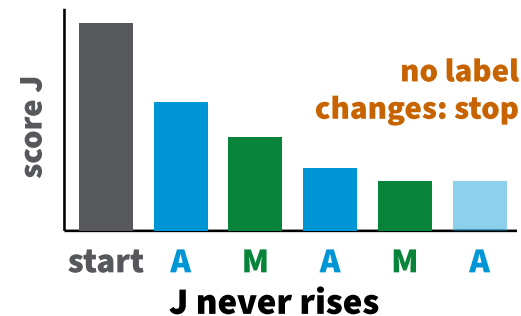
Neither k-means step raises J, so k-means always stops

↳ **Descent result.** $J(\text{labels}, \text{centres}) = \sum_n (x_n - \mu_{k(n)})^2$. One round: assign (new labels), then move (new centres).

$$\begin{aligned} J(\text{old labels}, \text{old centres}) &\geq J(\text{new labels}, \text{old centres}) && \text{assign: each point's term made smallest} \\ &\geq J(\text{new labels}, \text{new centres}) && \text{move: each group's sum smallest at its average} \end{aligned}$$

Why the average: for a fixed group G_k ,

$$\begin{aligned} \frac{d}{d\mu} \sum_{n \in G_k} (x_n - \mu)^2 &= -2 \cdot \sum_{n \in G_k} (x_n - \mu) && \text{derivative of each square} \\ 0 &= \sum_{n \in G_k} x_n - |G_k| \mu && \text{slope 0; split the sum, divide by } -2 \\ \mu &= \frac{1}{|G_k|} \cdot \sum_{n \in G_k} x_n && \begin{aligned} &\text{divide by } |G_k| > 0; \\ &\text{2nd derivative } 2|G_k| > 0: \text{ a minimum} \end{aligned} \end{aligned}$$



A = assign · M = move

Why it stops: after each move, J depends only on the labelling and strictly falls at every label change (ties keep the old label), so **no labelling repeats**; only K^N labellings exist.

In words: each step tightens the groups or leaves them alone, and k-means stops after finitely many rounds.

Why it matters: like EM's log-likelihood (Theorem 4) (p. 49), J only improves; but "stops" is not "best": a poor start can stop at a poor answer. In practice, use several starts or k-means++ (Arthur & Vassilvitskii, 2007), then run EM from k-means' centres.

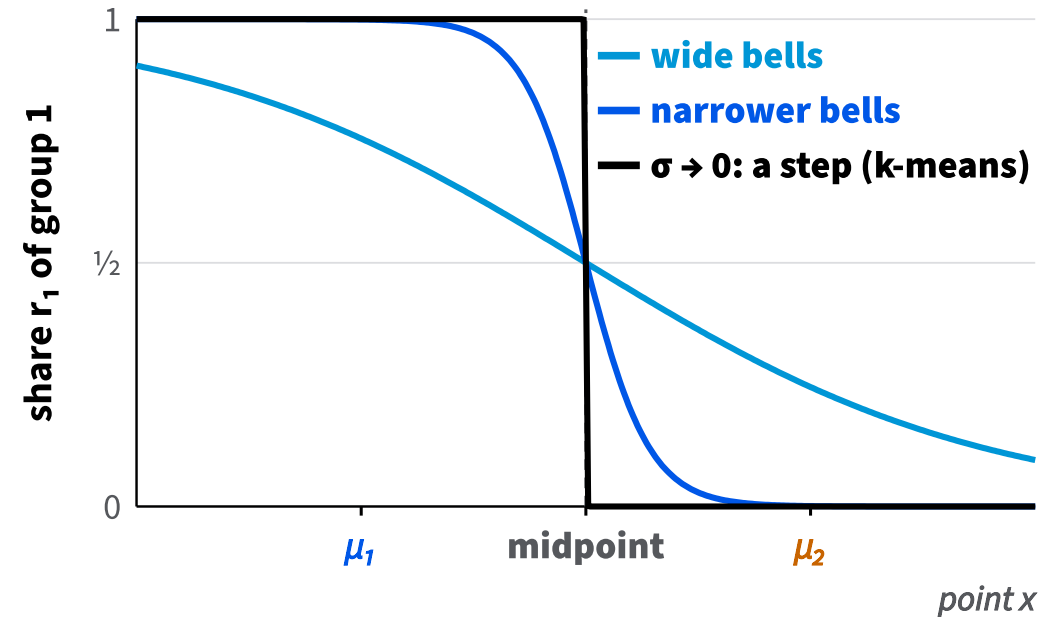
Each step lowers J or leaves it; only finitely many labellings exist, so k-means stops.

Theorem 5: k-means is EM with bells of width zero

Theorem 5 (k-means as a limit). Take a Gaussian mixture with equal weights π_k and one shared width σ , **held fixed** (not re-estimated), and let $\sigma \rightarrow 0$. Then EM's E-step becomes "assign each point to its nearest centre" and its M-step becomes "move each centre to the plain average of its points": EM becomes k-means.

In words: a share is a soft vote; shrinking every bell turns it into a hard vote for the nearest centre.

Why it matters: the most used clustering method is a special case of EM, so EM's two steps explain k-means, and EM is k-means that keeps its doubt.



nearer μ_1 : share $\rightarrow 1$ · nearer μ_2 : share $\rightarrow 0$
midpoint: $\frac{1}{2}$

Soft labels harden into hard ones; weighted averages become plain averages.

Proof: write the share as $1/(1 + e^{-u})$, then let σ shrink

↳ **Step 1: two groups.** Equal weights π_k and the factor $1/(\sqrt{2\pi} \cdot \sigma)$ cancel in the share. Distances $d_k = |x - \mu_k|$; $\Delta = d_2^2 - d_1^2$.

$$\begin{aligned} r_1 &= \frac{e^{-d_1^2/2\sigma^2}}{e^{-d_1^2/2\sigma^2} + e^{-d_2^2/2\sigma^2}} \\ &= \frac{1}{1 + e^{-d_2^2/2\sigma^2} / e^{-d_1^2/2\sigma^2}} \\ &= \frac{1}{1 + e^{-(d_2^2 - d_1^2)/2\sigma^2}} \\ &= \frac{1}{1 + e^{-u}}, \quad u = \frac{\Delta}{2 \cdot \sigma^2} \end{aligned}$$

E-step share

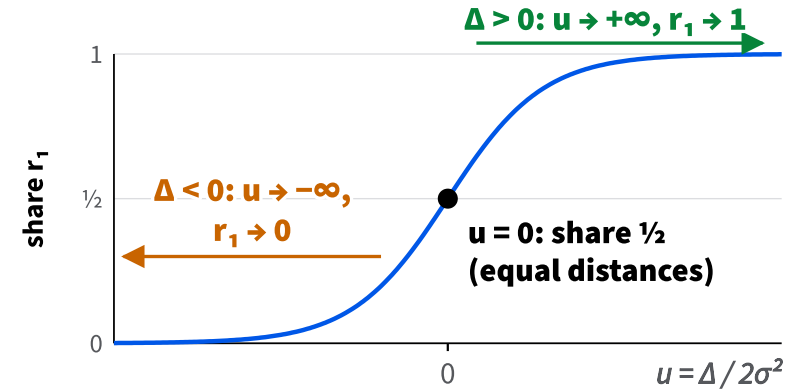
divide by the first height

$$e^a / e^b = e^{a-b}$$

$$\Delta = d_2^2 - d_1^2$$

$$\xrightarrow{\sigma \rightarrow 0} \begin{cases} 1/(1+0) = 1, & \Delta > 0 \\ 1/(1+\infty) = 0, & \Delta < 0 \\ \frac{1}{2} \text{ for every } \sigma, & \Delta = 0 \end{cases}$$

$$\begin{aligned} e^{-u} &\rightarrow 0 \text{ if } u \rightarrow +\infty; \\ e^{-u} &\rightarrow \infty \text{ if } u \rightarrow -\infty; \\ u &= 0 \text{ if } \Delta = 0 \end{aligned}$$



$\sigma \rightarrow 0$: $|u| = |\Delta| / 2\sigma^2$ grows without bound,
so the share is pushed to a hard 1 or 0

$\Delta > 0$ means $d_1 < d_2$: **centre 1 is nearer**, so **E-step** → **nearest centre** (and $r_1 \rightarrow 0$ when $\Delta < 0$).

In words: narrow bells make the distance gap Δ count $1/2\sigma^2$ times over, so the nearer centre takes the whole point.

Why it matters: only the distance gap Δ decides the share, so **the nearer centre wins outright** once the bells are narrow; the same division works for any number of groups K .

E-step half: the soft share hardens into a hard vote for the nearer centre.

Proof, step 2: the E-step share becomes the nearest centre

$$\begin{aligned}
 r_1 &= \frac{\pi_1 \cdot \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot e^{-d_1^2/2\sigma^2}}{\sum_{j=1}^K \pi_j \cdot \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot e^{-d_j^2/2\sigma^2}} \\
 &= \frac{e^{-d_1^2/2\sigma^2}}{\sum_{j=1}^K e^{-d_j^2/2\sigma^2}} \\
 &= \frac{e^{-d_1^2/2\sigma^2}}{e^{-d_1^2/2\sigma^2} + \sum_{j=2}^K e^{-d_j^2/2\sigma^2}} \\
 &= \frac{1}{1 + \sum_{j=2}^K e^{-d_j^2/2\sigma^2} / e^{-d_1^2/2\sigma^2}} \\
 &= \frac{1}{1 + \sum_{j=2}^K e^{-(d_j^2 - d_1^2)/2\sigma^2}} \\
 &\xrightarrow{\sigma \rightarrow 0} \frac{1}{1 + 0} = 1
 \end{aligned}$$

E-step: group 1's height
 $\div$ all heights (Bayes' rule)

equal weights $\pi_j = 1/K$ and
 the common $1/(\sqrt{2\pi} \cdot \sigma)$ cancel

take the $j = 1$ term
 out of the sum

divide top and bottom
 by the first height

$$e^a / e^b = e^{a-b}$$

each $d_j^2 - d_1^2 > 0$: each
 exponent $\rightarrow -\infty$, each term $\rightarrow 0$

↳ **Step 2: E-step $\rightarrow$ nearest centre, any K .** Fix $\sigma > 0$ first, then let $\sigma \rightarrow 0$. Fix one point $x = x_n$; $d_j = |x - \mu_j|$ is its distance to centre j . $r_1 = r_{n1}$ is the share of group 1 for this one point (not a point, and not the centre with the smallest μ). Number the centres so that centre 1 is the one nearest to x (any centre can be called 1); then $d_1 < d_j$ for every $j \neq 1$. The sum runs over the K centres, not over points; G_k (a set of points) appears only in step 3.

Divide by the nearest centre's height: every other term is a ratio that vanishes as $\sigma \rightarrow 0$.

Proof, step 2 (continued): every share becomes 1 or 0

↳ **Step 2 continued. Step 3: M-step with the 1-or-0 weights.** Assume no point is equidistant from two centres, and every G_k is non-empty ($|G_k| > 0$), where G_k is the set of points nearest centre k and $|G_k|$ = how many. Recall (p. 62) $r_1 = \frac{1}{1 + \sum_{j=2}^K e^{-(d_j^2 - d_1^2)/2\sigma^2}}$ for one point $x = x_n$.

Why for every j : centre 1 was chosen as the centre nearest to x , so $d_1 < d_j$ for each $j \geq 2$ (no ties assumed). Distances are ≥ 0 , so squaring keeps the order: $d_1^2 < d_j^2$, i.e. $d_j^2 - d_1^2 > 0$. **Example:** $x = 1$, centres at 1 and 4: $d_1 = 0$, $d_2 = 3$, $d_2^2 - d_1^2 = 9 > 0$, so $e^{-9/2\sigma^2} \rightarrow 0$.

So each exponent $-(d_j^2 - d_1^2)/2\sigma^2 \rightarrow -\infty$ and each of the $K - 1$ terms $\rightarrow 0$; a finite sum of terms that $\rightarrow 0$ also $\rightarrow 0$, so $r_1 \rightarrow 1$: the **Assign** step.

Consequence: $r_{n1} \rightarrow 1$ for $n \in G_1$; for a point nearer another centre, the same argument with that centre in place of 1 gives $r_{n1} \rightarrow 0$.

In words: narrow bells: the far bell's height falls much faster than the near one's, so their ratio $e^{-(d_j^2 - d_1^2)/2\sigma^2} \rightarrow 0$ and the nearest centre takes the whole point.

Every other centre is farther, so every other term vanishes: the share becomes 1 or 0.

Proof, step 3: weighted averages become plain averages

↳ Recall step 2 (p. 63): as $\sigma \rightarrow 0$, $r_{n1} \rightarrow 1$ for $n \in G_1$ and $r_{n1} \rightarrow 0$ otherwise.

M-step → plain average:

$$\begin{aligned}
 \mu_1 &= \frac{\sum_n r_{n1} \cdot x_n}{\sum_n r_{n1}} && \text{M-step} \\
 &\xrightarrow{\sigma \rightarrow 0} \frac{\sum_{n \in G_1} 1 \cdot x_n + \sum_{n \notin G_1} 0 \cdot x_n}{\sum_{n \in G_1} 1 + \sum_{n \notin G_1} 0} && \text{weights 1 or 0} \\
 &= \frac{\sum_{n \in G_1} x_n}{|G_1|} && \text{drop the zeros}
 \end{aligned}$$

The weights 1 or 0 come from the E-step; $|G_1| > 0$, so the division is safe. The same holds for every centre k : μ_k becomes the plain average of its set G_k , the **Move** step of k-means.

Both EM steps become the two k-means steps, for every K . ■

In words: with hard weights, a weighted average is the plain average of the points that carry weight 1.

Why it matters: k-means needs no new theory: it is EM's two steps in the limit of very narrow bells.

Both EM steps turn into the two k-means steps.

One EM round: share 0.989, then centres 1, 4 → 0.52, 4.48

↳ **Definition (p. 56)** applied to points 0, 1, 4, 5: centres 1 and 4, width $\sigma = 1$, weights $\pi_k = \frac{1}{2}$ (cancel); bell height $e^{-d^2/2}$, d = distance to the centre.

point x	bell 1 $e^{-(x-1)^2/2}$	bell 2 $e^{-(x-4)^2/2}$	share r of group 1	$r \cdot x$
0	$e^{-1/2}=0.607$	$e^{-8}=0.000335$	0.9994	0
1	$e^0=1$	$e^{-9/2}=0.0111$	0.9890	0.989
4	$e^{-9/2}=0.0111$	$e^0=1$	0.0110	0.044
5	$e^{-8}=0.000335$	$e^{-1/2}=0.607$	0.0006	0.003
total			2.000	1.036

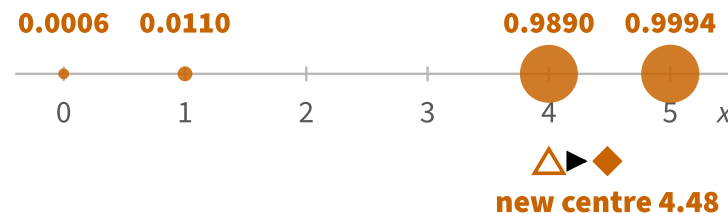
E-step, share = bell 1 ÷ (bell 1 + bell 2). Point 1: $r = \frac{1}{1 + 0.0111} = 0.989$, as HH gave coin A $\frac{9}{9+1} = 0.9$ (p. 28).

M-step, weighted average: $\mu_1 = \frac{1.036}{2.000} = 0.518 \approx 0.52$; the mirror $x \mapsto 5 - x$ swaps the groups, so $\mu_2 = 5 - 0.518 \approx 4.48$.

group 1: dot size = share r (number above)



group 2: dot size = share r (number above)



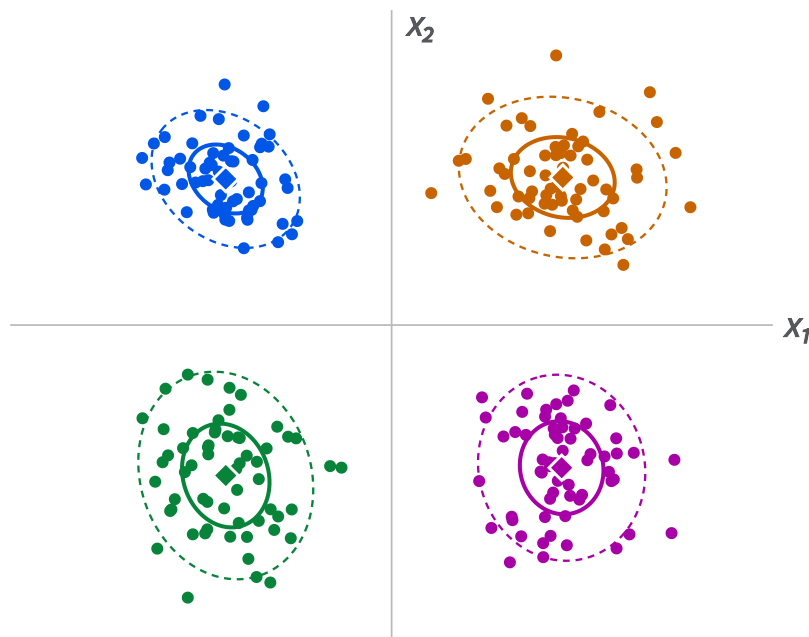
hollow: old centre

diamond: weighted average = new centre

Bayes' rule gives soft labels; weighted averages pull the centres to 0.52 and 4.48.

The same EM round finds four groups in 240 points

↳ Definition (p. 56) applied in the plane: $K = 4$ groups; each group's width (oval) is re-estimated too.



ovals: 1 and 2 widths from each hill's centre
diamonds: centres

Points (x_1, x_2) in the plane, four groups. Each bell is now a hill whose contours are ovals.

Step 1: each hill's height at each point.

Step 2 (E): colour each point by its shares; pale = unsure.

Step 3 (M): move each centre to its share-weighted average; refit each oval from the share-weighted spread of its points.

Average log-likelihood per point **risers every round**, from -4.309 to -2.901 at round 40 (Theorem 4 (p. 49): an EM round never lowers the likelihood).

round **40** • average log-likelihood per point **-2.901** • colour: the group with the largest share, paler when unsure

Recolour, move, repeat: clustering is EM.

k-means stops at J = 1; EM with narrow bells ends there too

↳ **Descent result (p. 59) applied:** k-means on 0, 1, 4, 5 from a **poor guess**, centres 0 and 1; J = sum of (point – its centre)². **Theorem 5 (p. 60) applied:** EM from the working guess, centres 1 and 4, with the bells narrowed.

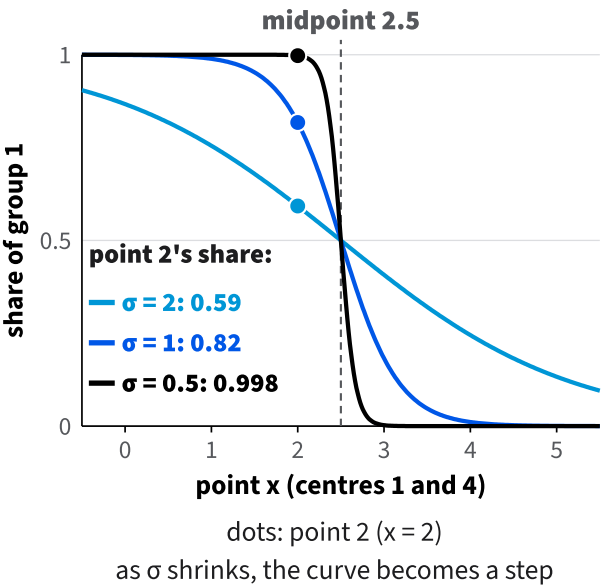
step	centres	groups	J
assign	0, 1	{0} · {1, 4, 5}	$0 + 0 + 9 + 16 = 25$
move	$0, \frac{1+4+5}{3} = \frac{10}{3}$	{0} · {1, 4, 5}	$(\frac{7}{3})^2 + (\frac{2}{3})^2 + (\frac{5}{3})^2 = \frac{49 + 4 + 25}{9} = 8.67$
assign	$0, \frac{10}{3}$	{0, 1} · {4, 5}	$1^2 + (\frac{2}{3})^2 + (\frac{5}{3})^2 = 4.22$
move	$\frac{0+1}{2}, \frac{4+5}{2} = 0.5, 4.5$	{0, 1} · {4, 5}	$4 \cdot 0.5^2 = 1$

Distances of 1, 4, 5 to the centre $\frac{10}{3}$: $\frac{7}{3}, \frac{2}{3}, \frac{5}{3}$.

A new point at $x = 2$ (distances 1 and 2 to centres 1 and 4) gets the share $\frac{e^{-1/2\sigma^2}}{e^{-1/2\sigma^2} + e^{-2/\sigma^2}}$, which **tends to 1** as $\sigma \rightarrow 0$ (the figure's dots: 0.59, 0.82, 0.998 at $\sigma = 2, 1, 0.5$).

With every share 1 or 0, EM's M-step gives $\frac{0+1}{2} = 0.5$ and $\frac{4+5}{2} = 4.5$ (groups {0, 1} and {4, 5}): **k-means' centres.**

All steps: Appendix, Section 5 (p. 123).

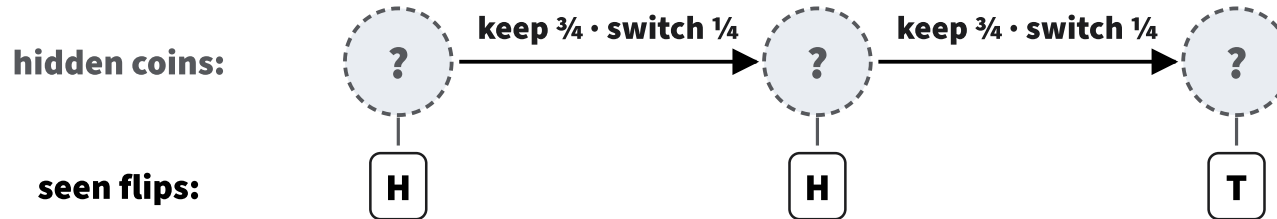


One family: EM is k-means that admits doubt; k-means is EM that refuses it.

Next: the coin may switch between flips



So far one hidden group per record. Now the friend may switch coins between flips: **one hidden coin per flip**, kept with chance $\frac{3}{4}$, switched with $\frac{1}{4}$.



Three flips, each with its own hidden coin. How many coin sequences are there?

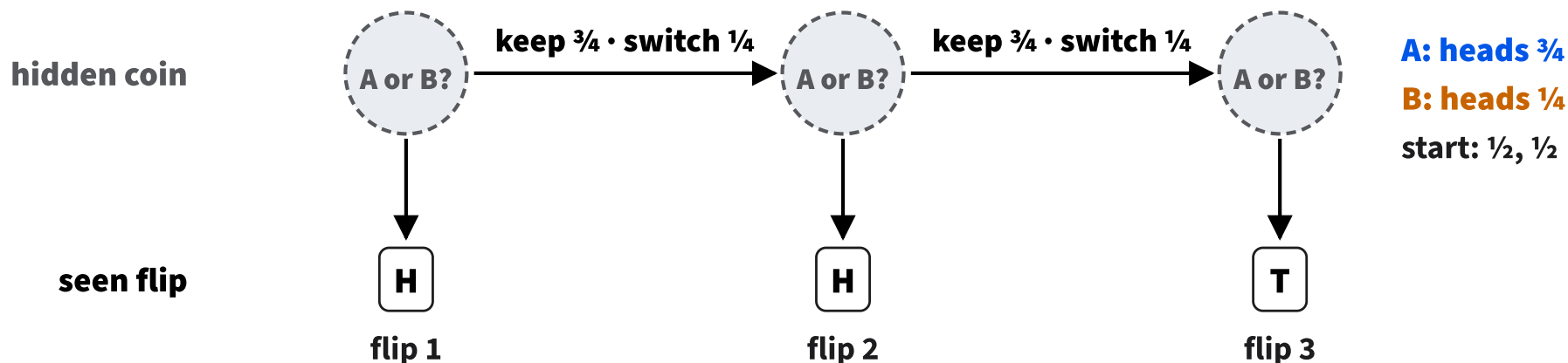
Answer. $8 = 2 \cdot 2 \cdot 2$. For 100 flips, 2^{100} : too many to list.

Hidden chains: the sum over sequences, done step by step.

If the coin can switch between flips, the coins form a hidden chain

Goal: the posterior of each flip's coin, and $p(\text{flips})$, without listing every sequence.

↳ In the two-coin game one hidden coin made both flips of a record. Now the friend may switch coins between flips (keep chance $\kappa = \frac{3}{4}$: keep $\frac{3}{4}$, switch $\frac{1}{4}$), so **every flip has its own hidden coin**.



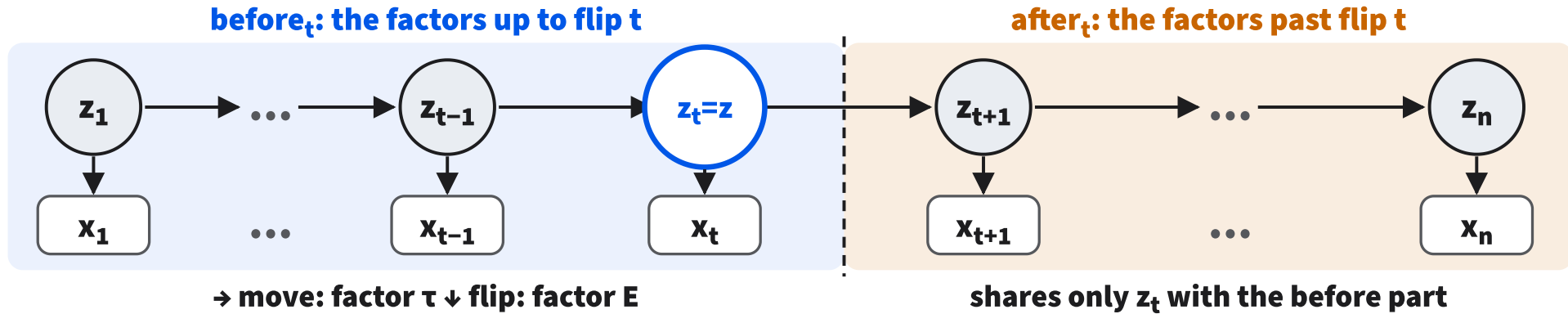
A **hidden Markov model**: each coin depends only on the coin before (Lecture 2's chain), and each flip only on its own coin. Early speech recognizers were built on it.

A hidden chain links the coins flip to flip; on H, H, T we will find flip 2's coin and $p(\text{H}, \text{H}, \text{T})$.

A hidden Markov model multiplies one factor per arrow

↳ Symbols: $z = (z_1, \dots, z_n)$ = the hidden coins, $x = (x_1, \dots, x_n)$ = the flips; $\pi(z) = \text{chance the first coin is } z$; $\tau(z' \rightarrow z) = \text{chance the coin moves from } z' \text{ to } z$: κ if $z = z'$ (keep), $1 - \kappa$ otherwise (switch), with keep chance $\kappa = \frac{3}{4}$ here; $E(z, x) = \text{chance coin } z \text{ shows } x$.

$$p(x, z) = \underbrace{\pi(z_1) \cdot E(z_1, x_1)}_{\text{start, flip 1}} \cdot \prod_{s=2}^n \underbrace{\tau(z_{s-1} \rightarrow z_s)}_{\text{move}} \cdot \underbrace{E(z_s, x_s)}_{\text{flip } s}$$



Cut at flip t : $p(x, z) = \text{before}_t \cdot \text{after}_t$, where

$$\text{before}_t = \pi(z_1) \cdot E(z_1, x_1) \cdot \prod_{s=2}^t \tau(z_{s-1} \rightarrow z_s) \cdot E(z_s, x_s), \quad \text{after}_t = \prod_{s=t+1}^n \tau(z_{s-1} \rightarrow z_s) \cdot E(z_s, x_s)$$

One factor per arrow; the part past flip t touches the earlier coins only through z_t .

Forward-backward works on any chain of n flips

↳ Recall (p. 70): $\pi(z)$ first coin, $\tau(z' \rightarrow z)$ move, $E(z, x)$ flip chance (one factor per arrow); flips $x = (x_1, \dots, x_n)$; z_t = hidden coin at flip t .

$\alpha_t(z)$: **forward**

chance of flips 1 to t **and** coin z at flip t

$$\alpha_1(z) = \pi(z) \cdot E(z, x_1), \quad \alpha_t(z) = \left(\sum_{z'} \alpha_{t-1}(z') \cdot \tau(z' \rightarrow z) \right) \cdot E(z, x_t)$$

$$\beta_n(z) = 1, \quad \beta_t(z) = \sum_{z'} \tau(z \rightarrow z') \cdot E(z', x_{t+1}) \cdot \beta_{t+1}(z')$$

$$p(x) = \sum_z \alpha_t(z) \cdot \beta_t(z), \quad p(z_t = z \mid x) = \frac{\alpha_t(z) \cdot \beta_t(z)}{p(x)}$$

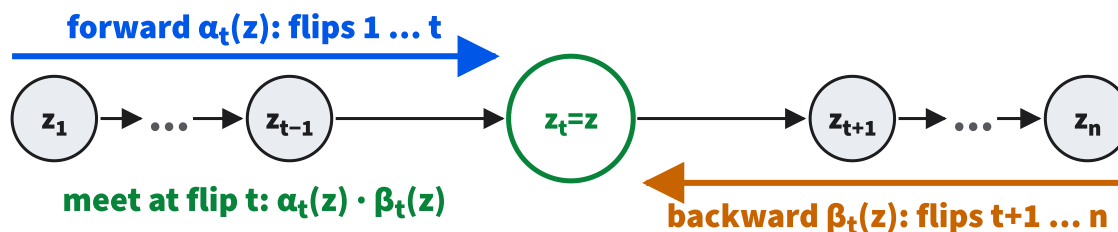
$\beta_t(z)$: **backward**

chance of flips $t+1$ to n , **given** coin z at flip t

Forward, left to right: add the ways in, then the flip.

Backward, right to left: after flip n nothing remains.

Combine at any flip t : add for the total, share for the posterior.



Each flip, each of K coins adds K ways in: about $n \cdot K^2$ steps, not K^n .

Proof, part 1: the cut gives $p(x, z_t=z) = \alpha_t(z) \cdot \beta_t(z)$

↳ Recall (p. 70): $p(x, z) = \text{before}_t \cdot \text{after}_t$ (factors up to flip t , factors past flip t); $z_{1:t-1} = (z_1, \dots, z_{t-1})$, $x_{1:t} = (x_1, \dots, x_t)$.

$$p(x, z_t=z) = \sum_{z_{1:t-1}} \sum_{z_{t+1:n}} p(x, z_{1:t-1}, z, z_{t+1:n})$$

(add over all the other coins)

$$= \sum_{z_{1:t-1}} \sum_{z_{t+1:n}} \text{before}_t(z_{1:t-1}, z) \cdot \text{after}_t(z, z_{t+1:n})$$

(one factor per arrow, split at flip t)

$$= \left(\sum_{z_{1:t-1}} \text{before}_t(z_{1:t-1}, z) \right) \cdot \left(\sum_{z_{t+1:n}} \text{after}_t(z, z_{t+1:n}) \right)$$

(before _{t} has no $z_{t+1:n}$, after _{t} no $z_{1:t-1}$:
distributive law)

$$= \left(\sum_{z_{1:t-1}} p(z_1) \cdot p(x_1 | z_1) \cdot \prod_{s=2}^t p(z_s | z_{s-1}) \cdot p(x_s | z_s) \right)$$

(write out before _{t} and after _{t} ;

$$\cdot \left(\sum_{z_{t+1:n}} \prod_{s=t+1}^n p(z_s | z_{s-1}) \cdot p(x_s | z_s) \right)$$

π, τ, E are these conditionals)

$$= \left(\sum_{z_{1:t-1}} p(x_{1:t}, z_{1:t-1}, z_t=z) \right) \cdot \left(\sum_{z_{t+1:n}} p(x_{t+1:n}, z_{t+1:n} | z_t=z) \right)$$

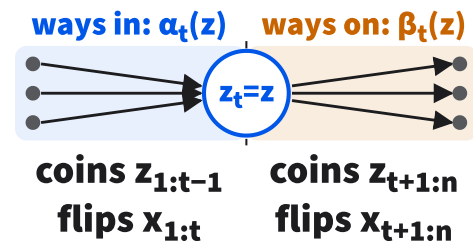
(left: chain rule. Right: chain rule + Markov:
given z_t , later coins and flips ignore $z_{1:t-1}, x_{1:t}$,
so before and after are independent given z_t)

$$= p(x_{1:t}, z_t=z) \cdot p(x_{t+1:n} | z_t=z) = \alpha_t(z) \cdot \beta_t(z)$$

(add out summed coins; definitions of α_t, β_t)

In words: a sequence through z at flip t = a way in • a way on; the ways on ignore how we arrived, so add each side, then multiply once.

Why it matters: adding over z gives $p(x) = \sum_z \alpha_t(z) \cdot \beta_t(z)$; dividing by $p(x)$ (Bayes' rule) gives $p(z_t=z | x)$.



Two short messages per coin replace the sum over all K^n sequences.

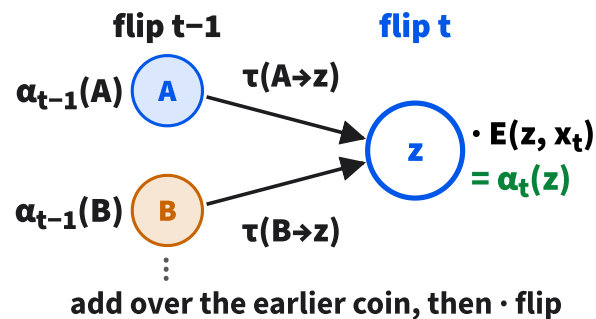
Proof, part 2: forward adds the ways in, then the flip

↳ Markov: coin t depends only on coin $t-1$, flip t only on coin t . Start: $\alpha_1(z) = p(x_1, z_1=z) = \pi(z) \cdot E(z, x_1)$.

$$\begin{aligned}
 \alpha_t(z) &= p(x_{1:t}, z_t=z) && \text{(definition)} \\
 &= \sum_{z'} p(x_{1:t-1}, z_{t-1}=z', z_t=z, x_t) && \text{(add over the coin before, } z_{t-1}=z') \\
 &= \sum_{z'} p(x_{1:t-1}, z_{t-1}=z') \cdot p(z_t=z \mid z_{t-1}=z', x_{1:t-1}) \cdot p(x_t \mid z_t=z, z_{t-1}=z', x_{1:t-1}) && \text{(chain rule)} \\
 &= \sum_{z'} p(x_{1:t-1}, z_{t-1}=z') \cdot p(z_t=z \mid z_{t-1}=z') \cdot p(x_t \mid z_t=z) && \text{(Markov: coin } t \text{ needs only coin } t-1; \text{ flip } t \text{ needs only coin } t) \\
 &= \sum_{z'} \alpha_{t-1}(z') \cdot \tau(z' \rightarrow z) \cdot E(z, x_t) && \text{(definitions of } \alpha_{t-1}, \tau, E) \\
 &= \left(\sum_{z'} \alpha_{t-1}(z') \cdot \tau(z' \rightarrow z) \right) \cdot E(z, x_t) && (E(z, x_t) \text{ does not depend on } z')
 \end{aligned}$$

In words: to stand on coin z at flip t , come from any coin z' , move, then show flip t .

Why it matters: each step reuses the K numbers α_{t-1} : K^2 products per flip, never the list of earlier sequences.



Forward: (old α $\cdot$ move, added over the old coin) $\cdot$ flip chance.

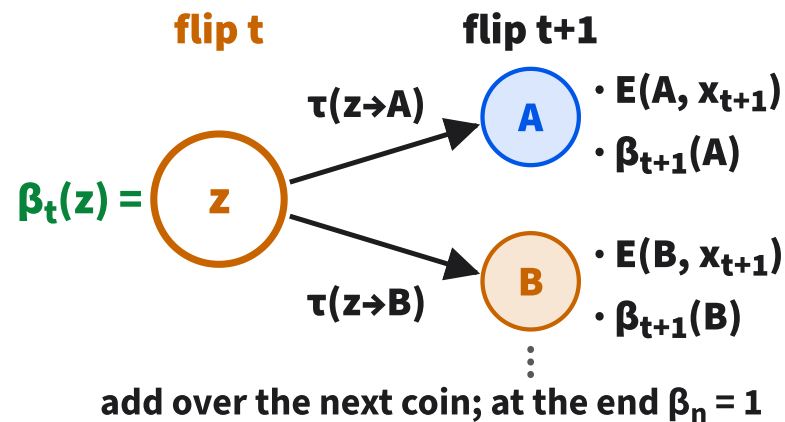
Proof, part 3: the backward message adds the ways on

↳ Markov: given the next coin z_{t+1} , the later flips forget z_t . Start: after flip n no flips remain, so $\beta_n(z) = 1$.

$$\begin{aligned}
 \beta_t(z) &= p(x_{t+1:n} \mid z_t = z) && \text{(definition)} \\
 &= \sum_{z'} p(z_{t+1} = z', x_{t+1}, x_{t+2:n} \mid z_t = z) && \text{(add over the next coin, } z_{t+1} = z') \\
 &= \sum_{z'} p(z_{t+1} = z' \mid z_t = z) \cdot p(x_{t+1} \mid z_{t+1} = z', z_t = z) \cdot p(x_{t+2:n} \mid z_{t+1} = z', x_{t+1}, z_t = z) && \text{(chain rule)} \\
 &= \sum_{z'} p(z_{t+1} = z' \mid z_t = z) \cdot p(x_{t+1} \mid z_{t+1} = z') \cdot p(x_{t+2:n} \mid z_{t+1} = z') && \text{(Markov: given } z_{t+1}, \text{ drop } z_t \text{ and } x_{t+1}) \\
 &= \sum_{z'} \tau(z \rightarrow z') \cdot E(z', x_{t+1}) \cdot \beta_{t+1}(z') && \text{(definitions of } \tau, E, \beta_{t+1})
 \end{aligned}$$

In words: from coin z at flip t , move to any coin z' , show flip $t+1$, then carry on from z' .

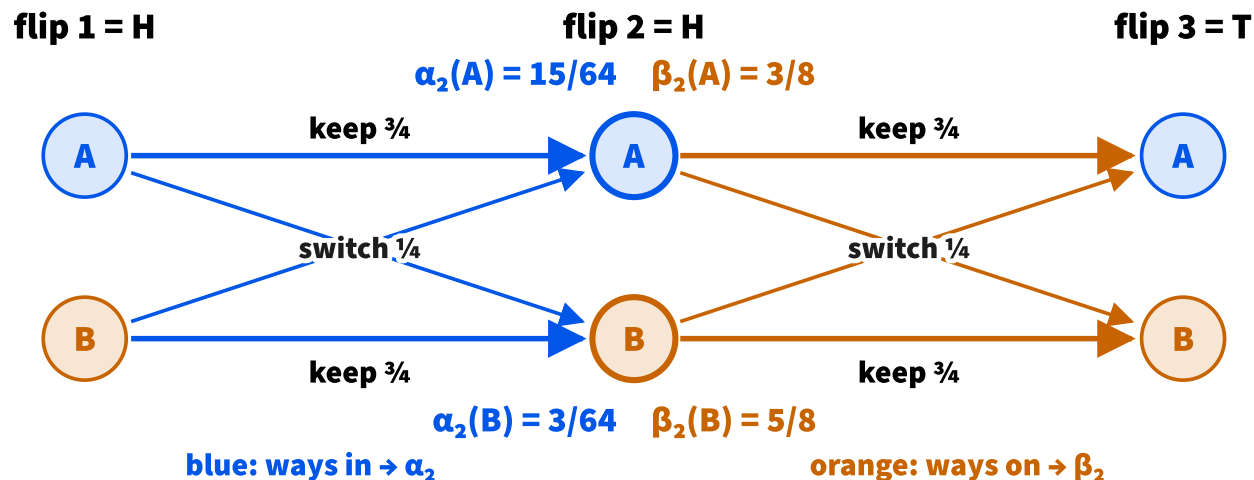
Why it matters: together with parts 1 and 2 this proves every line of the forward-backward theorem.



Backward: (move · next flip · next β), added over the next coin.

On H, H, T, forward · backward at flip 2 gives $p(\text{HHT}) = 15/128$

↳ Forward-backward theorem (p. 71) applied to the coins (first coin $\frac{1}{2}$; keep $\frac{3}{4}$, switch $\frac{1}{4}$; A shows H with $\frac{3}{4}$, B with $\frac{1}{4}$), cut at flip 2. The forward pass gives α_2 , the backward pass β_2 :



$$p(\text{HHT}) = \sum_z \alpha_2(z) \cdot \beta_2(z) = \underbrace{\frac{15}{64} \cdot \frac{3}{8}}_{\text{A at flip 2}} + \underbrace{\frac{3}{64} \cdot \frac{5}{8}}_{\text{B at flip 2}} = \frac{45}{512} + \frac{15}{512} = \frac{15}{128}$$

Share (Bayes' rule): coin A made flip 2 with chance $\frac{45}{45+15} = \frac{45}{60} = \frac{3}{4}$.

All steps: Appendix, Section 6 (p. 124).

Same as adding all $2^3 = 8$ sequences, but 2 numbers per coin per flip: Lecture 2's sum-product.

To learn the switching game, run the same EM as for two coins

↳ So far the rules were given. Now we see only the flips and must **learn** them: the heads chances θ_A, θ_B and the keep chance κ ; the start chance stays $\frac{1}{2}$.

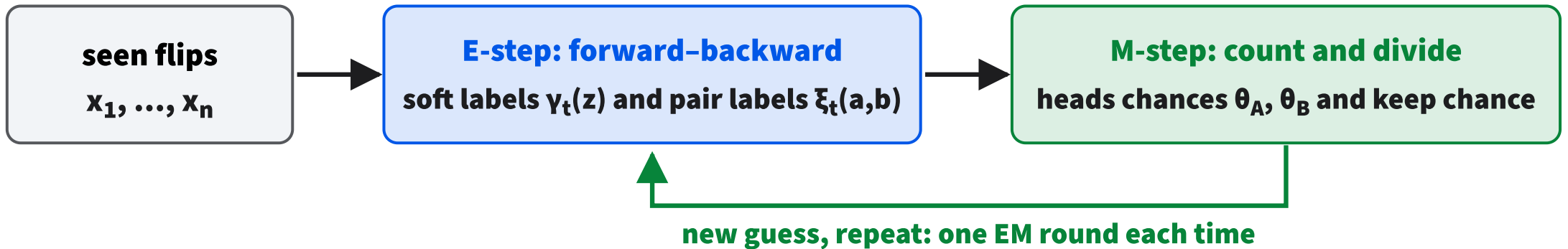
Why learn κ too? Nobody tells us how often the friend switches, and κ shapes the soft labels: with κ near 1 the coin rarely switches, so neighbouring flips share their coin; with $\kappa = \frac{1}{2}$ each flip is judged on its own. A wrong κ gives wrong soft labels, so EM must learn it along with θ_A, θ_B .

	two-coin game	switching game (hidden chain)
hidden	one coin per record	one coin per flip
E-step	$p(\text{coin} \mid \text{record})$ by Bayes (HH: 0.9) (p. 28)	$p(\text{coin at flip } t \mid \text{all flips})$ by forward-backward
extra soft label	none	$p(\text{coins at flips } t, t+1 \mid \text{all flips})$
M-step	heads to A ÷ flips to A	the same; keep chance = expected keeps ÷ moves

Same EM as for two coins, plus one new rule to learn: the keep chance κ .

Two soft labels: one flip's coin, and a neighbouring pair

Soft labels. $\gamma_t(z)$ = chance that coin z made flip t , given all the flips (one coin's soft label, like HH's 0.9 (p. 28)); $\xi_t(a, b)$ = chance that the coin is a at flip t and b at flip $t+1$, given all the flips (needed to count keeps: $\text{kept} = \xi_t(A, A) + \xi_t(B, B)$). On H, H, T: $\gamma_1(A) = 0.8$, $\gamma_2(A) = 0.75$, $\gamma_3(A) = 0.4$; kept between flips 1 and 2 with chance 0.8.



This EM for hidden chains is Baum–Welch (Baum et al., 1970), older than the name "EM".

Baum–Welch = forward–backward soft labels + counting

↳ Recall (p. 71): $\alpha_t(z)$ = chance of flips 1.. t and coin z at flip t ; $\beta_t(z)$ = chance of flips $t+1$.. n , given coin z at t ; $\tau(a \rightarrow b)$ = chance the coin moves from a to b ; $E(b, x)$ = chance coin b shows x ; θ_A = heads chance of coin A.

Theorem 6 (Baum–Welch; Baum, Petrie, Soules & Weiss, 1970). For a hidden Markov model with flips $x_1, \dots, x_n$, the rules to learn are the heads chances θ_A, θ_B and the **keep chance** κ (kappa): between two flips the friend keeps the same coin with chance κ and switches with chance $1 - \kappa$, i.e. $\tau(a \rightarrow a) = \kappa, \tau(a \rightarrow b) = 1 - \kappa$ ($a \neq b$); in our game $\kappa = \frac{3}{4}$ (keep $\frac{3}{4}$, switch $\frac{1}{4}$). One EM round is the E-step below and the M-step on the next slide, and it never lowers $p(x)$. Here the start chance stays $\frac{1}{2}$; with a separate chance per move, $\tau(a \rightarrow b) = \sum_{t < n} \xi_t(a, b) / \sum_{t < n} \gamma_t(a)$.

$$\gamma_t(z) = \frac{\alpha_t(z) \cdot \beta_t(z)}{p(x)}$$

$$\xi_t(a, b) = \frac{\alpha_t(a) \cdot \tau(a \rightarrow b) \cdot E(b, x_{t+1}) \cdot \beta_{t+1}(b)}{p(x)}$$

E-step, one coin: chance coin z made flip t , given all flips.

E-step, pairs: chance of coin a at t and b at $t+1$, given all flips.

E-step: forward–backward gives both soft labels, γ for one coin and ξ for a pair.

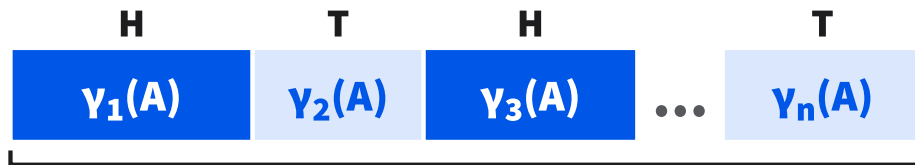
Baum–Welch, M-step: count with the soft labels

↳ Theorem 6 continued (p. 78): with the soft labels γ_t and ξ_t from the E-step, the M-step counts and divides.

$$\theta_A = \frac{\sum_{t=1}^n \gamma_t(A) \cdot [x_t=H]}{\sum_{t=1}^n \gamma_t(A)}, \quad \kappa = \frac{\sum_{t=1}^{n-1} (\xi_t(A, A) + \xi_t(B, B))}{n-1}$$

M-step: count and divide; $[x_t=H] = 1$ for heads, 0 for tails. κ : expected keeps $\div$ the $n-1$ moves ($\xi_t(A, A) + \xi_t(B, B)$ = chance the coin was kept between flips t and $t+1$).

Proof plan. The expected complete-data log-likelihood (coins and flips together) splits into $\gamma \cdot \ln(\text{heads chance})$ and $\xi \cdot \ln(\text{keep or switch chance})$ terms; counting maximizes each (Theorem 1) (p. 11); Theorem 4 (EM increases the likelihood) (p. 49): $p(x)$ never falls.



all blocks: $u + v = \sum_t \gamma_t(A)$, soft flips

solid: $u = \sum_t \gamma_t(A)[x_t=H]$, soft heads

$\theta_A = \text{solid length} \div \text{total length}$

	A at t+1	B at t+1	total
A at t	$\xi_t(A, A)$	$\xi_t(A, B)$	$\gamma_t(A)$
B at t	$\xi_t(B, A)$	$\xi_t(B, B)$	$\gamma_t(B)$
total	$\gamma_{t+1}(A)$	$\gamma_{t+1}(B)$	1

green = kept; the four cells add to 1

E-step: forward–backward soft labels. M-step: count with them (θ_B : the same with B).

Proof, E-step: cutting at a move gives the pair soft label

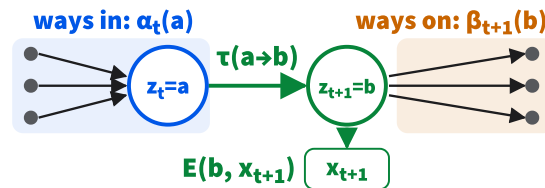
↳ **Why pairs:** keeping links two neighbouring coins, so we need each pair $(z_t, z_{t+1}) = (a, b)$; recall $p(x, z) = \text{one factor per arrow}$ (p. 70).

$$\begin{aligned}
 \xi_t(a, b) &= p(z_t=a, z_{t+1}=b \mid x) = \frac{p(x, z_t=a, z_{t+1}=b)}{p(x)} && \text{(definition; Bayes' rule)} \\
 &= \frac{1}{p(x)} \cdot \sum_{z_{1:t-1}} \sum_{z_{t+2:n}} \text{before}_t(z_{1:t-1}, a) \cdot \tau(a \rightarrow b) \cdot E(b, x_{t+1}) \cdot \text{after}_{t+1}(b, z_{t+2:n}) && \text{(add over other coins; split at the move)} \\
 &= \frac{1}{p(x)} \cdot \left(\sum_{z_{1:t-1}} \text{before}_t(z_{1:t-1}, a) \right) \cdot \tau(a \rightarrow b) \cdot E(b, x_{t+1}) \cdot \left(\sum_{z_{t+2:n}} \text{after}_{t+1}(b, z_{t+2:n}) \right) && \text{(distributive law; sums share no coins)} \\
 &= \frac{1}{p(x)} \cdot \underbrace{p(x_{1:t}, z_t=a)}_{\alpha_t(a)} \cdot \tau(a \rightarrow b) \cdot E(b, x_{t+1}) \cdot \underbrace{p(x_{t+2:n} \mid z_{t+1}=b)}_{\beta_{t+1}(b)} && \text{(brackets are marginals: } \alpha_t(a), \beta_{t+1}(b))
 \end{aligned}$$

In words: a pair (a at flip t , b at flip $t+1$) is weighted by all the ways to arrive at a , times moving $a \rightarrow b$ and showing flip $t+1$, times all the ways to go on from b ; divide by $p(x)$ to get a chance.

Check: if we do not care which coin comes next (add over b), we get back the one-flip label $\gamma_t(a)$ (since $\sum_b \tau(a \rightarrow b) \cdot E(b, x_{t+1}) \cdot \beta_{t+1}(b) = \beta_t(a)$).

Why it matters: forward-backward, run once, gives both labels γ and ξ .



Pair label: arrive at a , move to b and show the next flip, go on from b ; divide by $p(x)$.

Proof, M-step: start, move and flip terms separate

↳ EM's M-step maximizes the **expected complete-data log-likelihood** $Q(\theta)$; the posterior $q(z) = p_{\theta'}(z | x)$ from the current guess θ' is held fixed. $[\cdot] = 1$ if true, 0 if false.

$$\begin{aligned}
 Q(\theta) &= \sum_z p_{\theta'}(z | x) \cdot \ln p_{\theta}(x, z) = \sum_z q(z) \cdot \ln p_{\theta}(x, z) && \text{(the M-step objective)} \\
 &= \sum_z q(z) \cdot \left[\ln \pi(z_1) + \sum_{t=1}^{n-1} \ln \tau(z_t \rightarrow z_{t+1}) + \sum_{t=1}^n \ln E(z_t, x_t) \right] && \text{(log of a product = sum of logs)} \\
 &= \sum_z q(z) \cdot \ln \pi(z_1) + \sum_{t=1}^{n-1} \sum_z q(z) \cdot \ln \tau(z_t \rightarrow z_{t+1}) + \sum_{t=1}^n \sum_z q(z) \cdot \ln E(z_t, x_t) && \text{(swap the sums)} \\
 &= \sum_a \ln \pi(a) \cdot \sum_z q(z) \cdot [z_1=a] + \sum_{t=1}^{n-1} \sum_{a,b} \ln \tau(a \rightarrow b) \cdot \sum_z q(z) \cdot [z_t=a, z_{t+1}=b] && \text{(sort the sequences by the} \\
 &\quad + \sum_{t=1}^n \sum_c \ln E(c, x_t) \cdot \sum_z q(z) \cdot [z_t=c] && \text{coins each log factor uses)} \\
 &= \underbrace{\sum_a \gamma_1(a) \cdot \ln \pi(a)}_{\text{start}} + \underbrace{\sum_{t=1}^{n-1} \sum_{a,b} \xi_t(a, b) \cdot \ln \tau(a \rightarrow b)}_{\text{moves}} + \underbrace{\sum_{t=1}^n \sum_z \gamma_t(z) \cdot \ln E(z, x_t)}_{\text{flips}} && \begin{aligned} &(\sum_z q(z) \cdot [z_t=c] \\ &= p_{\theta'}(z_t=c | x) = \gamma_t(c), \\ &\sum_z q(z) \cdot [z_t=a, z_{t+1}=b] = \xi_t(a, b)) \end{aligned}
 \end{aligned}$$

In words: each log factor touches one or two coins, so it needs only γ or ξ , never all K^n sequences.

Why it matters: the three terms share no parameter, so the M-step maximizes each on its own. **Start term:** π is fixed at $\frac{1}{2}$ here, so it needs no update.

Each log factor becomes one term, weighted by the soft label of the coins it touches.

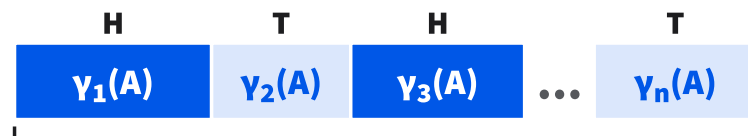
Proof, M-step: the flip terms give heads ÷ flips

↳ Recall Theorem 1 (counting) (p. 11): $u \cdot \ln \theta + v \cdot \ln(1 - \theta)$ is largest at $\theta = \frac{u}{u+v}$, heads ÷ flips. Coin A: $E(A, H) = \theta_A$, $E(A, T) = 1 - \theta_A$.

$$\begin{aligned}
 \text{flip terms of A} &= \sum_{t=1}^n \gamma_t(A) \cdot \ln E(A, x_t) && (\text{the } z=A \text{ flip terms; } z=B \text{ terms hold only } \theta_B) \\
 &= \underbrace{\left(\sum_t \gamma_t(A) \cdot [x_t=H] \right)}_{u : \text{ soft heads}} \cdot \ln \theta_A + \underbrace{\left(\sum_t \gamma_t(A) \cdot [x_t=T] \right)}_{v : \text{ soft tails}} \cdot \ln(1 - \theta_A) && (\text{H uses } \theta_A, \text{ T uses } 1 - \theta_A) \\
 \Rightarrow \theta_A &= \frac{u}{u+v} = \frac{\sum_t \gamma_t(A) \cdot [x_t=H]}{\sum_t \gamma_t(A) \cdot [x_t=H] + \sum_t \gamma_t(A) \cdot [x_t=T]} && (\text{Theorem 1}) \\
 &= \frac{\sum_{t=1}^n \gamma_t(A) \cdot [x_t=H]}{\sum_{t=1}^n \gamma_t(A)} && (\text{each flip is H or T})
 \end{aligned}$$

In words: credit each flip to coin A by its soft label, then divide heads by flips.

Why it matters: this is Theorem 6's θ_A (p. 79) (θ_B : the same with B).



all blocks: $u + v = \sum_t \gamma_t(A)$, soft flips

solid: $u = \sum_t \gamma_t(A)[x_t=H]$, soft heads

$\theta_A = \text{solid length} \div \text{total length}$

Soft counts in, Theorem 1 out: heads ÷ flips.

Proof, M-step: the move terms give keeps ÷ moves

↳ One keep chance κ for both coins: $\tau(a \rightarrow a) = \kappa, \tau(a \rightarrow b) = 1 - \kappa$ for $a \neq b$. Theorem 1 again (p. 11): $u \cdot \ln \kappa + v \cdot \ln(1 - \kappa)$ peaks at $\kappa = \frac{u}{u+v}$.

$$\begin{aligned}
 \text{move terms} &= \sum_{t=1}^{n-1} \sum_{a,b} \xi_t(a, b) \cdot \ln \tau(a \rightarrow b) && \text{(the moves part)} \\
 &= \underbrace{\sum_{t < n} (\xi_t(A, A) + \xi_t(B, B)) \cdot \ln \kappa}_{u: \text{ expected keeps}} + \underbrace{\sum_{t < n} (\xi_t(A, B) + \xi_t(B, A)) \cdot \ln(1 - \kappa)}_{v: \text{ expected switches}} && \text{(kept uses } \kappa, \text{ switched } 1 - \kappa) \\
 \Rightarrow \kappa &= \frac{u}{u+v} = \frac{\sum_{t < n} (\xi_t(A, A) + \xi_t(B, B))}{\sum_{t < n} \sum_{a,b} \xi_t(a, b)} && \text{(Theorem 1)} \\
 &= \frac{\sum_{t=1}^{n-1} (\xi_t(A, A) + \xi_t(B, B))}{n-1} && \text{(each move's four } \xi \text{ add to 1)}
 \end{aligned}$$

In words: new keep chance = expected keeps ÷ number of moves.

If each coin had its own move chances, the same counting gives

$\tau(a \rightarrow b) = \sum_{t < n} \xi_t(a, b) / \sum_{t < n} \gamma_t(a)$: expected moves $a \rightarrow b$ ÷ expected visits to a .

Why it matters: this is Theorem 6's M-step for κ (p. 79); by Theorem 4 (p. 49), $p(x)$ never falls. ■

	A at t+1	B at t+1	total
A at t	$\xi_t(A, A)$	$\xi_t(A, B)$	$\gamma_t(A)$
B at t	$\xi_t(B, A)$	$\xi_t(B, B)$	$\gamma_t(B)$
total	$\gamma_{t+1}(A)$	$\gamma_{t+1}(B)$	1

green = kept; the four cells add to 1

Soft counts in, Theorem 1 out: keeps ÷ moves.

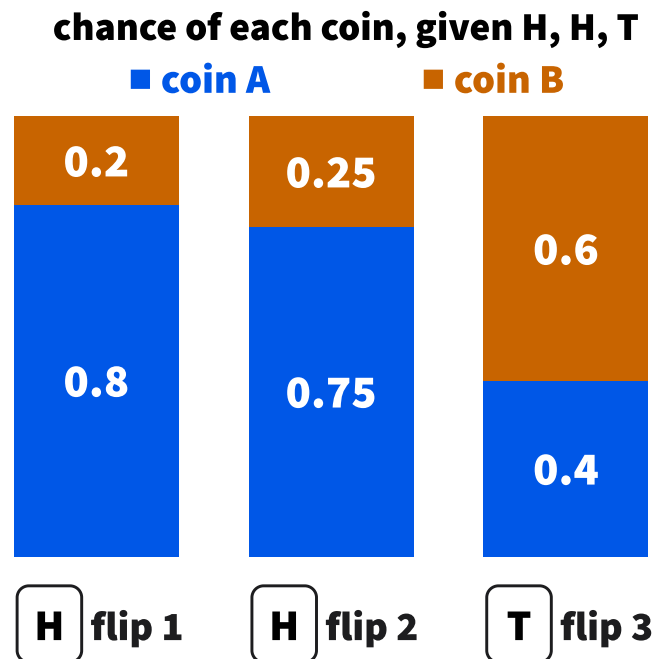
E-step: coin A's soft labels on H, H, T are 0.8, 0.75, 0.4

↳ Theorem 6's E-step (p. 78) applied (guess $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$, keep $\frac{3}{4}$): $\gamma_t(A) = \alpha_t(A) \cdot \beta_t(A) / p(\text{HHT})$, the chance coin A made flip t , given all flips. Forward-backward gives the products; recall $p(\text{HHT}) = \frac{15}{128} = \frac{240}{2048}$ (p. 75).

flip	1 = H	2 = H	3 = T
$\alpha_t(A) \cdot \beta_t(A) \cdot 2048$	192	180	96
$\gamma_t(A) = \text{that} \div 240$	0.8	0.75	0.4
$\gamma_t(B) = 1 - \gamma_t(A)$	0.2	0.25	0.6

Flip 3 is a tail, so coin B (tails $\frac{3}{4}$) takes over there.

All steps: Appendix, Section 6 (p. 124).



Soft labels for coin A at flips 1, 2, 3: 0.8, 0.75, 0.4. The M-step counts with them.

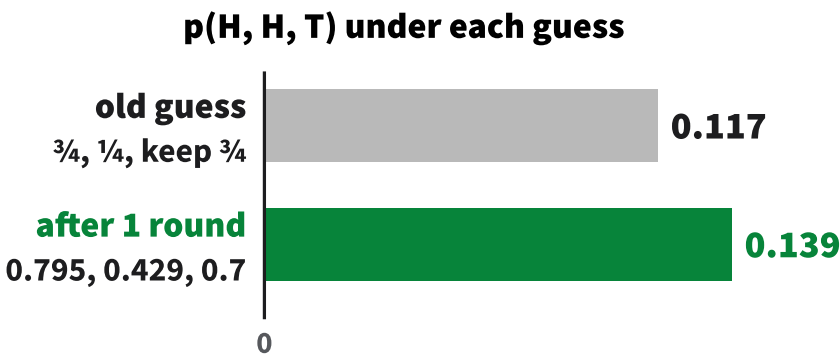
M-step: counting with soft labels raises $p(\text{HHT})$ to 0.139

↳ Theorem 6's M-step (p. 79) applied: recall coin A's soft labels 0.8, 0.75, 0.4 (p. 84) (B gets the rest), and the expected keeps 0.8 between flips 1 and 2 and 0.6 between flips 2 and 3 (from the pair labels ξ_t).

parameter	old	new = count ÷ total
θ_A	$\frac{3}{4}$	$\frac{0.8+0.75}{0.8+0.75+0.4} = \frac{1.55}{1.95} = 0.795$
θ_B	$\frac{1}{4}$	$\frac{0.2+0.25}{0.2+0.25+0.6} = \frac{0.45}{1.05} = 0.429$
keep κ	$\frac{3}{4}$	$\frac{0.8+0.6}{2} = 0.7$
$p(\text{HHT})$	$\frac{15}{128} = 0.117$	0.139 (forward pass)

θ : heads credited ÷ flips credited; κ : expected keeps ÷ 2 moves.

All steps: Appendix, Section 6 (p. 125).



One Baum–Welch round raises $p(\text{HHT})$, as Theorem 4 (EM increases the likelihood) promises.

Next: when even the shortcut is too big



Forward-backward works because each coin touches only its neighbours. A hidden **chain of thought** (100 words a language model writes before its answer, each from a 50,000-word vocabulary) has $50,000^{100} \approx 10^{470}$ possible values, and each word depends on all earlier ones: **no small messages**.

How can we still get a soft label, such as HH's $p(A \mid \text{HH}) = 0.9$ (p. 28), without adding up every way?

guess 0.5

replay the game at random, count

skip the record

Answer. Replay and count: Monte Carlo, as in Lecture 4.

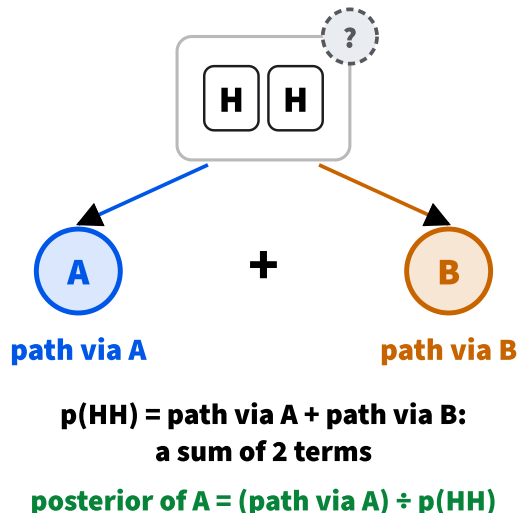
Replace the exact E-step by samples.

The E-step's sum can be impossible, yet running the model forward is cheap

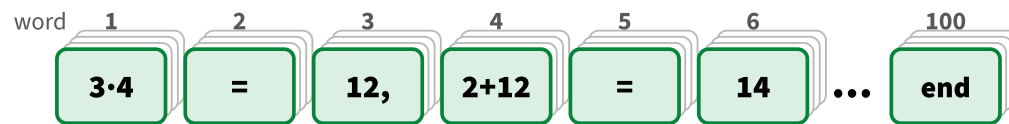
Goal: run EM when the hidden variable has far too many values to add up.

↳ Recall (p. 33): the E-step needs the **posterior** $p_{\theta}(z \mid x) = p_{\theta}(x, z) / \sum_{z'} p_{\theta}(x, z')$ of the hidden z given a record x such as HH, at the current guess θ . The sum has **one term per value** of z .

Coins: the hidden z has 2 values



Reasoning: the hidden z is a 100-word text



behind each word: a stack of 50,000 possible words

Sum over every z : $50,000 \cdot 50,000 \cdot \dots$ (100 times)
 $\approx 10^{470}$ terms, more than atoms in the visible universe

Run forward: pick one word at a time, left to right

100 easy steps; keep only the runs that reach the right answer, and the kept runs are posterior draws

Adding up every z can be hopeless; running forward and keeping the matches is cheap. Monte Carlo EM uses such draws.

Monte Carlo EM: draw m hidden values per record, then fit as if they were told

↳ Recall (p. 33): EM (expectation–maximization) needs the posterior $p_{\theta_t}(z \mid x)$ of the hidden z at each guess θ_t . When its sum over z is too large, **draw** z instead.

Definition (Monte Carlo EM; G. C. G. Wei and M. A. Tanner, J. Amer. Statist. Assoc., 1990). Records $x^{(1)}, \dots, x^{(N)}$; start from θ_0 ; choose m , the number of draws per record. For $t = 0, 1, 2, \dots$:

E-step: for each record n , draw m hidden values, $z^{(n,j)}$ = the j -th draw for record n (for example by replaying the model and keeping the matching runs):

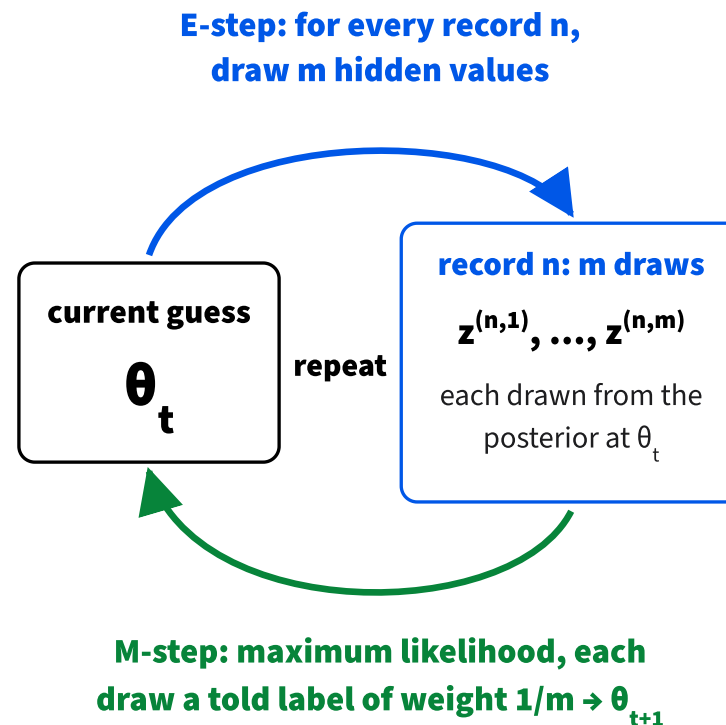
$$z^{(n,1)}, \dots, z^{(n,m)} \sim p_{\theta_t}(z \mid x^{(n)})$$

M-step: maximum likelihood with each draw as a told label of weight $\frac{1}{m}$:

$$\theta_{t+1} = \arg \max_{\theta} \sum_{n=1}^N \frac{1}{m} \cdot \sum_{j=1}^m \ln p_{\theta}(x^{(n)}, z^{(n,j)})$$

Why it matters: only draws are needed, **never the sum** over all z .

Each draw counts $1/m$; with many draws, the fraction equal to z nears its posterior: exact EM.



EM's M-step expectation becomes an average over m draws

↳ Recall Section 3 (p. 33) (EM = expectation–maximization): $q_t^{(n)}(z) = p_{\theta_t}(z | x^{(n)})$ is record n 's posterior (its soft label) at the guess θ_t .

Monte Carlo rule (law of large numbers): for any function g , $\mathbb{E}_{z \sim q}[g(z)] \approx \frac{1}{m} \cdot \sum_{j=1}^m g(z^{(j)})$, $z^{(j)} \sim q$ independent.

$$\sum_n \sum_z q_t^{(n)}(z) \cdot \ln p_{\theta}(x^{(n)}, z) \quad \text{EM's M-step objective}$$

$$= \sum_n \mathbb{E}_{z \sim q_t^{(n)}} [\ln p_{\theta}(x^{(n)}, z)] \quad \text{weights } q_t^{(n)}(z) \text{ sum to 1: an expectation}$$

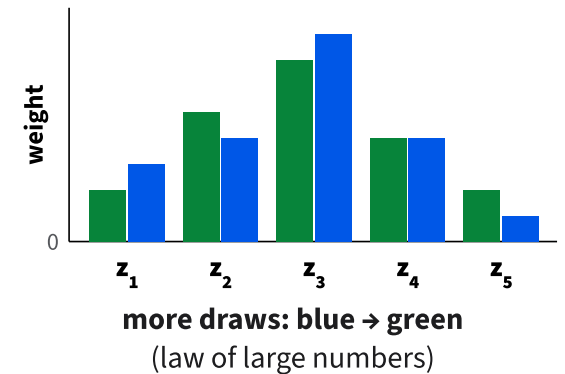
$$\approx \sum_n \frac{1}{m} \cdot \sum_{j=1}^m \ln p_{\theta}(x^{(n)}, z^{(n,j)}) \quad \text{Monte Carlo rule, } g(z) = \ln p_{\theta}(x^{(n)}, z), z^{(n,j)} \sim q_t^{(n)}$$

$$= \sum_n \sum_z \hat{q}^{(n)}(z) \cdot \ln p_{\theta}(x^{(n)}, z) \quad \text{group equal draws}$$

Line 3 is **Monte Carlo EM's M-step objective**; $\hat{q}^{(n)}(z) = \text{fraction of the } m \text{ draws equal to } z$.

two ways to weigh the values of z

■ posterior $q(z)$: the expectation
■ fraction of m draws: the average



In words: the same log-probability terms, with each posterior share $q_t^{(n)}(z)$ replaced by a fraction of draws $\hat{q}^{(n)}(z)$.

Why it matters: the M-step keeps EM's form, so whatever solved EM's M-step solves this one, with fractions in place of shares.

Same log-probability terms, new weights: each posterior share becomes a fraction of draws.

Draws act as told labels: Section 1's maximum likelihood

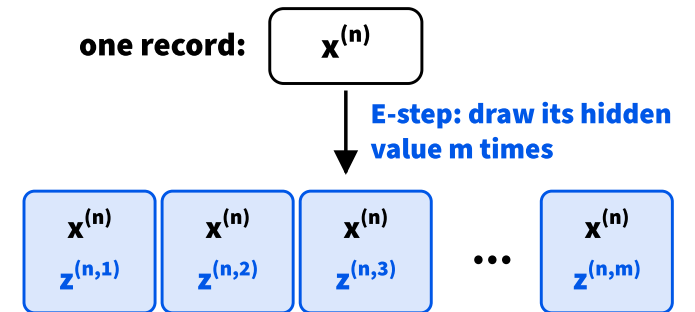
↳ Recall Section 1 (Theorem 1) (p. 11): when the friend tells every coin $z^{(n)}$, maximum likelihood maximizes $\sum_n \ln p_\theta(x^{(n)}, z^{(n)})$, and the answer is **count and divide**.

$$\begin{aligned}
 & \sum_{n=1}^N \frac{1}{m} \cdot \sum_{j=1}^m \ln p_\theta(x^{(n)}, z^{(n,j)}) && \text{Monte Carlo EM's M-step objective} \\
 &= \frac{1}{m} \cdot \sum_{\text{pairs } (n,j)} \ln p_\theta(x^{(n)}, z^{(n,j)}) && \text{one sum over the } N \cdot m \text{ pairs (record, draw)} \\
 &= \frac{1}{m} \cdot \ln \prod_{\text{pairs } (n,j)} p_\theta(x^{(n)}, z^{(n,j)}) && \text{a sum of logs is the log of a product} \\
 &= \frac{1}{m} \cdot \ln p_\theta(\text{pretend data}) && \text{the } N \cdot m \text{ pairs as independent told records}
 \end{aligned}$$

The factor $\frac{1}{m} > 0$ does not move the arg max: θ_{t+1} is **told-label maximum likelihood**.

In words: the drawn hidden values are used exactly as if a friend had told them.

Why it matters: no new optimizer. Any model we can fit with told labels, we can fit with Monte Carlo EM.



m told copies, each of weight 1/m

N records • m draws = Nm told pairs

M-step: fit as if the labels were told

(for the coins: count and divide)

Each draw is a told label, so the M-step is ordinary maximum likelihood on the pretend data.

Replay and keep: the kept runs are draws from the posterior

↳ Recall (p. 88): the E-step needs draws from the posterior $p_\theta(z \mid x) = p_\theta(x, z)/p_\theta(x)$, and $p_\theta(x) = \sum_{z'} p_\theta(x, z')$ is the sum that can be too large. Running the model **forward** needs no sum.

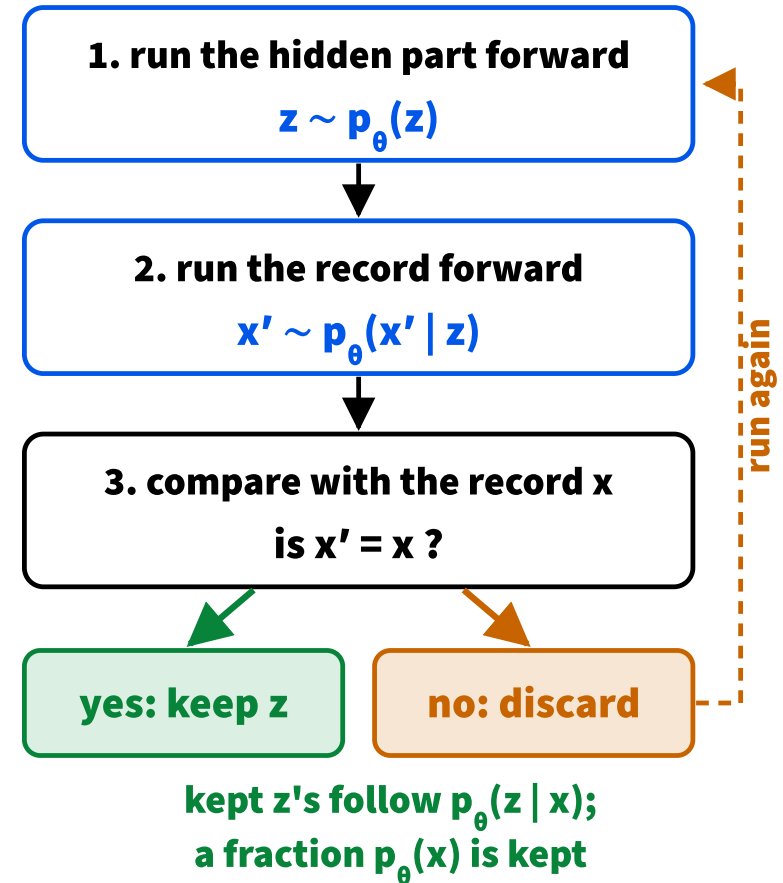
Proposition (replay and keep: rejection sampling). To draw the hidden z for a record x , repeat:

1. draw z from the model, $z \sim p_\theta(z)$;
2. generate a record from it, $x' \sim p_\theta(x' \mid z)$;
3. **keep** z only if $x' = x$.

The kept z 's are draws from the posterior $p_\theta(z \mid x)$.

Why: a value z is kept with chance $p_\theta(z) \cdot p_\theta(x \mid z)$ (be drawn, then reproduce x), which is proportional to its posterior.

Good: no sum over z is ever computed. **Cost:** only a fraction $p_\theta(x)$ of runs is kept, so **rare records waste most runs**.



Run forward, keep the matches: posterior draws, with no sum computed.

Proof: dividing by the keep rate gives the posterior

↳ **Proof of the proposition (replay and keep) (p. 91).** One run draws $z \sim p_\theta(z)$, then $x' \sim p_\theta(x' | z)$; it is **kept** when $x' = x$. For one run:

$$\begin{aligned}
 P(z \mid \text{kept}) &= \frac{P(z, \text{kept})}{P(\text{kept})} && \text{conditional probability} \\
 &= \frac{p_\theta(z) \cdot p_\theta(x \mid z)}{\sum_{z'} p_\theta(z') \cdot p_\theta(x \mid z')} && \text{draw } z; \text{ then } x' \text{ must equal } x; \text{ add over } z' \\
 &= \frac{p_\theta(x, z)}{\sum_{z'} p_\theta(x, z')} && \text{product rule} \\
 &= \frac{p_\theta(x, z)}{\underbrace{p_\theta(x)}} && \text{sum rule} \\
 &= \frac{p_\theta(x, z)}{P(\text{kept})} \\
 &= p_\theta(z \mid x) && \text{definition} \blacksquare
 \end{aligned}$$

Runs are independent, so the kept values are too.

In words: a value z survives in proportion to how well it explains x , which is exactly the posterior.

Three textbook rules take "draw, then match" to the posterior, with $p_\theta(x)$ only in the proof.

The three rules used:

conditional probability

$$P(z \mid \text{kept}) = P(z, \text{kept}) / P(\text{kept})$$

product rule

$$p_\theta(z) \cdot p_\theta(x \mid z) = p_\theta(x, z)$$

sum rule

$$\sum_{z'} p_\theta(x, z') = p_\theta(x)$$

Theorem 7: the noise in a fraction of m draws shrinks like $1/\sqrt{m}$

↳ Monte Carlo EM replaces each posterior share by a **fraction of m draws**. How far from the share can that fraction be?

Theorem 7 (Monte Carlo error). Let q = the exact soft label (chance of coin A). Draw the coin m times, each time A with chance q , and let the **drawn share** = (number of A's) / m . Then

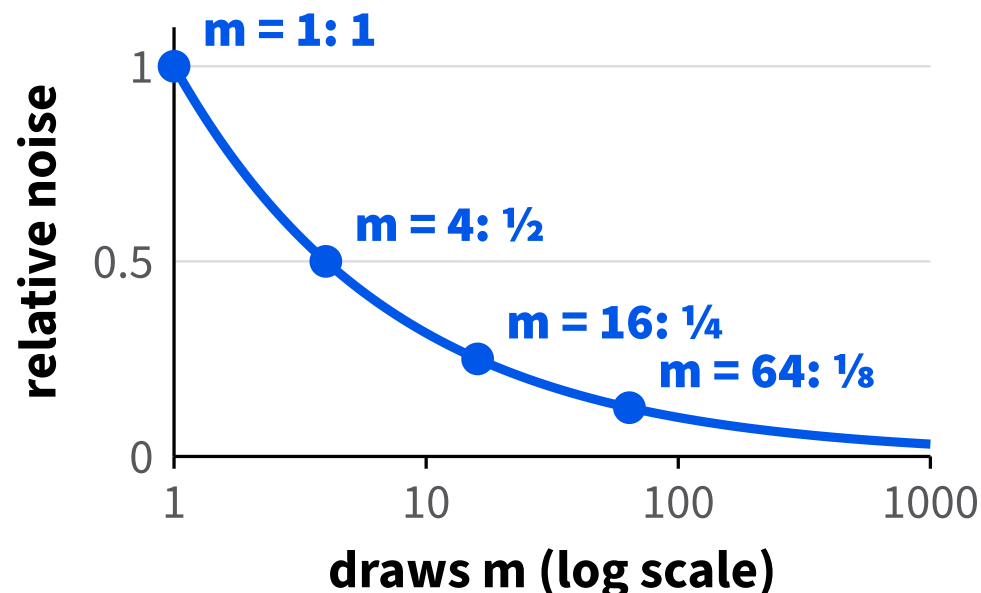
drawn share = q on average, typical error $\sqrt{q \cdot (1 - q) / m}$.

Four times the draws, half the error.

On HH ($q = 0.9$): 10 draws, e.g. A A A B A A A A A A, give share 0.9; repeating gives 0.8, 1.0, 0.9, ... around 0.9, typically off by $\sqrt{0.9 \cdot 0.1 / 10} \approx 0.095$. With 40 draws: ≈ 0.047 , half.

Why $\sqrt{m}$: averaging m independent draws divides the variance by m , so the typical error by $\sqrt{m}$. **Why it matters:** the M-step uses the drawn share, so more draws give a more precise EM step.

standard deviation $\div \sqrt{(q \cdot (1 - q))} = 1/\sqrt{m}$
any q : 4 times the draws, half the noise



Noise shrinks like $1/\sqrt{m}$: pick m for the precision you need; large m gives EM.

Proof of Theorem 7: indicators, then means and variances add

↳ **Setup.** Let $I_j = 1$ if draw j is that value, else 0; the fraction is $\frac{1}{m} \cdot \sum_j I_j$.

Mean:

$$\begin{aligned}\mathbb{E}\left[\frac{1}{m} \cdot \sum_j I_j\right] &= \frac{1}{m} \cdot \sum_j \mathbb{E}[I_j] && \text{linearity of expectation} \\ &= \frac{1}{m} \cdot m \cdot (1 \cdot q + 0 \cdot (1 - q)) = q && \mathbb{E}[I_j] = 1 \cdot q + 0 \cdot (1 - q) = q\end{aligned}$$

Standard deviation:

$$\begin{aligned}\text{sd} &= \sqrt{\text{Var}\left[\frac{1}{m} \cdot \sum_j I_j\right]} && \text{definition} \\ &= \sqrt{\frac{1}{m^2} \cdot \sum_j \text{Var } I_j} && \text{independent: variances add; } \frac{1}{m} \text{ out squared} \\ &= \sqrt{\frac{1}{m} \cdot \text{Var } I_1} && \text{same law for every } j: m \text{ equal terms} \\ &= \sqrt{\frac{1}{m} \cdot q \cdot (1 - q)} && \text{Var } I_j = \mathbb{E}[I_j^2] - q^2 = q - q^2 = q \cdot (1 - q) \quad (I_j^2 = I_j) \\ &= \sqrt{q \cdot (1 - q)/m} && \blacksquare\end{aligned}$$

In words: the fraction is right on average, and its noise falls like $1/\sqrt{m}$.

variance; 0 or 1

$$\begin{aligned}\text{Var } X &= \mathbb{E}[X^2] - (\mathbb{E}X)^2; I_j^2 = I_j, \text{ so} \\ \mathbb{E}[I_j^2] &= q\end{aligned}$$

scaling

$$\text{Var}(c \cdot X) = c^2 \cdot \text{Var } X$$

independent draws

$$\text{Var } \sum_j I_j = \sum_j \text{Var } I_j$$

The two chains give

$$\begin{aligned}\mathbb{E}\left[\frac{1}{m} \cdot \sum_j I_j\right] &= \frac{1}{m} \cdot m \cdot q = q \\ \text{Var}\left[\frac{1}{m} \cdot \sum_j I_j\right] &= \frac{1}{m^2} \cdot m \cdot q \cdot (1 - q) \\ &= \frac{q \cdot (1 - q)}{m}\end{aligned}$$

Mean q , standard deviation $\sqrt{q \cdot (1 - q)/m}$: right on average, noise $\propto 1/\sqrt{m}$.

Hard EM replaces each soft label by the single most likely hidden value

Definition (hard EM). At the guess θ_t , for each record n :

E-step: take the most likely hidden value (ties by a fixed rule)

$$\hat{z}^{(n)} = \arg \max_z p_{\theta_t}(z | x^{(n)}) = \arg \max_z p_{\theta_t}(x^{(n)}, z)$$

(same denominator $p_{\theta_t}(x^{(n)})$), and use the label 1 on $\hat{z}^{(n)}$, 0 elsewhere.

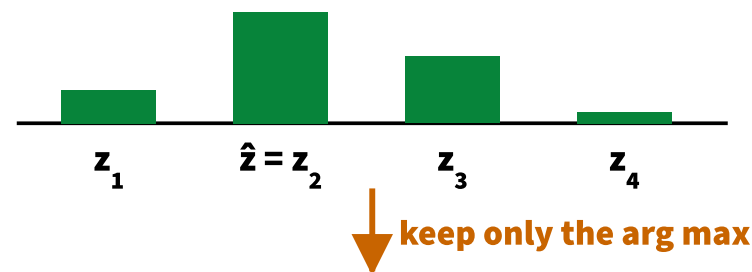
M-step: $\theta_{t+1} = \arg \max_{\theta} \sum_n \ln p_{\theta}(x^{(n)}, \hat{z}^{(n)})$, maximum likelihood with these told labels.

It is Monte Carlo EM with **one "draw" per record that is always the most likely value**, as in k-means (Section 5) (p. 58).

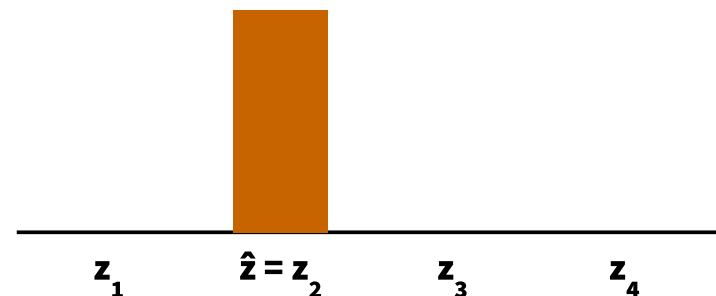
In words: each record goes entirely to its most likely hidden value.

Why it matters: no normalizing sum and no random draws are needed, but the label is **no longer the posterior**.

EM: soft label = the posterior of z



hard EM: weight 1 on $\hat{z}$, 0 elsewhere



Hard EM rounds each posterior to 0 or 1; it is cheap, but it throws the uncertainty away.

Theorem 4 does not cover hard EM: the log-likelihood can fall

↳ Recall Lemma 1 (p. 47): for any label q , $\ell(\theta) \geq \sum_z q(z) \cdot \ln \frac{p_\theta(x, z)}{q(z)}$ per record, with equality only when q is the posterior. Here $\ell(\theta) = \sum_n \ln p_\theta(x^{(n)})$.

$$\ell(\theta_{t+1}) \geq \sum_n \sum_z q^{(n)}(z) \cdot \ln \frac{p_{\theta_{t+1}}(x^{(n)}, z)}{q^{(n)}(z)}$$

Lemma 1, hard label $q^{(n)}$

$$= \sum_n \ln p_{\theta_{t+1}}(x^{(n)}, \hat{z}^{(n)})$$

$q^{(n)} = 1$ on $\hat{z}^{(n)}$; the 0 terms vanish

$$\geq \sum_n \ln p_{\theta_t}(x^{(n)}, \hat{z}^{(n)})$$

the M-step maximizes this sum

$$= \sum_n \left[\ln p_{\theta_t}(x^{(n)}) + \ln p_{\theta_t}(\hat{z}^{(n)} | x^{(n)}) \right]$$

product rule $p(x, z) = p(x) \cdot p(z | x)$

$$= \ell(\theta_t) - \sum_n (-\ln p_{\theta_t}(\hat{z}^{(n)} | x^{(n)}))$$

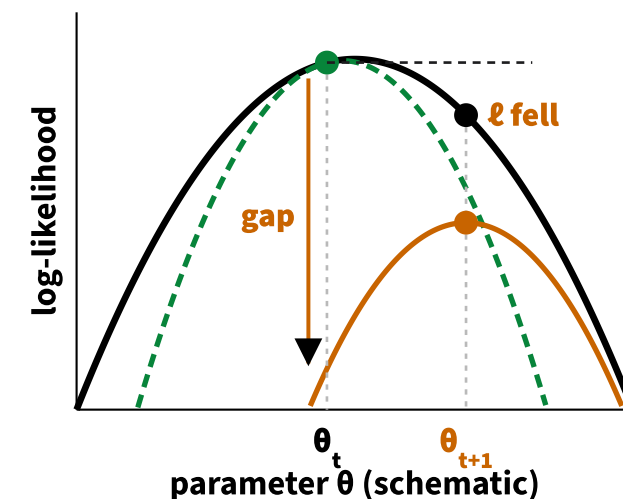
definition of ℓ ; orange = the gap

The **gap** is > 0 unless every posterior is certain. With EM's exact posterior it is 0, and the chain ends at $\ell(\theta_t)$: **Theorem 4 (p. 49)**.

In words: hard labels are not the posterior, so the bound no longer touches ℓ at the guess.

Why it matters: only $\ell(\theta_{t+1}) \geq \ell(\theta_t) - \text{gap}$ is guaranteed: hard EM can **end below its start**.

$\ell(\theta)$, the log-likelihood (black)
EM's bound (dashed): touches ℓ at θ_t
hard EM's bound: a gap at θ_t



Theorem 4's increase needs the exact posterior as the soft label; hard EM gives that up.

Replay the game and keep the runs that match the record: the kept runs follow the posterior

↳ **Proposition (replay and keep) (p. 91) applied to HH.** Replay the game at the guess $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$ (fair toss for a coin, flip it twice) and **keep** a run only if it shows HH. Shown: 32 replays at their expected counts; a fair toss sends $32 \cdot \frac{1}{2} = 16$ runs to each coin.

picked A
16 runs



A shows HH with chance $\frac{3}{4} \cdot \frac{3}{4} = 9/16 \rightarrow 9$ of A's 16 runs are kept

picked B
16 runs



B shows HH with chance $\frac{1}{4} \cdot \frac{1}{4} = 1/16 \rightarrow 1$ of B's 16 runs is kept

kept: $9 + 1 = 10$ runs; A made 9 of them $\rightarrow$ share $9/10 = 0.9$ = the posterior of A given HH

No sum was computed: we only ran the game forward and compared. This is **rejection sampling**. **Its cost:** $32 - 10 = 22$ of the 32 runs are thrown away. The keep rate $10/32 = 9/32 + 1/32$ is exactly $p(\text{HH})$, the sum of the two paths, which we never computed: the proposition's second claim. So **rare records waste most runs**.

Kept runs show A : B = 9 : 1, so each kept run's coin is a posterior draw: A with chance 0.9.

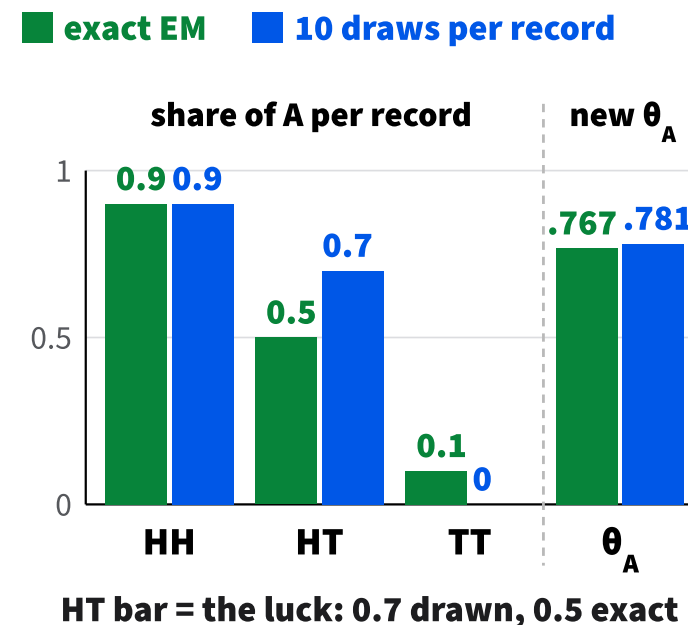
One Monte Carlo EM round with ten draws gives $\theta_A = 0.781$, close to EM's 0.767

↳ **Definition (Monte Carlo EM) (p. 88)** applied to the coins, $m = 10$. At $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$, we drew each record's coin 10 times with a computer's random numbers (A with chance 0.9, 0.5, 0.1, the posteriors) (p. 41). In this one run A came up 9, 7, 0 times; another run gives other counts. M-step: count and divide (Theorem 1) (p. 11) with these fractions.

record	fraction of A's	heads to A	flips to A
HH	$9/10 = 0.9$	$0.9 \cdot 2 = 1.8$	$0.9 \cdot 2 = 1.8$
HT	$7/10 = 0.7$	$0.7 \cdot 1 = 0.7$	$0.7 \cdot 2 = 1.4$
TT	$0/10 = 0$	$0 \cdot 0 = 0$	$0 \cdot 2 = 0$
total		2.5	3.2

$$\theta_A = \frac{\text{heads to A}}{\text{flips to A}} = \frac{2.5}{3.2} = 0.781 \quad \text{EM (recall): } \frac{2.3}{3.0} = 0.767$$

Coin B, from the remaining fractions 0.1, 0.3, 1: $\theta_B = 0.5/2.8 = 0.179$.



Then draw afresh at (0.781, 0.179) and repeat: EM with "share" replaced by "fraction of draws".

Theorem 7 on HT: ten draws give noise 0.158, and four times the draws halve it

↳ **Theorem 7 applied to HT.** At $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$ the posterior of A given HT is $q = \frac{1}{2}$ (recall) (p. 41); we drew $m = 10$ times.

$$\begin{aligned} \text{standard deviation} &= \sqrt{q \cdot (1 - q) / m} && \text{Theorem 7} \\ &= \sqrt{\frac{1}{2} \cdot \frac{1}{2} / 10} = 0.158 && \frac{1}{4} / 10 = 0.025, \sqrt{0.025} = 0.158 \end{aligned}$$

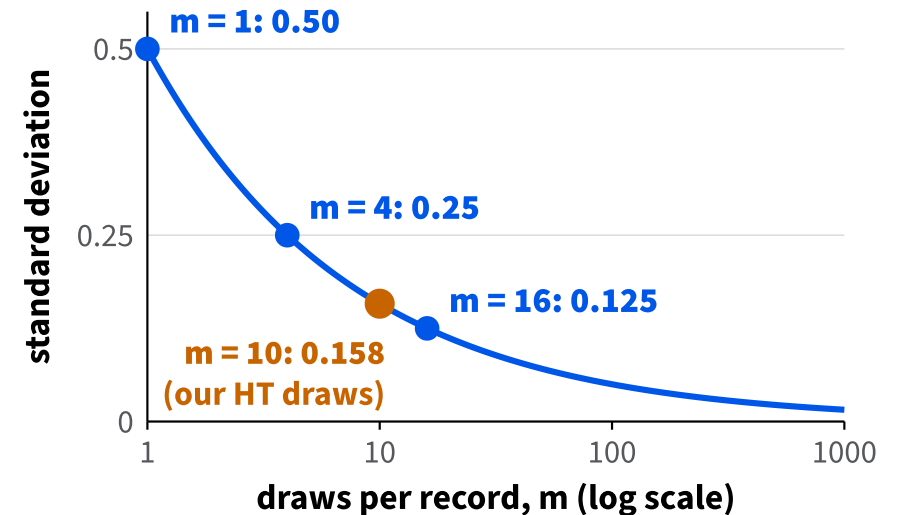
Ten draws typically miss by about 0.16, so HT's 0.7 (posterior 0.5) is **ordinary luck**, which the new guess inherits: $\theta_A = 0.781$, not EM's 0.767 (p. 42).

Hard EM instead: round, do not draw

Give each record wholly to its likelier coin: HH $\rightarrow$ A, HT $\rightarrow$ A (tie), TT $\rightarrow$ B. Counting then gives $\theta_A = \frac{3}{4}, \theta_B = 0$, and the labels stay A, A, B: hard EM is **stuck**. Its fit $\ell = -4.268$ is below the start's -4.000 (p. 28): rounding throws away information.

All steps: Appendix, Section 7 (p. 126).

HT, posterior $q = \frac{1}{2}$: standard deviation $0.5 / \sqrt{m}$
4 times the draws halves it: $0.50 \rightarrow 0.25 \rightarrow 0.125$



Draws add noise $\propto 1/\sqrt{m}$ that more draws remove; rounding to one coin can get stuck.

Next: the same records, three different answers. Which do we trust?

run on HH, HT, TT	start (θ_A, θ_B)	ends at	ℓ
EM (soft labels; recall Section 3) (p. 43)	$(\frac{3}{4}, \frac{1}{4})$	$(0.789, 0.211)$	-3.989 , best
hard EM	$(\frac{3}{4}, \frac{1}{4})$	$(\frac{3}{4}, 0)$, stuck	-4.268
EM , identical start	$(\frac{1}{2}, \frac{1}{2})$	$(\frac{1}{2}, \frac{1}{2})$, never moves	$3 \ln \frac{1}{4} = -4.159$

Identical start: each record has chance $\frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{4}$ (any two flips at $\frac{1}{2}$), so $\ell = 3 \cdot \ln \frac{1}{4}$; equal coins split every record $\frac{1}{2} : \frac{1}{2}$, so the guess never moves.

Swap the names: $\theta_A = 0.211$, $\theta_B = 0.789$. Is the fit better, worse, or the same?

the same

better

worse

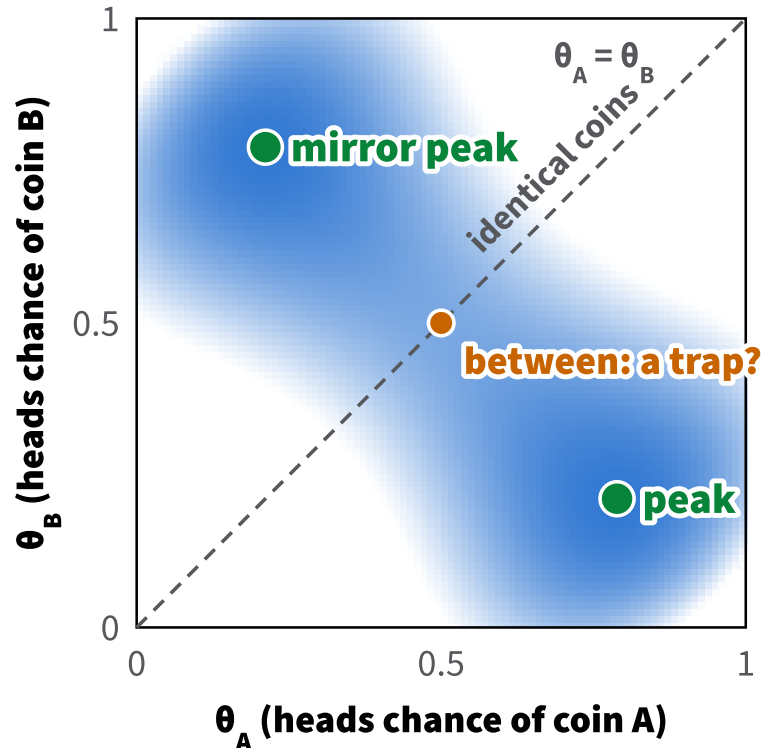
Answer. The same. $p(\text{HH}) = \frac{1}{2} \cdot 0.789^2 + \frac{1}{2} \cdot 0.211^2$ either way: the two paths just swap places, since each coin is picked with chance $\frac{1}{2}$. So every record's chance, and ℓ , is unchanged.

A fit can be stuck, or best yet not unique: which answer to trust is the next question.

The coin fit has two mirror-image peaks

ℓ : lower ℓ

higher ℓ



two equally high peaks, mirror images across the diagonal

Can a learned $\theta = (\theta_A, \theta_B)$ be trusted? Colour the **log-likelihood** ℓ of our records HH, HT, TT over every pair of heads chances.

Two equally high peaks appear, mirror images across the diagonal $\theta_A = \theta_B$ (identical coins), with an orange point between them.

Three questions for this section:

1. Why are the peaks mirror images?
2. Can EM get stuck between them?
3. Can the data always pick one answer?

First the general theory, then the key coin numbers.

Before trusting a fit, ask whether a different setting fits just as well.

Renaming the coins never changes the likelihood

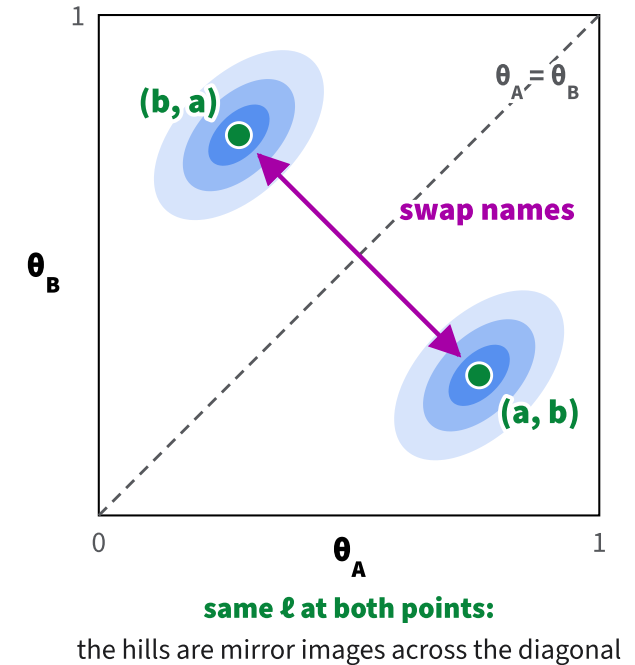
Label-switching lemma. If the friend picks each coin with chance $\frac{1}{2}$, then $\ell(\theta_A, \theta_B) = \ell(\theta_B, \theta_A)$ for every pair of heads chances.

Write $f(x | t)$ for the chance that one coin with heads chance t produces record x .

$$\begin{aligned}
 \ell(\theta_A, \theta_B) &= \sum_x \ln p_\theta(x) && \text{(independent records)} \\
 &= \sum_x \ln \left[\frac{1}{2} \cdot f(x | \theta_A) + \frac{1}{2} \cdot f(x | \theta_B) \right] && \text{(sum over hidden coin } z) \\
 &= \sum_x \ln \left[\frac{1}{2} \cdot f(x | \theta_B) + \frac{1}{2} \cdot f(x | \theta_A) \right] && \text{(addition commutes)} \\
 &= \ell(\theta_B, \theta_A) && \text{(line 2, names swapped)}
 \end{aligned}$$

In words: calling coin A "B" and coin B "A" leaves every record's probability unchanged.

Why it matters: every peak (a, b) has an equally high **mirror peak** (b, a) . With an unknown pick chance π , the same steps give $\ell(\pi, \theta_A, \theta_B) = \ell(1 - \pi, \theta_B, \theta_A)$.



Names are labels, not facts: the data see only the unordered pair of coins.

Identical coins stay identical under EM

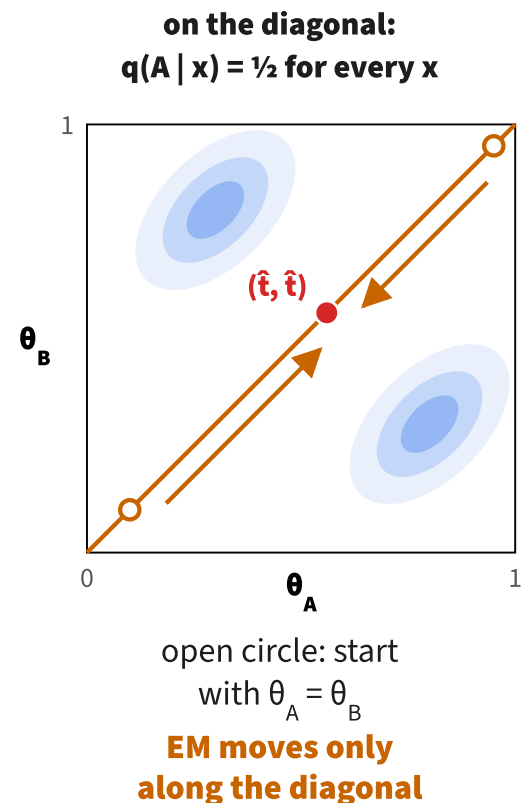
Setting. Pick chance $\frac{1}{2}$. Record x has h_x heads in n_x flips, so one coin with heads chance t gives it $f(x | t) = t^{h_x} \cdot (1 - t)^{n_x - h_x}$. Let $\hat{t} = \sum_x h_x / \sum_x n_x$: all heads over all flips.

Diagonal fact, part 1. From identical coins (t, t) , one EM step lands on $(\hat{t}, \hat{t})$; later steps stay there.

$$\begin{aligned}
 q(A | x) &= \frac{\frac{1}{2} \cdot f(x | t)}{\frac{1}{2} \cdot f(x | t) + \frac{1}{2} \cdot f(x | t)} = \frac{1}{2} && \text{(Bayes' rule, identical coins)} \\
 \theta_A^{\text{new}} &= \frac{\sum_x q(A | x) \cdot h_x}{\sum_x q(A | x) \cdot n_x} && \text{(M-step: credited heads } \div \text{ credited flips)} \\
 &= \frac{\frac{1}{2} \cdot \sum_x h_x}{\frac{1}{2} \cdot \sum_x n_x} = \hat{t} && (q(A | x) = \frac{1}{2} \text{ for every } x) \\
 &= \theta_B^{\text{new}} && (q(B | x) = \frac{1}{2} \text{ too: the same sums)} \\
 \Rightarrow (\hat{t}, \hat{t}) &\mapsto (\hat{t}, \hat{t}) && \text{(again identical coins: same steps, } t = \hat{t})
 \end{aligned}$$

In words: identical coins split every record half and half; both learn the pooled $\hat{t}$.

Why it matters: a start with $\theta_A = \theta_B$ never leaves the diagonal: EM cannot break the symmetry.



Identical starts stay identical: EM moves only along the diagonal.

Where EM stops on the diagonal, both slopes of ℓ are zero

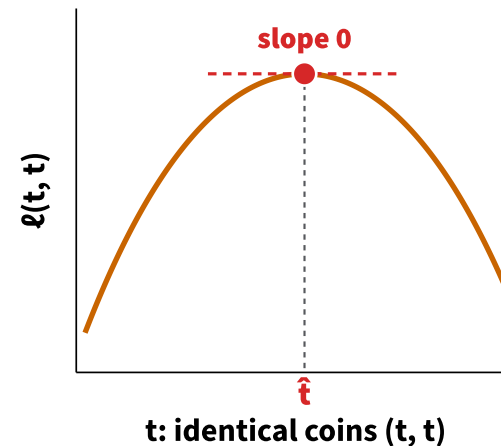
Recall (p. 102): $\ell = \sum_x \ln \left[\frac{1}{2} \cdot f(x | \theta_A) + \frac{1}{2} \cdot f(x | \theta_B) \right]$ with $f(x | t) = t^{h_x} \cdot (1 - t)^{n_x - h_x}$, one coin's chance of record x ; $\hat{t} = \sum_x h_x / \sum_x n_x$.

Diagonal fact, part 2. $(\hat{t}, \hat{t})$ is a **stationary point** of ℓ : $\partial \ell / \partial \theta_A = \partial \ell / \partial \theta_B = 0$ there.

$$\begin{aligned}
 \left. \frac{\partial \ell}{\partial \theta_A} \right|_{(t,t)} &= \sum_x \frac{\frac{1}{2} \cdot f'(x | \theta_A)}{\frac{1}{2} \cdot f(x | \theta_A) + \frac{1}{2} \cdot f(x | \theta_B)} \Big|_{(t,t)} && \text{(chain rule for } \ln; \text{ only A's term has } \theta_A) \\
 &= \frac{1}{2} \cdot \sum_x \frac{f'(x | t)}{f(x | t)} && \text{(at } (t, t) \text{ the two halves add to } f) \\
 &= \frac{1}{2} \cdot \frac{d}{dt} \sum_x \ln f(x | t) && (f'/f = (\ln f)') \\
 &= 0 \text{ at } t = \hat{t} && \text{(Theorem 1: best fit = heads/flips)}
 \end{aligned}$$

In words: on the diagonal each slope is half the slope of one coin's log-likelihood, which is flat at $\hat{t}$. The same steps give $\partial \ell / \partial \theta_B = 0$.

Why it matters: stationary is not the same as best: whether $(\hat{t}, \hat{t})$ is a peak or a trap depends on how ℓ curves across the diagonal.



one coin's ℓ ,
top at $\hat{t}$ (Theorem 1)

both slopes vanish there

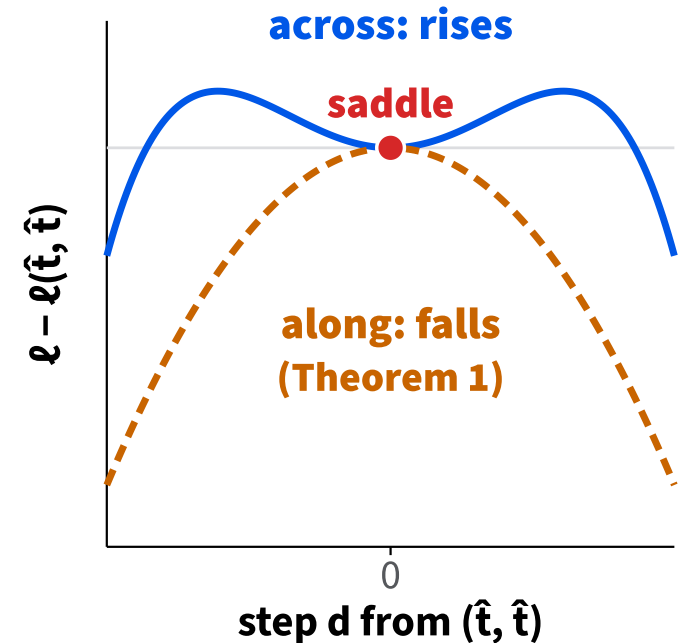
EM's resting point on the diagonal is flat in every direction, but not necessarily a peak.

Across the diagonal, ℓ can curve upward: a saddle

Saddle test. Along the diagonal, $\ell(t, t)$ is one coin's log-likelihood, largest at $t = \hat{t}$ (Theorem 1) (p. 11). Across it, ℓ rises if $\sum_x f''(x | \hat{t})/f(x | \hat{t}) > 0$; then $(\hat{t}, \hat{t})$ is a **saddle**.

$$\ell(\hat{t} + d, \hat{t} - d) \approx \ell(\hat{t}, \hat{t}) + \frac{d^2}{2} \cdot \sum_x \frac{f''(x | \hat{t})}{f(x | \hat{t})} \quad (\text{averaging } \pm d \text{ cancels the } d \text{ terms})$$

In words: along the diagonal it is a **top**, across it a **dip**. By the diagonal fact (p. 103), EM started there stops there anyway.



d^2 term: $\frac{1}{2} d^2 \sum f''/f$

sum $> 0 \Rightarrow$ saddle

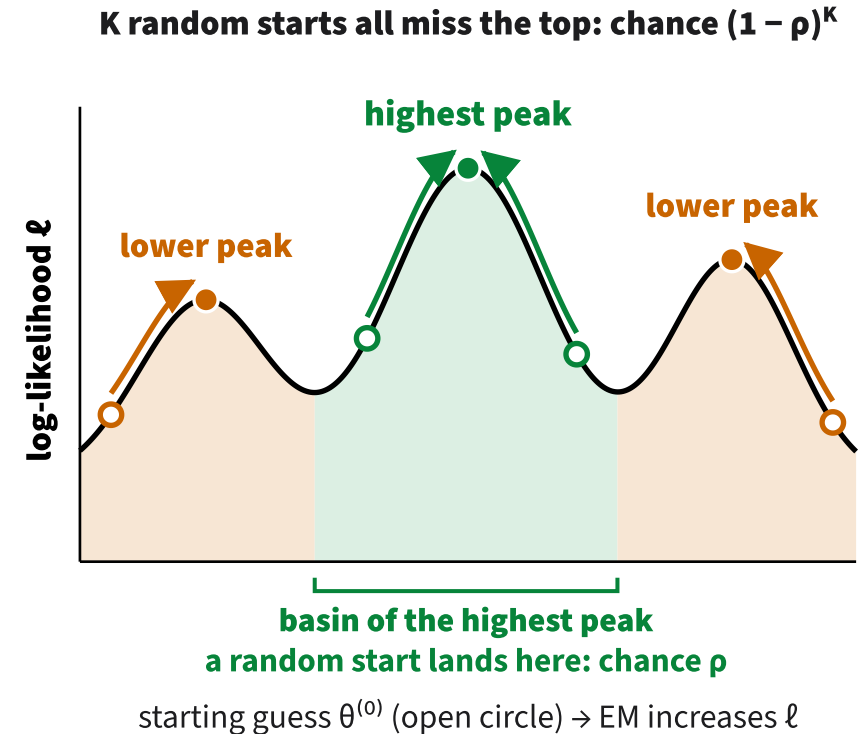
A saddle is a worse fit, and a symmetric start makes EM stop exactly there.

Random restarts reach the top with high probability

Theorem 4 (p. 49): ℓ never goes down, but EM may stop at a lower peak or a saddle.

Restart principle. Run EM from K independent random starts; keep the run with the largest ℓ .

If a random start reaches the top with chance $\rho > 0$, all K runs miss with chance $(1 - \rho)^K \rightarrow 0$.



Several random starts escape saddles and lower peaks; keep the largest ℓ .

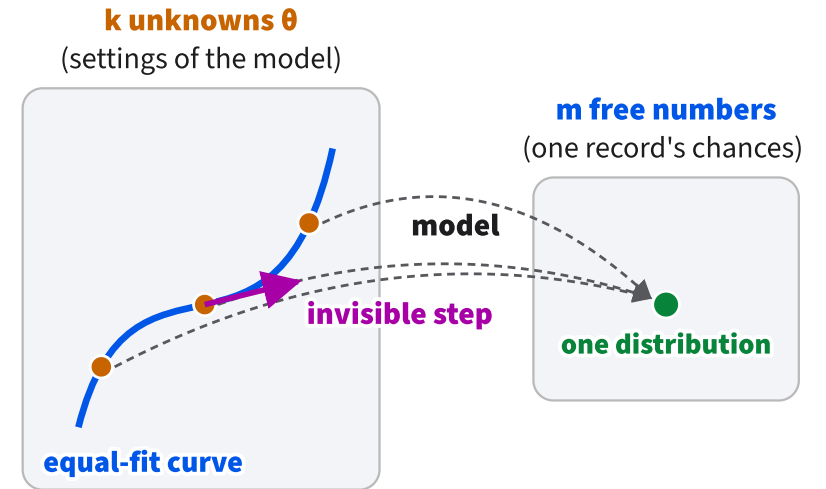
More unknowns than free numbers means no unique fit

↳ Question 3: can the data always pick one answer?

Theorem 8 (counting test). If a model has **more unknowns** than one record's distribution has **free numbers** (its chances, minus one because they sum to 1), then a **whole curve** of settings fits equally well.

Coins: 3 unknowns $\pi = p(A)$, θ_A , θ_B ; two flips give $p(0), p(1), p(2)$, summing to 1: **2 free numbers** < 3 .

Why: 3 knobs, but the data read only 2 dials. One direction of turning the knobs leaves both dials unchanged.



$k > m$: a whole curve of settings
lands on the same distribution of one record

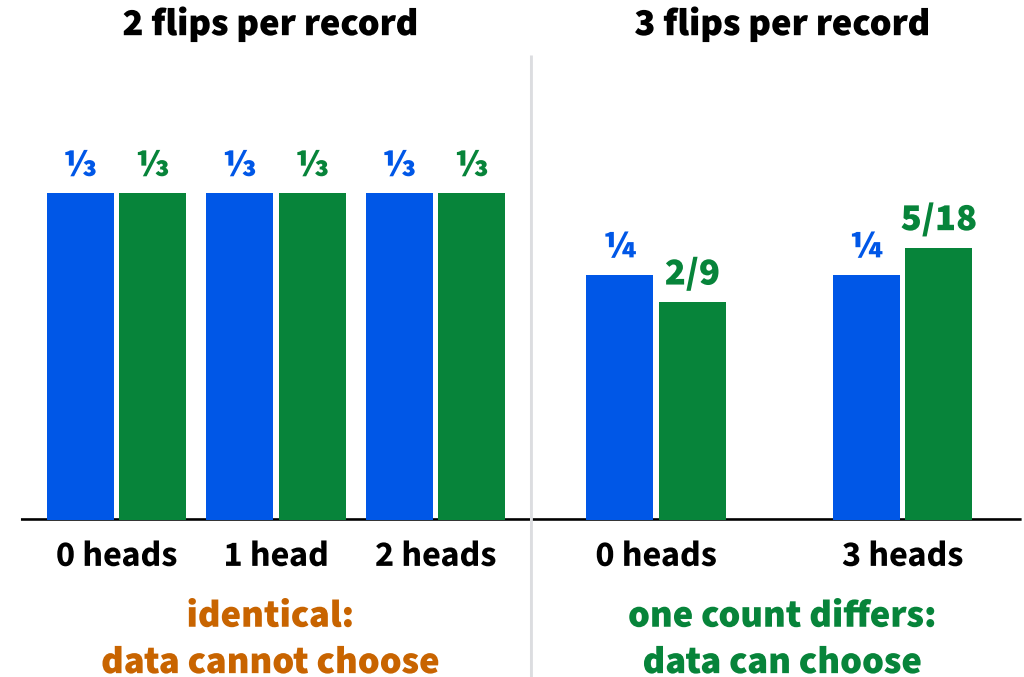
Count before you fit: no algorithm can choose between settings the data cannot tell apart.

With three flips per record, the two coins are pinned down

Theorem (Teicher). With **three or more flips** per record (and two really different coins), the data pin down π, θ_A, θ_B : only renaming $A \leftrightarrow B$ is left.

In words: two flips leave a curve of equal fits; three flips leave only a setting and its mirror.

Teicher, 1963; Allman, Matias & Rhodes, 2009.



■ **setting 1:** $\pi = \frac{1}{2}, \theta_A = 0.789, \theta_B = 0.211$

■ **setting 2:** $\pi = \frac{1}{4}, \theta_A = 1, \theta_B = \frac{1}{3}$

Observe more per record, and the curve of equal fits shrinks back to two mirror peaks.

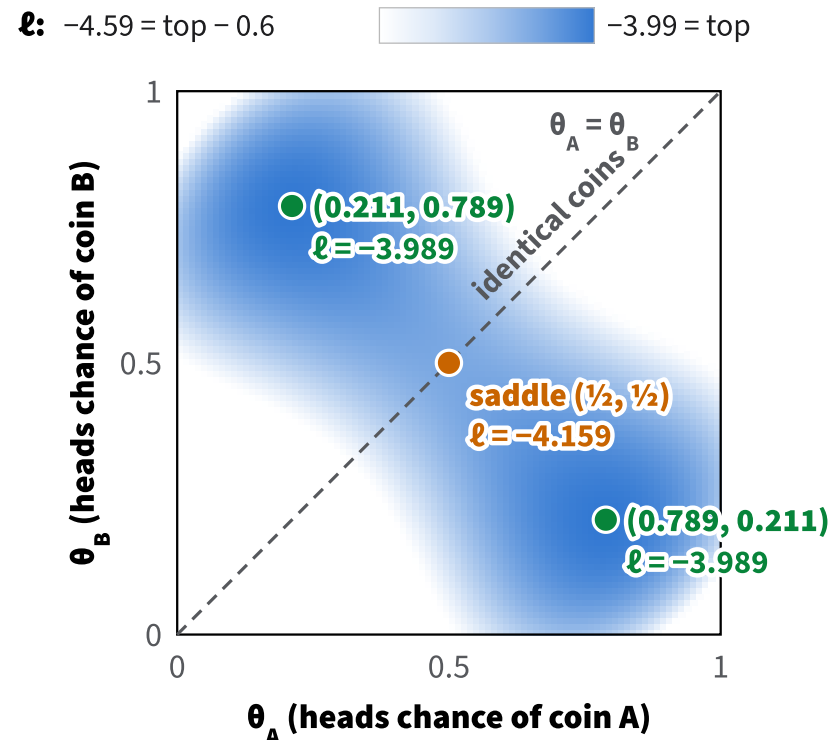
Both mirror peaks have $\ell = -3.989$; the saddle only -4.159

↳ EM's limit (p. 43): $\theta_A = 0.789$, $\theta_B = 1 - \theta_A = 0.211$, so $p(\text{HT}) = \theta_A \cdot \theta_B = \frac{1}{6}$ and $p(\text{HH}) = p(\text{TT}) = \frac{1}{2} \cdot (1 - \frac{2}{6}) = \frac{1}{3}$.

point (θ_A, θ_B)	$p(\text{HH})$	$p(\text{HT})$	$p(\text{TT})$	ℓ
peak $(0.789, 0.211)$	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{3}$	-3.989
mirror $(0.211, 0.789)$	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{3}$	-3.989
saddle $(\frac{1}{2}, \frac{1}{2})$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	-4.159

Peaks: $\ell = 2 \cdot \ln \frac{1}{3} + \ln \frac{1}{6} = -3.989$. Saddle: $\ell = 3 \cdot \ln \frac{1}{4} = -4.159$.

Why $(\frac{1}{2}, \frac{1}{2})$ is a saddle, step by step: Appendix, Section 8 (p. 127).



equal peaks, mirror images; the saddle between is lower

A start with $\theta_A = \theta_B$ stays at the worse saddle; restart off the diagonal.

Two different coin settings give exactly the same data

↳ Theorem 8 (p. 107) on the coins, two flips per record, $p(h)$ = chance of h heads.

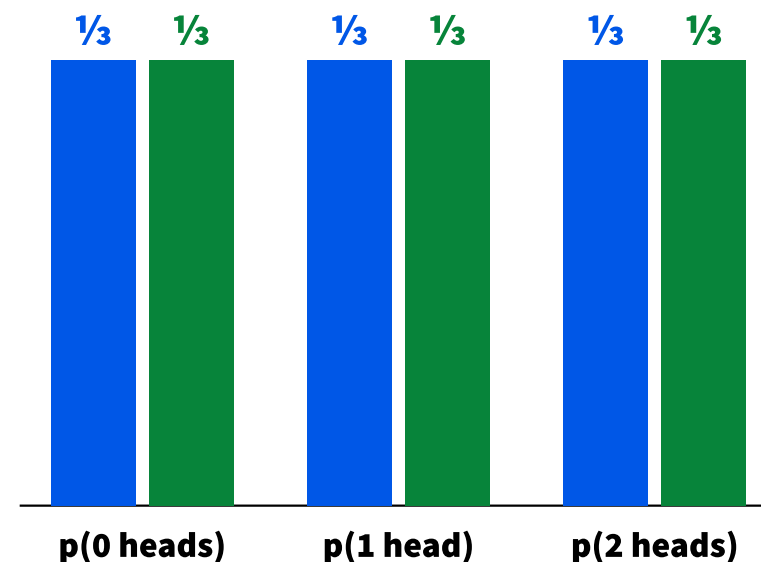
π, θ_A, θ_B	setting 1 (EM's fit): $\frac{1}{2}, 0.789, 0.211$	setting 2: $\frac{1}{4}, 1, \frac{1}{3}$
$p(2)$	$\frac{1}{2} \cdot (\theta_A^2 + \theta_B^2) = \frac{1}{3}$	$\frac{1}{4} \cdot 1 + \frac{3}{4} \cdot \frac{1}{9} = \frac{1}{3}$
$p(0)$	$= p(2) \quad (1 - \theta_A = \theta_B)$	$\frac{1}{4} \cdot 0 + \frac{3}{4} \cdot \frac{4}{9} = \frac{1}{3}$
$p(1)$	$1 - \frac{1}{3} - \frac{1}{3} = \frac{1}{3}$	$1 - \frac{1}{3} - \frac{1}{3} = \frac{1}{3}$

Same data, same $\ell = -3.989$: a **ridge** of equal fits. A **third flip** separates them: $p(3 \text{ heads}) = \frac{1}{4}$ vs $\frac{5}{18}$ (Teicher (p. 108)).

■ **setting 1:** $\pi = \frac{1}{2}, \theta_A = 0.789, \theta_B = 0.211$

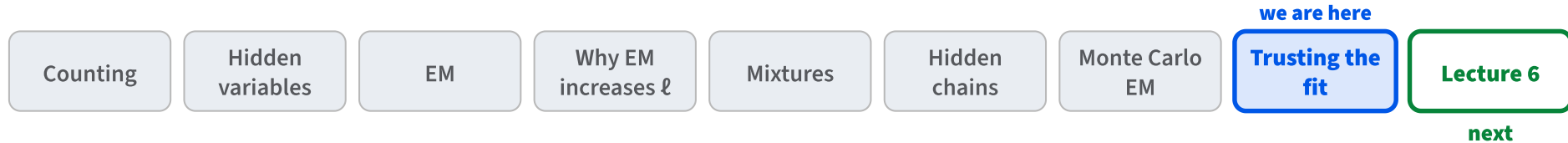
■ **setting 2:** $\pi = \frac{1}{4}, \theta_A = 1, \theta_B = \frac{1}{3}$

identical bars



Different knobs, identical readings: the data cannot choose, so no algorithm can.

Next lecture: when the exact posterior costs too much



Exact E-step

The expectation step (E-step) for coins, mixtures, chains: **sum** over every hidden value.

Sampled E-step

Monte Carlo EM: **draw** hidden values instead of summing.

Lecture 6

Pick the guess q (a distribution over the hidden values) from a simple family and **maximize** Lemma 1's bound (p. 47) $B(q, \theta) \leq \ln p_\theta(x)$.

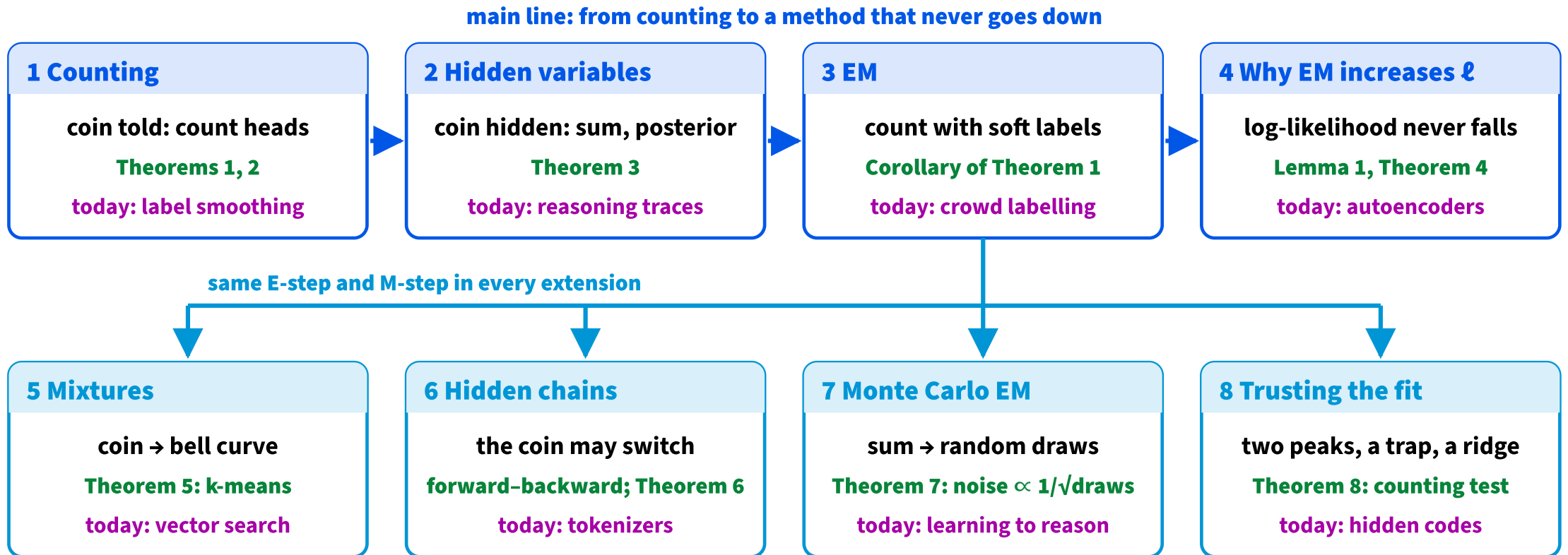
Coin game: the exact E-step sums 2 terms per record (coin A or B). With 16 hidden black-or-white pixels, how many?

32 65,536 16

Answer. $2^{16} = 65,536$: each pixel doubles the count. A 28×28 image has 784 pixels, so $2^{784} \approx 10^{236}$ terms, which **no computer can sum**. Lecture 6 approximates instead.

Today's bound $B(q, \theta) = \sum_z q(z) \cdot \ln \frac{p_\theta(x, z)}{q(z)}$ **becomes Lecture 6's main tool.**

One coin game, one method, four extensions



black: what changes · green: the result (proved or computed) · magenta: where it runs today

Exact EM never lowers the log-likelihood ℓ (Theorem 4); Sections 5–8 keep its two steps.

Count when the coin is told; when it is hidden, EM counts with soft labels

Section	Result
1 Counting	Theorem 1 (p. 11): $\hat{\theta} = \text{heads} \div \text{flips}$. Theorem 2 (Laplace) (p. 15): uniform prior $\Rightarrow$ posterior mean $(\text{heads} + 1) \div (\text{flips} + 2)$.
2 Hidden variables	$p_{\theta}(x) = \sum_z p_{\theta}(x, z)$ (p. 22); the posterior $p_{\theta}(z \mid x)$ is a path's share. Theorem 3 (Fisher) (p. 26): slope of $\ln p_{\theta}(x)$ = posterior-weighted path slopes.
3 EM	E : soft labels $q_t = p_{\theta_t}(z \mid x)$. M : count with them and divide (Corollary) (p. 37).
4 Why EM increases ℓ	Lemma 1 (p. 47): $\ln p_{\theta}(x) = B(q, \theta) + D_{\text{KL}}(q \parallel p_{\theta}(z \mid x)) \geq B$, equal at the posterior. Theorem 4 (p. 49): $\ell(\theta_{t+1}) \geq \ell(\theta_t)$.

E guesses the coin with the posterior, M counts with the guess, and ℓ never falls.

The same two steps for bells, chains and draws, plus when to trust the fit

Section	Result
5 Mixtures	E : each group's share (responsibility) of a point. M : share-weighted averages. Theorem 5 (p. 60) : width $\sigma \rightarrow 0$ gives k-means.
6 Hidden chains	Forward-backward (p. 71) : $\gamma_t(z) = \alpha_t(z) \cdot \beta_t(z) / p(x)$. Theorem 6 (Baum–Welch) (p. 78) : count with γ_t and pair labels ξ_t .
7 Monte Carlo EM	Proposition (p. 91) : kept runs are posterior draws. Theorem 7 (p. 93) : noise of the drawn share $= \sqrt{q \cdot (1 - q) / m}$.
8 Trusting the fit	Restart (p. 106) ; names can swap (p. 102) . Theorem 8 (p. 107) : more unknowns than free numbers $\Rightarrow$ a ridge . Teicher (p. 108) : three or more flips pin the coins (up to a swap).

Every extension keeps E (a share of a sum) and M (count with the shares).

Four questions: counting, posterior, soft counts, the increase

1. A coin shows 3 heads, 0 tails. Laplace's rule estimates its heads chance as...

1

4/5

3/4

Answer. 4/5: $\frac{h+1}{h+t+2} = \frac{3+1}{3+0+2}$. Plain counting says $3/3 = 1$: certainty from 3 flips.

2. $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$, fair pick. The record HHH came from coin A with chance...

9/10

27/28

3/4

Answer. 27/28: paths $\frac{1}{2} \cdot \left(\frac{3}{4}\right)^3 = \frac{27}{128}$ and $\frac{1}{2} \cdot \left(\frac{1}{4}\right)^3 = \frac{1}{128}$; A's share $27/(27+1)$.

3. Soft labels for coin A: HH 0.9, HT 0.5, TT 0.1. What θ_B does the M-step give?

0.7/3

2.3/3

0.1

Answer. 0.233. B's shares = 1–A's: 0.1, 0.5, 0.9.
 Heads: $0.1 \cdot 2 + 0.5 \cdot 1 + 0.9 \cdot 0 = 0.7$.
 Flips (2 per record): $0.1 \cdot 2 + 0.5 \cdot 2 + 0.9 \cdot 2 = 3$.
 $\theta_B = 0.7/3$ (heads ÷ flips, Corollary of Theorem 1) (p. 37).

4. One exact EM round takes ℓ from -5.20 to -5.31 . What happened?

a bug

normal

converged

Answer. A bug: Theorem 4 (p. 49) says $\ell(\theta_{t+1}) \geq \ell(\theta_t)$ in every exact round; ℓ fell by 0.11.

Bayes for the label, counting for the coin, and ℓ must never fall.

References (1 of 3): counting, hidden variables, EM, the bound

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Start here: Dempster, Laird and Rubin (1977) named EM and proved it increases the likelihood.

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For one book on today's lecture, read Bishop, Chapter 9: mixtures, k-means and EM.

References (3 of 3): draws, reasoning, identifiability

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Each paper runs expectation-maximization's two steps at scale: infer, then count.

The appendix follows: the few calculations the lecture skipped

The lecture showed one key calculation per idea. The appendix keeps **only the steps behind results the slides state without their arithmetic**, each as a short recap and a three-step table, two slides at most per section, in lecture order.

1 • Counting

complete in the main flow

2 • Hidden variables

complete in the main flow

3 • EM

why EM stops at 0.789

4 • Why EM increases the likelihood

better guess, higher floor;
one round goes up

5 • Mixtures

narrow bells give hard
labels

6 • Hidden chains

$p(\text{HHT}) = 15/128$; keep
chance 0.7

7 • Monte Carlo EM

hard EM gets stuck lower

8 • Trusting the fit

identical coins sit on a
saddle

Every number in the lecture is derived on its own slide or in the appendix.

Why EM stops at 0.789

Quick recap. θ_A, θ_B = how often coins A and B show heads. EM's rounds (p. 43) settle at $\theta_A = 0.789$. The data has three records, HH, HT, TT: each is **one third** of the data.

Step	Number	Why
1. Target	$p(\text{HH}) = \frac{1}{3}$	The best fit gives each record its share in the data.
2. Mirror coins	$\theta_B = 1 - \theta_A$	The data looks the same with heads and tails swapped.
3. Solve	$\theta_A = \frac{3+\sqrt{3}}{6} = 0.789$	The only value above $\frac{1}{2}$ that makes HH one third.

$$\frac{1}{2} \cdot \theta_A^2 + \frac{1}{2} \cdot (1 - \theta_A)^2 = \frac{1}{3} \Rightarrow \theta_A = 0.789, \theta_B = 0.211$$

EM stops where the model's shares equal the data's shares.

A better guess of the coin gives a higher floor

Quick recap. **Guess** $q(A)$ = how sure we are that HH came from coin A. **Floor** B = the lower estimate of the score $\ln p(\text{HH}) = -1.163$ that this guess builds (Lemma 1 (p. 47), the floor on HH (p. 52)). The true answer (posterior) is 0.9.

Step	Number	Why
1. Coin-flip guess $q(A) = \frac{1}{2}$	$B = -1.674$	Far from 0.9, so the floor sits low: gap 0.51.
2. Encoder guess $q(A) = 0.8$	$B = -1.208$	Close to 0.9, so the floor is near the score: gap 0.04.
3. Any guess	$B \leq -1.163$	Jensen's inequality: the floor is never above the score.

$$-1.674 < -1.208 < -1.163$$

Every guess gives a floor; the closer to the posterior, the smaller the gap.

One EM round on the coins: the score goes up

Quick recap. **Score** $\ell(\theta)$ = log-likelihood of HH, HT, TT. **Bound** B = a lower estimate of the score, built from the soft labels (Lemma 1 (p. 47)). One round moves the guess $\theta_0 = (\frac{3}{4}, \frac{1}{4})$ to $\theta_1 = (0.767, 0.233)$ (p. 53).

Step	Number	Why
1. Old score	$\ell(\theta_0) = -4.000$	Start here. With the old guess's labels, the bound equals the score.
2. Bound after the M-step	$B = -3.996$	The M-step picks θ_1 to make the bound as high as possible.
3. New score	$\ell(\theta_1) = -3.993$	The score is never below the bound.

$$-4.000 \overset{\text{equal}}{=} -4.000 \overset{\text{M-step}}{\leq} -3.996 \overset{\text{score} \geq \text{bound}}{\leq} -3.993$$

Equal, then up, then above: the score cannot fall (Theorem 4 (p. 49)).

Narrow bells turn EM's soft share into k-means' hard choice

Quick recap. The k-means example (p. 67): two bells (Gaussians) at 1 and 4, of width σ . The point 2 is 1 away from the near bell and 2 away from the far bell. **Share** = near height $\div$ both heights (EM's soft label).

Step	Number	Why
1. Wide, $\sigma = 2$	share 0.59	Both bells still reach the point, so it is shared.
2. Medium, $\sigma = 1$	share 0.82	The far bell falls off faster than the near one.
3. Narrow, $\sigma = 0.5$	share 0.998	The far bell is almost zero at the point.

$$\text{share} = \frac{e^{-1/(2\cdot\sigma^2)}}{e^{-1/(2\cdot\sigma^2)} + e^{-4/(2\cdot\sigma^2)}} \rightarrow 1 \text{ as } \sigma \rightarrow 0$$

As $\sigma \rightarrow 0$, the soft share becomes k-means' hard label.

Messages from both ends give $p(\text{HHT}) = 15/128$

Quick recap. Flips H, H, T from a hidden chain of coins (p. 75): keep the coin with $\frac{3}{4}$, switch with $\frac{1}{4}$; A shows H with $\frac{3}{4}$, B with $\frac{1}{4}$. **Forward** α = chance of the flips so far; **backward** β = chance of the flips still to come.

Step	Number	Why
1. Forward to flip 2	$\alpha_2 = (\frac{15}{64}, \frac{3}{64})$	Each flip adds up both ways of arriving at a coin.
2. Backward to flip 2	$\beta_2 = (\frac{3}{8}, \frac{5}{8})$	The last flip T is likelier from coin B.
3. Join at flip 2	$p(\text{HHT}) = \frac{15}{128}$	Past times future, summed over the two coins.
4. Soft labels for A	0.8, 0.75, 0.4	Each flip's $\alpha \cdot \beta$ for A, divided by $p(\text{HHT})$.

$$\frac{15}{64} \cdot \frac{3}{8} + \frac{3}{64} \cdot \frac{5}{8} = \frac{15}{128}$$

Two numbers per coin per flip replace the list of all $2^3 = 8$ coin sequences.

The new keep chance is 0.7, and the fit improves to 0.139

Quick recap. The M-step (p. 85) sets the **keep chance** κ to the average chance that the coin stayed the same between flips (pair labels, Theorem 6 (p. 78)). Start: $\kappa = \frac{3}{4}$, $p(\text{HHT}) = \frac{15}{128} = 0.117$ (p. 124).

Step	Number	Why
1. Kept, flips 1 → 2	0.8	H then H: the same coin explains both well.
2. Kept, flips 2 → 3	0.6	H then T: a switch to B is more plausible.
3. New keep chance	$\kappa = 0.7$	The average of the two moves.
4. New fit	$p(\text{HHT}) = 0.139$	Forward pass again with $\kappa = 0.7$, $\theta_A = 0.795$, $\theta_B = 0.429$.

$$\kappa = \frac{0.8+0.6}{2} = 0.7, \quad 0.117 \rightarrow 0.139$$

One Baum–Welch round raises the fit, as Theorem 4 (p. 49) promises.

Hard EM gets stuck below where it started

Quick recap. Hard EM (p. 99) gives each record only its likelier coin, not a soft label. Round 1 moved the guess to $(\theta_A, \theta_B) = (\frac{3}{4}, 0)$: coin B now always shows tails. **Score** ℓ = log-likelihood of HH, HT, TT; it started at -4.000 .

Step	Number	Why
1. Labels in round 2	A, A, B	B cannot show heads, so HH and HT go to A; TT goes to B.
2. Next guess	$(\frac{3}{4}, 0)$ again	Same labels give the same counts: EM is stuck.
3. Score there	$\ell = -4.268$	Lower than the start's -4.000 .

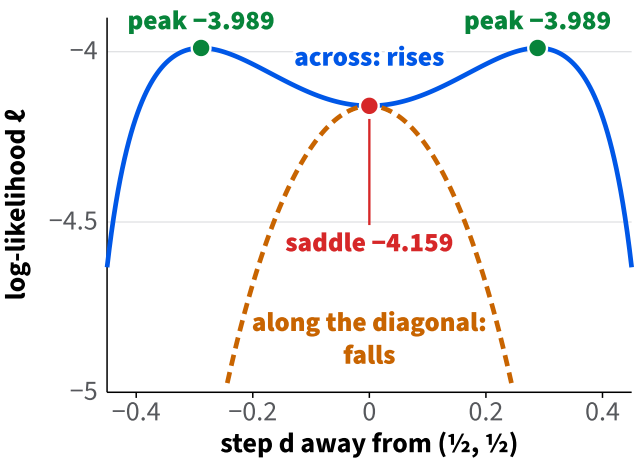
$$\ell = \ln \left(\frac{9}{32} \cdot \frac{3}{32} \cdot \frac{17}{32} \right) = -4.268 < -4.000$$

Hard labels are not the posterior, so Theorem 4 does not protect the score.

Two identical coins sit on a saddle

Quick recap. Two mirror peaks (p. 109); between them sit two identical fair coins, $(\theta_A, \theta_B) = (\frac{1}{2}, \frac{1}{2})$. **Saddle** (p. 105) = highest in one direction, lowest in another. **Along**: move both coins the same way; **across**: move them apart.

Step	Number	Why
1. Middle	$\ell = -4.159$	Each record has chance $\frac{1}{4}$: $3 \cdot \ln \frac{1}{4}$.
2. Along	$3 \cdot \ln(\frac{1}{4} - d^2)$	Any move lowers ℓ : highest at the middle.
3. Across	$\approx -4.159 + 4 \cdot d^2$	Any move raises ℓ : lowest at the middle.



the same point is lowest across and highest along

Highest along, lowest across: a saddle, a worse fit than the mirror peaks.